SITE AUTOMATION

# Settings Reference

Every setting of every job: its default, what it does, and where to set it

Site Automation - read it next to the setup guide | site-automation 0.37.0 | October 2026

**What this is.** Every setting the site-automation jobs read - the value it has when you set nothing, and what it does - with how to use one: where to put it (a template's Variables, `playbooks/group_vars/all.yml`, a group or a host), which place wins, and how to see what a job used.

**How to read it**

Keep it open next to the *Operations Runbooks Setup Guide*: each job's chapter there says which settings matter for it; this one lists them all, role by role.

Rendered from `docs/VARIABLES.md` and `docs/VARIABLES_REFERENCE.md` (generated from `roles/*/defaults/main.yml`) in the site-automation repository (version 0.37.0).

Host names and addresses in this guide are placeholders. Copy YAML from the repository files, not from the PDF.

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# 1. Settings: what they are, where they go, and who wins

A **variable** (also called a *setting*) is a named value the runbooks read: `check_disk_warn_pct: 85` means "warn when a disk is 85 % full". You change a variable to change how a check behaves, without touching any code. This page answers three questions:

1. **What can I set?** (find the name and the default)
2. **Where do I put it?** (for every host, for one group, for one host, for one run)
3. **What happens when the same setting is in two places?** (who wins)

You do not need to read it all. Do the two-minute version first.

## The two-minute version

1. **Find the setting.** Open VARIABLES_REFERENCE.md, press Ctrl+F and search for a word (`disk`, `certificate`, `mariadb`). Every setting is there with its default.
2. **Put the line in one file.**
   - Every host: `playbooks/group_vars/all.yml`
   - Hosts in one AAP group: `playbooks/group_vars/<group name>.yml`
   - One host: `playbooks/host_vars/<host name>.yml`
3. **Commit and Sync Changes** in VS Code. Run the check.

That is all for a normal change. The rest of this page explains the details and the exceptions.

**For one job template:** `aap/extra_vars/<playbook>.yml` (the list) has every setting that template reads, switched off, with its default and what it does. Copy the file into the template's **Variables** box (YAML selected), delete the `#` in front of the settings you change, and change their values. Switch on only what you change: a template's Variables beat `all.yml`.

## How to use a setting

Every job's chapter in these docs ends its settings part with an **All settings** link: the list of that job's settings in VARIABLES_REFERENCE.md, each with its default and what it does. Pick the place by **who the setting is for**:

| The setting is about | Put it in | Example |
|---|---|---|
| hosts (thresholds, what a check looks at) | `playbooks/group_vars/all.yml` (every host), `playbooks/group_vars/<group>.yml`, `playbooks/host_vars/<host>.yml` | `check_disk_warn_pct: 90` |
| one job template (who gets its email, its rules) | the job template's **Variables** box in AAP | `report_email_to: ops@yoursite.mil` on the certificate report template |
| one run | a **survey** question (its variable name = the setting's name) or **Prompt on launch** | a survey question `health_checks`, multiple choice `daily` / `weekly` |

Three examples:

1. playbooks/group_vars/all.yml - every host: disks warn at 90 %, a stricter audit log partition

```
check_disk_warn_pct: 90
check_disk_overrides:
  /var/log/audit: {warn: 70, crit: 85}

# 2. the "Certificate report" template's Variables - this job only: email it, stay green
report_email_to: [pki-team@yoursite.mil]
site_fail_on: []

# 3. playbooks/group_vars/win_web.yml - the Windows web servers only: IIS must run too
win_check_services_required_extra: [W3SVC, WAS]
```

- **A list replaces the default list** (it does not add to it): repeat the entries you keep. Details: "Three shapes of value" below.
- **Check what a job used:** in AAP, the job's **Details** tab shows the template's and the launch's variables; the job's output prints the decisions it made (thresholds in the finding texts, `REFUSED:` lines, the settings each report names in its email); its **artifacts** (Details > Artifacts) carry the results. A setting that seems ignored: If a setting does nothing.

## 1. What you can set

Every setting lives, with its default and an explanation, in the role that uses it: `roles/<role>/defaults/main.yml`. Two easy ways to read them:

- **VARIABLES_REFERENCE.md**, the same information as one searchable page (generated from the role files, so it is never out of date). Start here.
- In VS Code: **Ctrl+Shift+F** (search in all files), type the setting or a word, and look at the hits under `roles\...\defaults\main.yml`.

Setting names start with the thing they belong to:

| Starts with | Belongs to | Example |
|---|---|---|
| `check_disk_`, `check_services_`, `check_mariadb_` ... | one health check (`roles/check_<name>`) | `check_disk_warn_pct` |
| `win_check_` | one Windows health check | `win_check_disk_warn_pct` |
| `site_act_` | ACT (GenAI) | `site_act_level` |
| `site_` (other) | reporting and settings | `site_fail_on` |
| `patch_`, `win_patch_`, `automatic_restarts` | patching | `patch_never_patch_groups` |
| `servicenow_` | ServiceNow tickets | `servicenow_assignment_group` |
| `watch_` | service watch | `watch_containers`, `watch_discover` |
| `podman_discover_` | finding the podman containers (the `containers` check and service watch share it) | `podman_discover_ignore` |
| `poam_` | POA&M | `poam_stigman_api` |
| `stigman_` | STIG Manager deployment | `stigman_scope` |

Check names and what each finding means: RUNBOOKS.md.

### Three shapes of value

YAML

```
check_disk_warn_pct: 85                    # a number
check_fapolicyd_required: true             # true or false (lower case)
check_services_required: [sshd, crond]     # a list: [a, b] or one "- item" per line
check_disk_overrides:                      # a set of names with values (a "map")
  /var/log/audit: {warn: 70, crit: 85}
```

**A list replaces the default list. It does not add to it.** If the default is `[sshd, crond, rsyslog]` and you write `[chronyd]`, the hosts now require **only** `chronyd`. Copy the default list from the reference and add your entries to it. (Some settings have a separate `..._extra` version that *adds*, for example `check_services_required_extra`. The reference says which.)

### Job variables (per run, not per host)

These choose what a run does. Put them in a job template's **Variables** box (*Extra variables*) or in a survey question, never in a settings file:

| Variable | Meaning | Where it appears |
|---|---|---|
| `target` | which hosts or group to run on (default: all) | most playbooks |
| `health_checks` | which checks: `daily`, `weekly`, `all`, or a list | Health check |
| `ts_area`, `ts_service`, ... | what to troubleshoot | Troubleshoot |
| `use_act`, `site_act_level` | ask ACT, and what it may do | Health check, Troubleshoot |
| `site_fail_on` | which findings fail the job: `[critical]` or `[critical, warning]` | any check job |
| `patch_serial`, `win_patch_serial` | how many hosts are patched at the same time (1). **Only** here: in `all.yml` or a group's or host's variables it does nothing | Patch hosts, Windows patch |
| `stigman_replace_secrets` | `true` for ONE launch: replace the database secrets when the database is created again (SECRETS.md) | STIG Manager - deploy |

## 2. Where do I put a setting?

| Place | Who gets it | How you change it | Good for |
|---|---|---|---|
| `roles/<role>/defaults/main.yml` | everyone | do **not** edit (updates replace it) | the built-in default |
| `playbooks/group_vars/all.yml` | every host | VS Code, commit, sync | site-wide values: the LDAP server, the SIEM, thresholds |
| `playbooks/group_vars/<group>.yml` | the hosts of that AAP group | VS Code | one kind of server: STIG Manager containers, MariaDB, AAP itself |
| `playbooks/host_vars/<host>.yml` | that one host | VS Code (create the folder and file) | one odd server: a bigger disk limit for `/var/log/audit` |
| **AAP: inventory > Variables** | every host | AAP web UI | values you prefer to keep in AAP |
| **AAP: group > Variables** | the hosts of that group | AAP web UI | same, per group |
| **AAP: host > Variables** | that host | AAP web UI | same, per host |
| **Job template > Variables** (*Extra variables*) | every run of that template | AAP web UI | "this template always uses `site_act_level: diagnose`" |
| **Survey question / prompt on launch** | this launch only | asked at launch | a choice made each time |
| **AAP credential** | the job | AAP web UI | **secrets only** (SECRETS.md) |

**File names matter.** `playbooks/group_vars/<group>.yml` applies to the group with **exactly that name** in your AAP inventory, and `playbooks/host_vars/<host>.yml` to the host with exactly that name. A file for a group your inventory does not have is silently never used.

**A few settings belong to one host** - a network device's address, for example. They are at the end of the Settings Reference (settings of one host): set them in the host's Variables or `playbooks/host_vars/<host>.yml`, never in a template's Variables (there one value would be used for every host).

The files shipped in `playbooks/group_vars/` and the group each expects:

| File | Group name it applies to | If your group has another name |
|---|---|---|
| `all.yml` | every host | (always works) |
| `aap.yml` | `aap` | copy it to `aap_hosts.yml` if your group is `aap_hosts` (the example inventory's name) |
| `mariadb.yml` | `mariadb` | the MariaDB check also runs for a group named `mariadb_hosts`, but the **settings** come from the file named after the group: copy it to `mariadb_hosts.yml` |

| File | Group name it applies to | If your group has another name |
|---|---|---|
| `sat.yml`, `idm.yml`, `logstash.yml`, `stigman.yml`, `rhel8_all.yml` | the same name | copy or rename |
| `netapp.yml` | `netapp` | copy to `netapp_console_hosts.yml` if that is your group |

(The built-in safety lists cover both spellings of the AAP and NetApp group names, so AAP stays protected whatever you call it.)

**Updating the repository never overwrites your settings files.** The update script (`scripts/update-from-release.ps1` / `.sh`) protects `playbooks/group_vars/`, `playbooks/host_vars/`, `inventories/site/`, `poam/poam.csv` and anything in `.site-local`. If a new release adds a settings file you do not have, the script adds it **once** (the preview lists it as `YOURS`) and never changes it again. A setting that the release's `all.yml` mentions and yours does not is listed under `NEW SETTINGS`, so you can copy it over if you want it.

### Adding "extras" to a host or a group in AAP

Where the boxes are in AAP 2.7 (older versions word a label differently):

- **Inventory**: **Automation Execution › Infrastructure › Inventories** › your inventory › **Edit inventory** › **Variables**.
- **Group**: the same inventory › **Groups** tab › the group › **Edit group** › **Variables**.
- **Host**: the same inventory › **Hosts** tab › the host › **Edit host** › **Variables**.
- **Job template**: **Automation Execution › Templates** › the template › **Edit template** › **Extra variables** (the template's *Variables* box).

The **Variables** box on a host or group takes the same YAML as a settings file. Typical entries:

a group's Variables box: hosts in this group need a bigger disk margin

```
check_disk_overrides:
  /var/lib/pgsql: {warn: 70, crit: 85}
```

a host's Variables box: its MariaDB containers belong to a rootless account

```
check_mariadb_container_user: svc_podman
```

a group's Variables box: never check or watch the test containers, or anything of one user

```
podman_discover_ignore: ['test-.*', 'devuser/.*']
```

Connection settings (`ansible_host`, `ansible_user`, `ansible_python_interpreter`, `ansible_connection: winrm`) belong here too. The *login and password* never do: they are credentials (SECRETS.md).

**Prefer the project's settings files** (this repository), because they are reviewed and versioned in Git and updates never overwrite them. Use the AAP boxes when AAP is where your team already manages the inventory. Either is fine; the rule is **one setting, one place**.

## 3. Who wins when the same setting is in two places?

Strongest first. This order was **proven by running a test playbook** on ansible-core 2.16 and 2.21 with the same value set in every place and removing the winner one place at a time (the same result on both versions).

| # | Place | Notes |
|---|---|---|
| 1 | **Extra variables**: job template *Variables*, a survey answer, a prompt on launch, `-e` on the command line | beats everything below, including the built-in protection further down |
| 2 | `playbooks/host_vars/<host>.yml` | this project, one host |
| 3 | the **host's** Variables box in AAP | |
| 4 | `playbooks/group_vars/<group>.yml` | a child group beats its parent group |
| 5 | `playbooks/group_vars/all.yml` | |
| 6 | the **group's** Variables box in AAP | a deeper group beats its parent |
| 7 | the **inventory's** Variables box in AAP | |
| 8 | `roles/<role>/defaults/main.yml` | the built-in default |

Things that surprise people:

- **A file in this project beats the AAP group and inventory boxes**, even when the box is on a more specific group. If you type a value in an AAP box and see no effect, look for the same setting in `playbooks/group_vars/`. Every job prints a notice, `NOTE - settings in this project's files win over the same setting in AAP's inventory Variables box`, listing which files define which settings. (Silence it with `site_settings_notice: false`.) Not shown for `playbooks/host_vars/`.
- **A host's own Variables box (3) beats a group file (4/5)**, but a `playbooks/host_vars` file (2) beats the host's box.
- **A survey answer or template extra variable beats everything**: it is the one place that can override the rest for a single run.
- Two sibling groups (neither is inside the other) that set the same variable for the same host: do not rely on the result. Put it in a host file, or in one group.
- An inventory **sourced from the project** (`inventories/site/`, SETUP_AAP.md step 5): AAP imports its `group_vars/all.yml` into the inventory's Variables box and each `group_vars/<group>.yml` into that group's box. So they rank as places 7 and 6, and the project's `playbooks/` files beat them. Keep a setting in one of the two, not both.
- **Updated from 0.4.1 or older?** Your `playbooks/group_vars/all.yml` is yours, so the update left it as it was: it still sets `site_act_diagnose_only_groups`, `patch_never_patch_groups`, `patch_never_reboot_groups`, `site_act_provider` and `site_act_models` as real (uncommented) lines. They beat the same settings in AAP's inventory and group boxes. If you want those set in AAP (or the new built-in defaults), put `#` in front of those lines. See the upgrade notes in `CHANGELOG.md` (0.5.0).

### Built-in protection: settings that cannot be weakened from a file

Some lists are protected in code, so a mistake in a settings file cannot expose the AAP host:

| What | Built in | What your file can do |
|---|---|---|
| Never patched, never rebooted | `aap`, `aap_hosts` (`roles/patch/vars/main.yml`) | your `patch_never_patch_groups` / `patch_never_reboot_groups` list **replaces** the shipped one (`sat`, `netapp`, ...): repeat the names you keep. `aap` and `aap_hosts` stay protected whatever you write |
| ACT never fixes by itself | `aap`, `aap_hosts` (`roles/site_act/vars/main.yml`) | your `site_act_diagnose_only_groups` list **replaces** the shipped one (`sat`, `idm`, `netapp`, ...): repeat the names you keep. `aap` and `aap_hosts` stay protected whatever you write |

Only an extra variable (place 1) can override these, and nothing in these docs asks you to do that.

## Examples

**Everyone: warn earlier on disks.** `playbooks/group_vars/all.yml`:

```yaml
check_disk_warn_pct: 80
check_disk_crit_pct: 92
```

**One group: the STIG Manager hosts need these containers watched.**
`playbooks/group_vars/stigman.yml` (shipped; already has `watch_containers`).

**One host: a database server whose audit partition must stay emptier.**
`playbooks/host_vars/db01.example.mil.yml` (the file name is the host's name in AAP):

```yaml
check_disk_overrides:
  /var/log/audit: {warn: 70, crit: 85}
```

**One template: the "Troubleshoot" template always uses Ask Sage.** In the template's **Variables**:

```yaml
site_act_provider: asksage
```

**A list you want to extend.** Where the role has a `..._extra` setting, use it: it *adds*.

```yaml
check_services_required_extra: [tomcat]        # required on top of the built-in five
```

Where it has none, look up the default (`check_services_required` is `[sshd, crond, rsyslog, chronyd, auditd]`) and write the whole list with your addition:

```yaml
check_services_required: [sshd, crond, rsyslog, chronyd, auditd, tomcat]
```

## How to verify a setting is in effect

1. **Run the check on one host**: **Health check** with `target` = that host and `health_checks` = the check. The finding text quotes the limit it used (`/var is 96% full (critical at 92%)`), so you see the value that won.
2. **Break it on purpose** (or lower the limit) on a test host, and confirm the finding appears. A check that never fires proves nothing.
3. **Read the job's notice** (`NOTE - settings in this project's files ...`): it names every settings file that set something, so you can spot a file that overrides an AAP box.
4. **From the command line** (optional, in a terminal with Ansible and the repository): create a temporary file `playbooks/show_settings.yml`, **do not commit it**, and run it:

YAML

```yaml
- name: Show the settings a host gets
  hosts: "{{ target | default('all') }}"
  gather_facts: false
  tasks:
    - ansible.builtin.import_role: {name: check_disk}    # the role that owns the setting
      when: false                                        # loads its defaults, runs nothing
    - ansible.builtin.debug:
        msg: "{{ item }} = {{ lookup('vars', item, default='(not set)') }}"
      loop: [check_disk_warn_pct, check_disk_crit_pct]
```

shell

```
ansible-playbook -i inventories/site/hosts.yml playbooks/show_settings.yml -e
  target=db01.example.mil
```

It shows the value from the role default plus the files in this project. (Values typed into AAP's Variables boxes exist only inside AAP; there, use steps 1-3.)

## If a setting does nothing

| Symptom | Likely cause |
|---|---|
| You set it in AAP and nothing changes | the same setting is in `playbooks/group_vars/`: that file wins (table above) |
| You set it in a file and nothing changes | the file name does not match the group or host name exactly; a spelling or capital letter |
| A list lost entries | a list replaces the default: repeat the entries you keep |
| YAML error at the start of the job | indentation (spaces, never tabs), a missing space after `:`, or a value with `:` inside it that needs quotes |
| A survey answer is ignored | the survey question's **variable name** is not exactly the setting's name |
| `patch_serial` / `win_patch_serial` in a settings file changes nothing | Ansible picks how many hosts run at the same time before it reads any settings file: put it in the template's Variables, or a survey |

# 2. Every setting and its default

This page lists **every setting a role reads**, with the value it has when you set nothing, and what the role's own comment says. It is generated from `roles/<role>/defaults/main.yml`, so it is never out of date. Where to put a setting, and which place wins when it is in two, is in VARIABLES.md. Values are shortened to one line here: the role file has the full one.

A setting that is a list (`[a, b]`) **replaces** the default list when you set it; it does not add to it. Repeat the entries you want to keep.

**Contents:** aap_ee, aap_email_question, ad_lockout, ad_report, cert_inventory, check_accounts, check_auditd, check_certs, check_containers, check_disk, check_fapolicyd, check_logging, check_mariadb, check_mounts, check_network, check_patching, check_performance, check_selinux, check_services, check_sso, check_stig, check_time, check_trust, fapolicyd_containers, horizon, log_analysis, net_certs, net_config, net_perf, ontap_report, patch, poam, podman_discover, service_watch, servicenow, site_act, site_email, site_findings, stigman_stack, troubleshoot, vmware_vm, win_act, win_check_accounts, win_check_audit, win_check_certs, win_check_disk, win_check_eventlog, win_check_network, win_check_patching, win_check_performance, win_check_security, win_check_services, win_check_time, win_check_trust, win_patch, win_troubleshoot, settings of one host

## aap_ee

File: `roles/aap_ee/defaults/main.yml`

> **Note**
> SET THE EXECUTION ENVIRONMENT (playbooks/aap_ee.yml): puts one execution environment on many job templates at once, through AAP's API - after you add a new EE to AAP (docs/EXECUTION_ENVIRONMENT.md, chapter 6). Run it with Check first: it lists every template it would change, from which EE to which. Later rebuilds need no run: change the EE's Image in AAP (the next tag), and every template on it follows.

| Setting | Default | What it does |
|---|---|---|
| `aap_ee_name` | (empty) | The execution environment to put on the templates: its name in AAP (Automation Execution > Infrastructure > Execution Environments), or its ID when two have that name. |
| `aap_ee_project` | (empty) | Which job templates (at least one of the three; never every template in AAP): the templates of this AAP project (its name): your site-automation project |
| `aap_ee_from` | `[]` | Only the templates on one of these execution environments now (names or IDs); none = those with no execution environment set, which use the project's or organization's default (left as they are otherwise). Empty = any. |
| `aap_ee_templates` | `[]` | more job templates, by name or ID - whatever they are on |
| `aap_ee_exclude` | `[]` | templates left out, by name or ID (jobs that stay on another EE) |

| Setting | Default | What it does |
|---|---|---|
| aap_ee_ca_path | (empty) | a CA file for AAP's HTTPS certificate ("" = what the execution environment trusts) |

## aap_email_question

File: roles/aap_email_question/defaults/main.yml

**Note**
THE "ALSO EMAIL THE REPORT TO" QUESTION (playbooks/aap_email_question.yml): adds a text box to the survey of every report template and of the workflows that run them, through AAP's API, so whoever launches one can send that run's report to more people (report_email_also, added to the template's report_email_to). Run it once, and again after you add report templates. docs/EMAIL_REPORTS.md, "Also email the report to".

| Setting | Default | What it does |
|---|---|---|
| aap_email_question_state | present | present = add the question; absent = take it out again |
| aap_email_question_playbooks | - act_fix_approved.yml - ad_health_report.yml - ad_lockout.yml - cert_report.yml - databa… | Which templates: the job templates whose playbook is one of these (the file name - your template names do not matter). Every playbook that can email a report is here; a test keeps the list complete. |
| aap_email_question_templates | [] | more job templates or workflows, by name or ID |
| aap_email_question_exclude | [] | templates left out, by name or ID |
| aap_email_question_workflows | true | also the workflows that run them (a job in a workflow is never asked its own survey) |
| aap_email_question_project | (empty) | only job templates of this AAP project (its name); empty = any project |
| aap_email_question_ca_path | (empty) | a CA file for AAP's HTTPS certificate ("" = what the execution environment trusts) |
| aap_email_question_text | Also email the report to | The question, as whoever launches a template sees it. A text box of several lines (a pasted list keeps its line breaks), not required, empty by default. |
| aap_email_question_description | (worked out by the role; see its file) | The help text under the question. |

## ad_lockout

File: `roles/ad_lockout/defaults/main.yml`

**Note**
Account lockout (playbooks/ad_lockout.yml): is the account locked out, and WHAT is locking it out - the computers its bad passwords came from, from the Security log of every domain controller. Read-only, unless ad_lockout_unlock. A PowerShell script on every DC (WinRM, the Windows Machine credential): the account needs to read the DCs' Security logs - see docs/ACTIVE_DIRECTORY.md, "Account lockout".

| Setting | Default | What it does |
|---|---|---|
| `ad_lockout_user` | (empty) | The account to look at, as the service desk is given it: jdoe, CORP\jdoe or jdoe@corp.example.mil (a survey question, NOT required). Several: one per line, or separated by commas. "" = every account locked out now, each with the computer locking it out. |
| `ad_lockout_hours` | `24` | the Security log entries of the last this-many hours (raise it for a lockout of last week) |
| `ad_lockout_max_users` | `25` | at most this many accounts (asked for, or locked out now) in one run |
| `ad_lockout_unlock` | `false` | true = unlock the account (on the PDC emulator) after reading - only an account asked for by name, and only when it is locked out. A dry run (Check) says what it would unlock. It locks out again as soon as the source computer tries its old password again: the report says what to fix first. |
| `ad_lockout_check_sources` | `false` | true = also look on each source computer that is a host of this job's inventory (Windows, WinRM, the same credential - it needs administrator rights there): services and scheduled tasks running as the account, its sessions (a disconnected Remote Desktop session), processes and mapped drives. Read-only. |
| `ad_lockout_check_max` | `5` | look on at most this many source computers per account (the worst first) |
| `ad_lockout_exchange_servers` | `[]` | Your mail servers (Exchange / ActiveSync, by name or address, * and ? allowed, e.g. [EXCH*, 192.0.2.25]): bad passwords coming through one of them are a phone or a mail app with the old password. |
| `ad_lockout_max_events` | `2000` | ---- How much is read ---------- read at most this many Security log entries per account and DC (the newest) |
| `ad_lockout_resolve_max` | `50` | resolve at most this many addresses to names per DC (reverse DNS) |
| `ad_lockout_max_rows` | `50` | rows per table; the job's artifacts (ad_lockout) have all |

## ad_report

File: `roles/ad_report/defaults/main.yml`

> **Note**
> Active Directory health report (playbooks/ad_health_report.yml). Read-only: a PowerShell script on every domain controller (WinRM, the Windows Machine credential) and one more on one DC per domain. The account needs to read AD and run dcdiag / repadmin on the DCs: see docs/ACTIVE_DIRECTORY.md.

| Setting | Default | What it does |
|---|---|---|
| ad_report_hours | 24 | ---- Changes ---------- the changes of the last this-many hours (match it to the schedule: a daily job = 24) |
| ad_event_hours | 24 | the event log entries of the last this-many hours |
| ad_security_events | true | read each DC's Security log for WHO made a change (needs auditing of account and security group management) |
| ad_max_events | 2000 | read at most this many entries per event log and DC |
| ad_changes_max | 500 | look at most at this many changed objects per kind (created, changed accounts, GPO links) |
| ad_privileged_groups_extra | [DnsAdmins] | Privileged groups watched besides the built-in ones (Domain / Enterprise / Schema Admins, Administrators, Account / Server / Print / Backup Operators, Group Policy Creator Owners, Key Admins, Cert Publishers): names or SIDs of this domain's groups. |
| ad_groups_should_be_empty | [] | privileged groups that must have no member, e.g. [Schema Admins] - amber when they have one |
| ad_privileged_change_severity | warning | a member added to / removed from a privileged group: warning, critical or info |
| ad_schema_change_severity | warning | the schema version changed: warning, critical or info |
| ad_dc_change_severity | info | a domain controller added or removed: warning, critical or info |
| ad_dcdiag_skip | [SystemLog] | ---- What is read on each DC ---------- dcdiag tests to skip (SystemLog fails on any error in the System log: the Events table covers it) |
| ad_dcdiag_dns | true | also dcdiag's DNS tests that only read (basic, forwarders, delegation, record registration) |
| ad_trust_verify | true | verify outbound and two-way trusts (nltest /sc_verify) |

| Setting | Default | What it does |
|---|---|---|
| ad_repl_warn_hours | 6 | ---- Thresholds: at or above "warn" = amber, at or above "crit" = red ---------- a replication link with no success for this long (inter-site schedules: keep above their interval) |
| ad_repl_crit_hours | 24 | no success for this long (or a failing link) -> red |
| ad_repl_queue_warn | 50 | replication requests waiting in a DC's inbound queue |
| ad_time_warn_seconds | 5 | a DC's clock against the PDC emulator (the PDC: against its NTP source) |
| ad_time_crit_seconds | 60 | clock offset -> red (Kerberos fails at 300) |
| ad_time_sync_warn_hours | 24 | Windows Time last synchronized longer ago than this |
| ad_disk_warn_pct | 15 | free space of the database, log and SYSVOL volumes, % (at or below) |
| ad_disk_crit_pct | 5 | free % -> red |
| ad_cert_warn_days | 30 | the LDAPS / KDC certificate expires within this many days |
| ad_cert_crit_days | 7 | expires within this many days -> red |
| ad_backup_warn_days | 7 | the last system-state backup of a partition is older than this |
| ad_backup_crit_days | 30 | older than this -> red (always red past the tombstone lifetime) |
| ad_krbtgt_max_days | 180 | the krbtgt password is older than this -> amber (the STIG asks for 180) |
| ad_rid_warn_pct | 10 | RID pool left, % (at or below) |
| ad_rid_crit_pct | 2 | RID pool left -> red |
| ad_no_client_site_warn | 1 | this many client addresses in no AD subnet -> amber (fewer: blue) |
| ad_report_max_rows | 50 | rows per table (problems first); the job's artifacts have all |
| ad_report_fail | false | The job stays green when it finds problems: the report and email say what. true = mark it FAILED when there is a critical finding or a DC could not be read, for workflows. |

## cert_inventory

File: roles/cert_inventory/defaults/main.yml

**Note**
CERTIFICATE INVENTORY: every certificate report (Linux, Windows, VMware, network devices) can also write what it found - every certificate, with the same fields from each - for a dashboard (Power BI ...). Two ways, each on by itself: files (CSV + JSON) and a SQL Server table. A Check run writes nothing. The fields: id, source, host, location, subject, issuer, sans, not_after, days, status, tier (root / intermediate / ca-signed / self-signed), trust_store, in_use, public_key, sig_alg, serial, sha1, sha256, scanned_at, job (docs/CERT_INVENTORY.md).

| Setting | Default | What it does |
|---|---|---|
| `cert_inventory_files` | `false` | ---- files: <source>-certificates.csv and .json (source: linux, windows, vmware, net), replaced each run ---- true = write the files |
| `cert_inventory_dir` | `/var/lib/site-automation/cert-inventory` | where: on the AAP node the job runs on (a path AAP exposes to jobs: docs/CERT_INVENTORY.md), or on cert_inventory_host |
| `cert_inventory_host` | (empty) | an inventory host to write them on instead (a file server, or a Linux host with the share mounted); its credential must be on the job template |
| `cert_inventory_history` | `false` | true = also a dated copy (<source>-certificates-YYYY-MM-DD.csv) next to them, for trends |
| `cert_inventory_sql` | `false` | The login comes from the "SQL Server (certificate inventory)" credential (aap/credential_types/cert_inventory_sql.yml). The execution environment needs pymssql and community.general (the all-in-one EE has both). true = add / update the rows |
| `cert_inventory_sql_server` | (empty) | the SQL Server (name or address) |
| `cert_inventory_sql_port` | `1433` | its port |
| `cert_inventory_sql_database` | (empty) | the database (SQL Server 2016 or later: the rows go in as JSON, OPENJSON) |
| `cert_inventory_sql_table` | `dbo.cert_inventory` | schema.table |
| `cert_inventory_sql_create` | `false` | true = create the table (and its view <table>_current) when it is not there - the account then needs CREATE TABLE |

## check_accounts

File: `roles/check_accounts/defaults/main.yml`

**Note**
ACCOUNTS: the account problems auditors and attackers both look for, plus the one that quietly breaks automation - a service account password about to expire.

| Setting | Default | What it does |
|---|---|---|
| `check_accounts_min_uid` | `1000` | people's accounts start here |
| `check_accounts_inactive_days` | `35` | the STIG wants unused accounts disabled after 35 days |
| `check_accounts_ignore_inactive` | `[]` | Accounts that may stay unused (break-glass, service accounts that never log in interactively) |
| `check_accounts_nopasswd_allowed` | `[]` | sudoers entries allowed to use NOPASSWD, e.g. the account AAP connects as: [svc_aap, "%wheel"] |
| `check_accounts_expiry_watch` | `[]` | Accounts whose password/account expiry matters - above all the account AAP logs in with. e.g. [svc_aap] |
| `check_accounts_expiry_warn_days` | `14` | a watched account's password or account expiring within this many days -> warning (3 days or less -> critical) |

## check_auditd

File: `roles/check_auditd/defaults/main.yml`

**Note**
AUDITD: the audit system records who did what (the STIG depends on it). Running, enabled in the kernel, rules loaded, no events lost, room for its log, and (optionally) forwarding. Also how MUCH it records: too much rotates the evidence away within hours, fills the disk (then auditd stops the host or puts it in single-user mode, as the STIG sets it up), drops records on the way to the SIEM, and uses up the SIEM licence.

| Setting | Default | What it does |
|---|---|---|
| `check_auditd_min_rules` | `5` | fewer loaded rules than this = the STIG rules are missing |
| `check_auditd_log_warn_pct` | `75` | the audit log's filesystem |
| `check_auditd_log_crit_pct` | `90` | % used of that filesystem -> critical (warning from check_auditd_log_warn_pct) |
| `check_auditd_remote_required` | `false` | true = au-remote (audisp-remote) must be active |

| Setting | Default | What it does |
|---|---|---|
| check_auditd_exclude_accounts | [] | Accounts whose audit events are expected in bulk and do NOT count toward the volume warning, e.g. the account a vulnerability scanner logs in with (a credentialed scan runs thousands of commands with sudo). Names or UIDs; a host that does not have the account skips it. The report still says how much they made, and the... (all of it: the role file) |
| check_auditd_volume_hours | 24 | look at the audit records of the last N hours |
| check_auditd_volume_warn_mb_day | 200 | warning above this many MB of audit records a day (excluded accounts not counted) |
| check_auditd_retention_warn_hours | 24 | With max_log_file_action ROTATE, auditd keeps num_logs files of max_log_file MB and deletes the oldest. Warning when what is left on disk reaches back fewer hours than this. The STIG wants a week (168) when the records are NOT sent off the host as they are written (RHEL-08-030660). 0 = off. |
| check_auditd_volume_timeout | 120 | The breakdown (which account, which rule, which program) reads the audit log files of the window at low CPU and disk priority, and only when the volume or the retention is over its limit. It stops after this many seconds on a host with very large logs (and says so). |

## check_certs

File: roles/check_certs/defaults/main.yml

> **Note**
> CERTS: certificates that expire soon - as files on the host, inside Java keystores, and as served by TLS endpoints (what a browser or client actually sees).

| Setting | Default | What it does |
|---|---|---|
| check_certs_warn_days | 30 | a certificate expiring within this many days -> warning (and amber in the certificate report) |
| check_certs_notice_days | 60 | the certificate report colours these blue (an early heads-up; not a finding) |
| check_certs_crit_days | 14 | within this many days (or expired) -> critical |

| Setting | Default | What it does |
|---|---|---|
| check_certs_files | [] | Certificate files (PEM) to check. Globs work. Every certificate in a file is checked (a chain file with an expiring intermediate is a problem too). e.g. [/etc/stigman/*.crt, /etc/pki/tls/certs/server.crt] |
| check_certs_discover | true | Also look for certificates in the usual places. Files with many certificates are CA bundles (hundreds of roots, noisy) and are skipped unless listed in check_certs_files. Text (PEM) and binary (DER) certificates are both read; the names in any case (.crt, .CRT, .Pem ...). |
| check_certs_discover_dirs | [/etc/pki/tls/certs, /etc/pki/nginx, /etc/nginx, /etc/httpd/conf.d, /etc/stigman, /etc/st… | the folders searched (two levels deep) for *.crt, *.pem, *.cer, *.der (/etc/stigman/nginx/tls: STIG Manager's nginx certificate, three levels down) |
| check_certs_discover_extra_dirs | [] | more folders searched the same way, added to the ones above (no need to repeat them), e.g. [/home, /root, /opt/app/certs] |
| check_certs_discover_exclude | 'ca-bundle\|ca-certificates\|/localhost\.crt$' | regular expression: found files whose path matches are skipped |
| check_certs_discover_max_per_file | 10 | a found file with more certificates than this is a CA bundle: listed as not checked (list it in check_certs_files to check each) |
| check_certs_placeholder_subject | '(^\|,)O=SomeOrganization(,\|$)' | Regular expression on the subject of a package's placeholder certificate: postfix's /etc/pki/tls/certs/postfix.pem (O=SomeOrganization), self-signed for a year at install and never renewed. A found one is listed as not checked (no finding, no ticket) and none gets a CSR. A real certificate put in its place is checked;... (all of it: the role file) |
| check_certs_endpoints | [] | TLS endpoints, checked FROM this host, e.g. the site's own web server: check_certs_endpoints: - {name: STIG Manager, host: 127.0.0.1, port: 443, servername: stigman.example.mil} |

| Setting | Default | What it does |
|---|---|---|
| check_certs_keystores | [] | Java keystores (Tomcat, the ServiceNow MID Server ...). A JKS keystore lists without a password; a PKCS12 one needs it - give the NAME of an environment variable that holds it (set by an AAP credential), never the password itself: check_certs_keystores: - {path: /opt/servicenow/mid/agent/security/agent_keystore, passw... (all of it: the role file) |
| check_certs_discover_ports | false | ---- Finding them by themselves (the certificate report turns both on; the health check does not) ---- Listening TCP ports that answer TLS: each probed once, on the address it listens on, with `openssl s_client` (check_certs_probe_timeout seconds each, at most check_certs_discover_ports_max ports). Not these ports: pl... (all of it: the role file) |
| check_certs_discover_ports_skip | [22, 25, 53, 80, 110, 111, 123, 143, 323, 389, 631, 2049, 3306, 5355, 5432, 6379, 11211,… | ports never probed (plain-text services: ssh, smtp, dns, http, ldap, databases ...) |
| check_certs_discover_ports_max | 25 | probe at most this many listening ports (the rest are listed as not probed) |
| check_certs_probe_timeout | 3 | seconds for each port probe |
| check_certs_discover_keystores | false | Java keystores (*.jks, *.keystore, *.p12, *.pfx) in these folders, and the ones running Java programs use (-Djavax.net.ssl.keyStore=...). A keystore that needs a password is listed as not checked: no password is ever guessed - give it in check_certs_keystores (password_env). The JDK's own trust stores (cacerts) are sk... (all of it: the role file) |
| check_certs_discover_keystore_dirs | [/etc, /opt, /srv, /usr/share/tomcat, /var/lib/tomcat, /var/lib/pki] | the folders searched for keystores |

| Setting | Default | What it does |
|---|---|---|
| cert_report_discover | true | The certificate report and the CSR jobs (playbooks/cert_report.yml, cert_renew_csr.yml) switch on the port and keystore search above for their run, and the report the trust store too; the health check's certs check does not. false = those two jobs look only in the folders above and check_certs_files. |
| check_certs_discover_keystore_depth | 5 | how many folder levels deep the search goes |
| check_certs_discover_keystore_exclude | '/(cacerts\|jssecacerts)$\|/jre/lib/security/\|/lib/security/' | regular expression: keystore paths skipped (the JDK's trust stores) |
| check_certs_discover_keystore_max | 40 | read at most this many found keystores per host |
| check_certs_discover_trust | false | The host's trust store: the CA certificates it trusts, put in by update-ca-trust (the DoD roots ...). Every file in these folders, whatever its name. The certificate report lists them with their dates - never a finding, a ticket or a CSR: a CA renews its own certificate (put its new one in the folder and run update-ca... (all of it: the role file) |
| check_certs_trust_dirs | [/etc/pki/ca-trust/source/anchors] | the folders read (not the folders below them) |
| cert_skip_hosts | [] | ---- Leaving hosts out (the certificate report and the renewal; not the health check) ---- Hosts whose certificates something else renews - Satellite and its Capsules (satellite-installer, never by hand), appliances - are left out of the certificate report and never get a CSR: listed as not checked, no ticket. Name th... (all of it: the role file) |
| cert_skip_groups | [sat] | AAP inventory groups, e.g. [sat] (the health check still watches the Satellite web UI: group_vars/sat.yml) |

| Setting | Default | What it does |
|---|---|---|
| cert_renew_dir | /etc/pki/tls/renew | ---- Renewal: a new key and CSR for each certificate that expires soon (playbooks/cert_renew_csr.yml) ---- The certificates the report flags (within check_certs_warn_days - 30 days or less - or expired) get a new key and a CSR with the same subject and names (subjectAltName), on the host; the CSR is attached to the ce... (all of it: the role file) |
| cert_renew_skip_hosts | [] | Hosts whose certificates are renewed by hand (IdM, AAP ...): in the certificate report and its tickets as usual - each ticket says it is renewed by hand - but the renewal makes no CSR for them (listed, with why). The same forms as cert_skip_hosts / cert_skip_groups (which leave a host out of the report too). e.g. [idm... (all of it: the role file) |
| cert_renew_skip_groups | [] | e.g. [idm] |
| cert_renew_extra_dirs | false | true = a CSR for certificates found in check_certs_discover_extra_dirs (/home, /root ...) too; false = they are in the report only (a file there also under check_certs_discover_dirs, or listed in check_certs_files, is renewed) |
| cert_renew_key_bits | 2048 | RSA key size, at least 2048 (a bigger old key keeps its size; an EC certificate gets a new key on the same curve) |
| cert_renew_digest | sha256 | the CSR's signature: sha256, sha384 or sha512 |
| cert_renew_max | 10 | at most this many NEW CSRs per host in one run (the rest come the next run) |
| cert_renew_only | (empty) | regular expression: only certificates whose place matches - a file's path, a keystore's path, 'TLS port 8443, ...' (empty = every one), e.g. '^/etc/pki/tls/certs/' |
| cert_renew_issuer | (empty) | regular expression: only certificates whose issuer matches (empty = any CA), e.g. 'O=U\.S\. Government' |

| Setting | Default | What it does |
|---|---|---|
| cert_renew_exclude | '/(katello\|foreman\|foreman-proxy\|candlepin\|pulp\|puppet\|puppetlabs\|rhsm\|qpid)/\|/etc/pki/(c… | Certificates something else renews - never a CSR from here. The path (Satellite, Capsule, Puppet, subscription-manager on every registered host) and the issuer (Satellite's own CA, Puppet's CA). Check the report's Where column for your hosts' paths and add yours. |
| cert_renew_exclude_issuer | 'O=Katello\|Puppet CA' | regular expression on the issuer: Satellite's own CA (O=Katello), Puppet's CA - never a CSR from here |
| cert_renew_self_signed | false | Self-signed certificates (made on the host by an application or a person, not by a PKI) are reported and get a ticket, but no CSR - unless this is true: then the PKI is asked for a real certificate in their place (a STIG wants PKI certificates). Never a CA's own certificate (a root: it signs certificates), never one o… (all of it: the role file) |
| cert_renew_install_notes | /etc/stigman/: >- STIG Manager: put the issued certificate (with its chain) and the new k… | What to do with the issued certificate when something else deploys the file (a copy by hand would be overwritten): the start of the file's path -> the next step written on the ticket instead of "install it in place of the file". Add your own. |
| cert_renew_attach_ext | csr | The CSR's attachment on the ticket: <common name>-<8 characters of the CSR's SHA-256>.<ext> (a new CSR, a new name). An instance that refuses .csr attachments (its allowed extensions, or "Invalid file type: text/plain"): txt names every CSR .txt from the start. Refused, it is tried again as .txt (cert_renew_attach_ret… (all of it: the role file) |
| cert_renew_attach_type | text/plain | the attachment's Content-Type (a CSR is Base64 text) |
| cert_renew_attach_retry_txt | true | a refused attachment is tried again as <name>.txt (text/plain); false: straight to the work note |

| Setting | Default | What it does |
|---|---|---|
| cert_install_services | [] | ---- Install: the issued certificate in place of the old one (playbooks/cert_install_check.yml -> approval -> playbooks/cert_install_approved.yml) ---- With cert_renew_servicenow: catalog, the PKI puts the issued certificate on the request that carries its CSR. The install check finds it there (the certificate whose p... (all of it: the role file) |
| cert_install_known_services | [] | The services whose configuration is searched for the certificate's file - [{dir, service, test}] (test: the command that checks the configuration before the reload). Empty = httpd (/etc/httpd, apachectl configtest), apache2, nginx (nginx -t), haproxy (haproxy -c), postfix (postfix check), dovecot (doveconf -n), slapd,... (all of it: the role file) |
| cert_install_never_reload | [] | services (globs) the job never reloads, e.g. ['slapd*']: their certificates are installed by a person |
| cert_install_key_files | {} | The old key's file, when it cannot be found - neither in the configuration that names the certificate, nor once in the usual folders (several files with the old key: held, to be named here): {certificate file: key file}, e.g. {/etc/pki/tls/certs/web01.crt: /etc/pki/tls/private/web01.key} |
| cert_install_key_dirs | [] | where else to look for the old key (empty = /etc/pki/tls/private, /etc/ssl/private, the services' folders ...) |
| cert_install_trust_check | true | the chain must build up to a root (the host's trust store, or the PKI's root sent with it); false = install without |
| cert_install_verify_wait | 15 | seconds the port has, after the reload, to serve the new certificate (then: put back) |

| Setting | Default | What it does |
|---|---|---|
| cert_restore_backup | (empty) | ---- Restore: an old certificate back from its backup (playbooks/cert_restore.yml) ---- Each install copies the files it replaces to cert_renew_dir/backup/<date>-<certificate>/. The restore puts them back, reloads the service, checks the port serves the old certificate again, and the renewal waits again (the next inst... (all of it: the role file) |
| cert_restore_only | (empty) | or: the newest backup of each certificate whose name or subject matches (a regular expression), e.g. 'web01' |

## check_containers

File: roles/check_containers/defaults/main.yml

**Note**
CONTAINERS: the podman containers on the host - root's (rootful) and every user's (rootless) - found by themselves (roles/podman_discover: which ones exist, which systemd unit runs each one, which should be running). Read-only. Which containers and users are looked at, and which are left out: the podman_discover_* settings.

| Setting | Default | What it does |
|---|---|---|
| check_containers_required | [] | Containers that MUST exist and run on these hosts: a container name, or OWNER/NAME for one owner (root/web, alice/web). A name alone matches that container of any owner. Also matches a Quadlet file name without .container, e.g. [stigman-mysql, stigman-api, alice/web]. |
| check_containers_recent_hours | 24 | A container that stopped with an error (exit code not 0) this many hours ago or less is reported (containers:crashed). Older crashes are history and are not reported. |
| check_containers_restart_warn | 3 | More restarts than this (podman's restart count, or its systemd unit's) = a restart loop. |

## check_disk

File: roles/check_disk/defaults/main.yml

**Note**
DISK: how full each LOCAL filesystem is (space and inodes), and - from the history of earlier runs - how many days until it is full. Network filesystems (NFS) are the mounts check's job: a hung NFS server can make `df` hang, so this check never touches them.

| Setting | Default | What it does |
|---|---|---|
| check_disk_warn_pct | 85 | % used -> warning |
| check_disk_crit_pct | 95 | % used -> critical |
| check_disk_inode_warn_pct | 85 | inodes: too many small files fill a disk too |
| check_disk_inode_crit_pct | 95 | % of inodes used -> critical (warning from check_disk_inode_warn_pct) |
| check_disk_overrides | {} | Different limits for particular mount points, e.g. keep the audit log partition emptier: check_disk_overrides: /var/log/audit: {warn: 70, crit: 85} |
| check_disk_ignore | - '^/(proc\|sys\|dev\|run)(/\|$)' - '^/etc/(hosts\|hostname\|resolv\.conf)$' # bind-mounted fil… | Mount points to skip (regular expressions). |
| check_disk_exclude_types | [tmpfs, devtmpfs, squashfs, overlay, iso9660, efivarfs, nsfs] | filesystem types never checked (df -x) |
| check_disk_forecast | true | Days-until-full forecast. Each run adds a sample (date, bytes used) per mount to a small file on the host; once there is at least a day of history, the growth rate predicts when the filesystem fills. Check mode (a dry run) reads the history but does not add to it. |
| check_disk_forecast_warn_days | 14 | full within this many days at the current growth -> warning |
| check_disk_forecast_crit_days | 3 | full within this many days -> critical |
| check_disk_history_file | /var/lib/site-health/disk-history.json | where the forecast keeps its samples, on the host |
| check_disk_history_keep_days | 30 | samples older than this are dropped (the forecast uses what is left) |
| check_disk_timeout | 20 | seconds for df |

## check_fapolicyd

File: `roles/check_fapolicyd/defaults/main.yml`

> **Note**
> FAPOLICYD (application allow-listing): running, enforcing, configuration and trust database valid, and recent denials - which is what you see when it blocks a program.

| Setting | Default | What it does |
|---|---|---|
| `check_fapolicyd_required` | `false` | true = every host in the play must have fapolicyd. false = only check hosts that have it. |
| `check_fapolicyd_expect_enforcing` | `true` | permissive = 1 means it only logs, never blocks |
| `check_fapolicyd_check_trustdb` | `true` | Compare the trust database with the files on disk (can take a minute on big hosts; files changed outside RPM show up here, e.g. after a manual update). |
| `check_fapolicyd_denials_since` | `yesterday` | ausearch -ts value |
| `check_fapolicyd_denials_warn` | `1` | this many denials or more since check_fapolicyd_denials_since -> warning |

## check_logging

File: `roles/check_logging/defaults/main.yml`

> **Note**
> LOGGING: are logs kept, and do they reach the central log server (SIEM)?

| Setting | Default | What it does |
|---|---|---|
| `check_logging_journal_persistent` | `true` | the journal must survive a reboot (/var/log/journal) |
| `check_logging_remote` | `[]` | The central log server(s) this host forwards to, e.g. check_logging_remote: - {name: SIEM, host: siem.example.mil, port: 6514} |
| `check_logging_timeout` | `5` | seconds to wait for each log server's TCP port |
| `check_logging_queue_warn_mb` | `100` | rsyslog keeps messages it cannot deliver in /var/lib/rsyslog. A big queue = the log server has been unreachable for a while. |
| `check_logging_errors_since` | `"-24h"` | Error-priority journal messages in the last day. Many = something is complaining a lot. |
| `check_logging_errors_warn` | `500` | this many error messages or more -> warning |

## check_mariadb

File: `roles/check_mariadb/defaults/main.yml`

**Note**
DATABASE health for MariaDB AND MySQL (read-only; role name kept for compatibility): running, accepting connections, connection headroom, queries that run too long, replication (MariaDB multi-source, MySQL REPLICA/SLAVE), Galera / Percona XtraDB Cluster, MySQL Group Replication, recent errors in its log, and room on its disk. It runs status queries only - no data is read and nothing is changed. The engine and version are detected; see docs/MARIADB.md.

| Setting | Default | What it does |
|---|---|---|
| `check_mariadb_enabled` | (worked out by the role; see its file) | Which hosts: members of any of the inventory groups mariadb_hosts, mysql_hosts, database_hosts, mariadb or mysql, or any host that names a container below (or turns discovery on). Set this true/false to override for a host or group. |
| `check_mariadb_container` | (empty) | MariaDB or MySQL in podman containers (e.g. the ServiceNow database). The client inside each container is used (podman exec). Name them in one of these ways; all are checked, each as its own instance: check_mariadb_container: mariadb one container check_mariadb_containers: [mariadb, snow-db] several containers on the... (all of it: the role file) |
| `check_mariadb_containers` | `[]` | several containers on one host, e.g. [mariadb, snow-db] (see above) |
| `check_mariadb_container_discover` | `auto` | auto, true or false: find database containers by their image (see above) |
| `check_mariadb_server_paths` | `[/usr/sbin/mariadbd, /usr/libexec/mariadbd, /usr/sbin/mysqld, /usr/libexec/mysqld, /usr/b…` | Where a database server installed on the host lives (any one present = a host install, for `auto`). |
| `check_mariadb_container_image_regex` | `'(mariadb\|mysql\|percona)'` | regular expression: a container whose image matches is a database (discovery) |
| `check_mariadb_container_user` | (empty) | ROOTLESS podman: the containers belong to a user (e.g. AAP's own MariaDB, or an application account), and only that user can see them. Name the user and the check runs podman AS that user. Empty = root's containers. |

| Setting | Default | What it does |
|---|---|---|
| check_mariadb_service | (empty) | The systemd unit (host installs). Empty = the first of mariadb, mysqld, mysql that exists. |
| check_mariadb_client_args | [] | How to log in. Without a credential: as root through the local socket - MariaDB's default for a host install on RHEL (no password needed when the job runs with sudo). A database in a CONTAINER usually gives root a password (the official images do), and MySQL host installs usually do too, so there attach the "MariaDB m... (all of it: the role file) |
| check_mariadb_conn_warn_pct | 80 | connections in use, % of max_connections |
| check_mariadb_conn_crit_pct | 95 | connections in use, % of max_connections -> critical |
| check_mariadb_threads_running_warn | 32 | queries executing at this moment |
| check_mariadb_long_query_seconds | 300 | a query running longer than this is "long" |
| check_mariadb_replica_lag_warn | 300 | seconds a replica is behind its primary |
| check_mariadb_replica_lag_crit | 1800 | seconds behind its primary -> critical |
| check_mariadb_galera_size | 0 | expected Galera / XtraDB Cluster size (0 = do not check the size) |
| check_mariadb_group_size | 0 | expected MySQL Group Replication members (0 = do not check the size) |
| check_mariadb_restart_warn_minutes | 60 | restarted less than this ago -> warning |
| check_mariadb_bufferpool_hit_warn | 95 | % of reads served from memory (only judged after 1M reads) |
| check_mariadb_error_since_days | 1 | [ERROR] lines in its log this recent |
| check_mariadb_disk_warn_pct | 85 | the filesystem holding the data directory |
| check_mariadb_disk_crit_pct | 95 | % used of the data directory's filesystem -> critical |
| check_mariadb_performance | true | ---- PERFORMANCE (all warnings) ---------- false = only the health checks above. Counters that count since the server started (slow queries, temporary tables, joins, lock waits, deadlocks ...) are judged as an average since that start, and only once it has run chec... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| check_mariadb_perf_min_uptime_hours | 2 | judge the averages since the start only after this many hours |
| check_mariadb_slow_per_hour_warn | 60 | queries slower than long_query_time, per hour |
| check_mariadb_tmp_disk_pct_warn | 25 | % of temporary tables that went to disk (judged after 1000 of them) |
| check_mariadb_full_join_per_hour_warn | 100 | joins without a usable index (Select_full_join), per hour |
| check_mariadb_table_lock_wait_pct_warn | 1 | % of table locks (MyISAM / Aria) that had to wait (judged after 10000) |
| check_mariadb_row_lock_avg_ms_warn | 500 | average InnoDB row lock wait, ms (judged after 100 waits) |
| check_mariadb_deadlocks_per_hour_warn | 1 | InnoDB deadlocks per hour (judged after 10 of them) |
| check_mariadb_innodb_waits_per_hour_warn | 1 | waits for a free buffer pool page / a full log buffer, per hour (after 10) |
| check_mariadb_aborted_connects_per_hour_warn | 60 | failed connection attempts per hour |
| check_mariadb_table_open_per_hour_warn | 600 | tables opened per hour while the table cache is full |
| check_mariadb_open_files_warn_pct | 85 | files open, % of open_files_limit |
| check_mariadb_lock_wait_seconds | 30 | a session waiting this long for a lock, right now |
| check_mariadb_long_trx_seconds | 600 | an InnoDB transaction open this long, right now |
| check_mariadb_history_length_warn | 1000000 | InnoDB history list length (old row versions not yet purged) |

## check_mounts

File: `roles/check_mounts/defaults/main.yml`

> **Note**
> MOUNTS: filesystems that should be mounted but are not, local filesystems that went read-only (the kernel does that after disk errors), and network mounts (NFS/CIFS) that hang. Diagnose only: this check never mounts, unmounts or remounts anything.

| Setting | Default | What it does |
|---|---|---|
| check_mounts_network_types | [nfs, nfs4, cifs, smb3] | filesystem types counted as network mounts (checked for hangs and fullness) |
| check_mounts_local_types | [xfs, ext4, ext3, btrfs, vfat] | filesystem types checked for having gone read-only |
| check_mounts_timeout | 5 | seconds before a network mount counts as hung |
| check_mounts_warn_pct | 85 | Network mounts: how full (same idea as the disk check) |
| check_mounts_crit_pct | 95 | % used of a network mount -> critical (warning from check_mounts_warn_pct) |
| check_mounts_ignore | [] | fstab mount points that are allowed to be unmounted (regular expressions) |

## check_network

File: roles/check_network/defaults/main.yml

> **Note**
> NETWORK: can this host reach what it depends on? A default route, names that must resolve, and TCP ports that must answer (from THIS host, which is what matters when a service breaks).

| Setting | Default | What it does |
|---|---|---|
| check_network_require_default_route | true | no default route -> finding (false for an isolated host) |
| check_network_resolve_self | true | this host's own full name (hostname -f) resolves |
| check_network_dns_names | [] | e.g. [ldap.example.mil, servicenow.example.mil] |
| check_network_tcp | [] | TCP services this host needs, e.g. check_network_tcp: - {name: LDAP, host: ldap.example.mil, port: 636} - {name: SIEM (syslog over TLS), host: siem.example.mil, port: 6514} |
| check_network_timeout | 5 | seconds for each name lookup and TCP port test |

## check_patching

File: roles/check_patching/defaults/main.yml

> **Note**
> PATCHING (readiness, read-only): pending security updates, a reboot still needed after the last updates, and how long ago the host was last updated. Installs nothing - patching itself is playbooks/patch_hosts.yml.

| Setting | Default | What it does |
|---|---|---|
| `check_patching_max_age_days` | `35` | last package update older than this -> warning |
| `check_patching_security_warn` | `1` | this many pending security advisories -> warning |
| `check_patching_critical_is_critical` | `true` | a pending Critical-severity advisory -> critical |
| `check_patching_timeout` | `300` | seconds for dnf to talk to the repositories |
| `check_patching_reboot_check` | `true` | report a reboot still needed (needs-restarting -r, newer kernel installed than running) |

## check_performance

File: `roles/check_performance/defaults/main.yml`

> **Note**
> PERFORMANCE: is the host overloaded right now, or was something killed for lack of memory?

| Setting | Default | What it does |
|---|---|---|
| `check_performance_load_warn` | `1.5` | 15-minute load average PER CPU (1.0 = every CPU busy) |
| `check_performance_load_crit` | `3.0` | load per CPU -> critical |
| `check_performance_mem_avail_warn_pct` | `10` | memory still available for programs, % of total |
| `check_performance_mem_avail_crit_pct` | `5` | available memory below this % -> critical |
| `check_performance_swap_warn_pct` | `50` | swap in use, % of swap |
| `check_performance_swap_crit_pct` | `80` | swap in use above this % -> critical |
| `check_performance_iowait_warn` | `20` | % of CPU time waiting for disk (average over the sample) |
| `check_performance_iowait_crit` | `40` | I/O wait above this % -> critical |
| `check_performance_steal_warn` | `10` | % of CPU taken by the hypervisor (VM starved) |
| `check_performance_sample_seconds` | `5` | how long vmstat samples |

| Setting | Default | What it does |
|---|---|---|
| check_performance_oom_since | "-24h" | look for out-of-memory kills this far back |
| check_performance_zombie_warn | 20 | this many zombie processes or more -> warning |

## check_selinux

File: roles/check_selinux/defaults/main.yml

**Note**
SELINUX: enforcing (the STIG requires it), the config file agrees (so a reboot does not change it), no locally added permissive domains, and recent denials (AVCs) - which is what you see when SELinux blocks an application.

| Setting | Default | What it does |
|---|---|---|
| check_selinux_expected_mode | enforcing | the mode SELinux must be in (and the config file must say); anything else -> critical |
| check_selinux_avc_since | yesterday | ausearch -ts: recent (10 min), today, yesterday, this-week |
| check_selinux_avc_warn | 1 | this many denials or more -> warning |
| check_selinux_booleans | {} | Booleans that must have a value, e.g. {httpd_can_network_connect: "on"} |
| check_selinux_relabel_paths | [] | Paths whose file labels are compared with the policy (a dry-run restorecon). Mislabeled files are a common cause of denials after files are copied or moved. Keep it to small trees. e.g. [/etc/nginx, /var/www, /etc/stigman] |

## check_services

File: roles/check_services/defaults/main.yml

**Note**
SERVICES: systemd units that failed, services that must be running (and start at boot), and services stuck restarting over and over.

| Setting | Default | What it does |
|---|---|---|
| check_services_required | [sshd, crond, rsyslog, chronyd, auditd] | Must be running on every host. A host where one is not installed gets a warning (set check_services_required in the inventory to change the list for a group of hosts). |
| check_services_required_extra | [] | More for a group of hosts, e.g. in group_vars/mariadb_hosts.yml: [mariadb] |

| Setting | Default | What it does |
|---|---|---|
| check_services_ignore_failed | [] | Failed units to ignore (regular expressions), e.g. a known-broken vendor unit. |
| check_services_restart_warn | 3 | More restarts than this since the unit started = a restart loop |

## check_sso

File: roles/check_sso/defaults/main.yml

**Note**
SSO (read-only): can domain users still log in? A Linux host joined to Active Directory or IdM (sssd, realmd, winbind) keeps working on cached logins for a while after it loses its domain - then logins fail. This check finds it before then: sssd (or winbind) running, each sssd domain online, the host's keytab still gets a ticket from a KDC (its machine account works), the AD join is valid (adcli testjoin), domain users resolve, and how often sssd went offline or could not reach a KDC recently. A host that is not joined to any domain is skipped.

| Setting | Default | What it does |
|---|---|---|
| check_sso_hours | 24 | how far back the logs are read (sssd going offline, KDC unreachable) |
| check_sso_offline_warn | 1 | sssd went offline this many times in that window (and is online now) -> warning |
| check_sso_kdc_warn | 3 | "cannot contact any KDC" this many times -> warning |
| check_sso_keytab | /etc/krb5.keytab | the host's keytab |
| check_sso_keytab_test | true | get a ticket with it (kinit -k into memory: nothing is written) - proves the machine account works |
| check_sso_testjoin | true | adcli testjoin on hosts joined to Active Directory with adcli / realmd |
| check_sso_test_users | [] | domain users that must resolve (getent passwd), e.g. [svc_monitor@example.mil]; empty = none |
| check_sso_sssd_dir | /etc/sssd | sssd's configuration (sssd.conf, conf.d/): there = sssd is set up |
| check_sso_smb_conf | /etc/samba/smb.conf | Samba's: security = ads there = winbind is set up for Active Directory |
| check_sso_sssd_log_dir | /var/log/sssd | sssd's own logs, read with the journal |
| check_sso_timeout | 30 | seconds for each command that asks the domain (kinit, adcli, getent) |

## check_stig

File: `roles/check_stig/defaults/main.yml`

**Note**
STIG (drift): scans the Linux host against the DISA STIG with OpenSCAP - the SCAP Security Guide's STIG profile for its RHEL version (RHEL 9: V2R9, RHEL 10: V1R2 in scap-security-guide 0.1.82) - keeps each scan's report on the host, and compares the result with the host's checklist in STIG Manager (read-only: this check never writes to STIG Manager) and with the host's last scan. A rule STIG Manager has as Not a Finding that fails now is drift; a rule that passed in the last scan and fails now is new. playbooks/stig_drift_report.yml runs it; the health check runs it only when asked by name (health_checks: [stig] - in no shortcut: a scan takes minutes). docs/STIG_DRIFT.md.

| Setting | Default | What it does |
|---|---|---|
| `check_stig_install` | `false` | ---- The scan, on the host ---------- true: install openscap-scanner and scap-security-guide when missing (dnf, from the host's own repositories); false: a host without them is listed as not scanned |
| `check_stig_oscap` | (empty) | the oscap program; empty = found in PATH (/usr/bin/oscap) |
| `check_stig_profile` | `stig` | the profile: stig, stig_gui (a host with a desktop), or a full profile id (xccdf_...) |
| `check_stig_content_dir` | `/usr/share/xml/scap/ssg/content` | where scap-security-guide puts its datastreams (ssg-rhel<major>-ds.xml) |
| `check_stig_datastream` | (empty) | another datastream on the host instead (e.g. a DISA SCAP benchmark, with check_stig_profile its profile id); empty = the SCAP Security Guide's for the host's RHEL version |
| `check_stig_datastream_src` | (empty) | a datastream on the AAP side (in your repository, from playbooks/: files/stig/ssg-rhel9-ds.xml) copied to the host first - newer content than the host's repositories have; empty = none |
| `check_stig_tailoring` | (empty) | a tailoring file on the host (your site's changes to the profile, made with SCAP Workbench); empty = none |
| `check_stig_tailoring_src` | (empty) | or a tailoring file on the AAP side (from playbooks/), copied to the host first; empty = none |
| `check_stig_timeout` | `1800` | seconds the scan may take (a full scan takes one to five minutes) |

| Setting | Default | What it does |
|---|---|---|
| check_stig_dir | /var/lib/site-automation/stig | ---- Kept on the host ---------- each scan's HTML report, results (gzip) and summary (JSON), and latest-report.html / latest-summary.json; root-only. A dry run keeps nothing |
| check_stig_keep | 10 | how many scans are kept on the host (the oldest are deleted); the last one is what the next scan is compared with |
| check_stig_keep_report | true | keep oscap's HTML report of each scan (about 4 MB) |
| check_stig_keep_results | true | keep the XCCDF results of each scan (gzip, about 1 MB): what STIG Manager or eMASS can import |
| check_stig_stigman_api | (empty) | ---- STIG Manager (read-only) ---------- The job logs in as a Keycloak client with the client credentials grant - your autostig client, or another one - from the "STIG Manager API" credential (STIGMAN_TOKEN_URL, STIGMAN_CLIENT_ID, STIGMAN_CLIENT_SE... (all of it: the role file) |
| check_stig_stigman_token_url | (empty) | Keycloak's token endpoint (https://keycloak.example.mil/realms/stigman/protocol/openid-connect/token); empty = the credential's |
| check_stig_stigman_client_id | (empty) | the client, e.g. autostig; empty = the credential's |
| check_stig_stigman_client_secret | (empty) | its secret (vault it); empty = the credential's |
| check_stig_stigman_scope | (empty) | scopes to ask for, e.g. "stig-manager:collection:read"; empty = the client's default scopes |
| check_stig_stigman_validate_certs | true | check the HTTPS certificates of STIG Manager and Keycloak |
| check_stig_stigman_ca_path | (empty) | a CA bundle on the AAP side to check them with (the DoD roots); empty = the execution environment's |
| check_stig_stigman_timeout | 60 | seconds per request |

| Setting | Default | What it does |
|---|---|---|
| check_stig_stigman_collection | (empty) | Finding each host's asset. Set the collection per group or per host (group_vars, host_vars, or the inventory's variables in AAP): the asset is looked for only there. Empty = every collection the client can read; a host whose name matches assets in two collections is listed and not compared until its collection is set.... (all of it: the role file) |
| check_stig_stigman_collections | [] | with no collection set: search only these collections ([] = every one the client can read) |
| check_stig_stigman_asset | (empty) | this host's asset name in STIG Manager; empty = the inventory name, then its short name (web01 for web01.example.mil); the asset's FQDN field matches too |
| check_stig_stigman_benchmarks | {"8": RHEL_8_STIG, "9": RHEL_9_STIG, "10": RHEL_10_STIG} | the STIG Manager benchmark (STIG) for each RHEL version |
| check_stig_stigman_statuses | [saved, submitted, accepted] | the reviews that count as the checklist (a rejected review does not); [accepted] = only what was accepted |
| check_stig_stigman_missing_severity | warning | a host not found in STIG Manager (or its collection not readable): warning; critical; or none = listed in the report only |
| check_stig_severity | {high: critical, medium: warning, low: warning} | ---- Findings ---------- the finding for a drifted or new fail, by the rule's CAT (high = CAT I) |
| check_stig_list_max | 200 | at most this many rows in the report's table of fails STIG Manager has no review for |

## check_time

File: roles/check_time/defaults/main.yml

> **Note**
> TIME: is the clock synchronized, and by how much is it off? A clock that is off by minutes breaks Kerberos/AD logins (5 minutes), TLS, log correlation and audit evidence.

| Setting | Default | What it does |
|---|---|---|
| check_time_offset_warn | 0.5 | seconds |

| Setting | Default | What it does |
|---|---|---|
| check_time_offset_crit | 5.0 | seconds off -> critical (warning from check_time_offset_warn) |
| check_time_min_sources | 1 | time servers chrony must be able to reach |

## check_trust

File: roles/check_trust/defaults/main.yml

**Note**
TRUST (read-only): which trust stores on the host still hold a certificate authority that was compromised or that the browsers dropped - the system bundles, p11-kit's live view and blocklist, Java cacerts, Python's certifi bundles, Node's built-in roots, the bundles inside running podman containers, certificate files an application carries - and the software with its own trust logic (Go programs, Node, NSS) below the version that fixed it. The list of certificates to look for ships with the check (roles/check_trust/vars/main.yml: October 2026, the eight eMudhra intermediates and their roots); add your own in check_trust_extra. The health check runs it in the weekly set; playbooks/trust_report.yml runs it on every host, vCenter and ESXi too.

| Setting | Default | What it does |
|---|---|---|
| check_trust_extra | [] | More certificates to look for, besides the shipped list: name, sha256 (the certificate's fingerprint, with or without colons), spki_sha256 (its public key, optional), kind (root / intermediate), severity (critical / warning / notice), why. Get the fingerprints with: openssl x509 -in the.pem -noout -fingerprint -sha256... (all of it: the role file) |
| check_trust_severity | {} | another severity for a listed name: {"AAA Certificate Services": warning} once you decide to blocklist it; notice = listed only, no finding |
| check_trust_software_severity | warning | the finding for a program below the fixed version (a Go program, Node, NSS): warning or critical |
| check_trust_bundles | [/etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem, /etc/pki/ca-trust/extracted/openssl/c… | The system: the CA bundles every OpenSSL / Go / Python client reads (the ones that exist are read), p11-kit's live view (trust extract: what NSS and GnuTLS clients see, and the blocklist), the blocklist folders. the system CA bundles (RHEL, Debian / Ubuntu, Alpine, SUSE); the same file under two names is read once |

| Setting | Default | What it does |
|---|---|---|
| check_trust_blocklist_dirs | [/etc/pki/ca-trust/source/blocklist, /usr/share/pki/ca-trust-source/blocklist] | p11-kit blocklist folders: a certificate there is distrusted (listed as blocked, no finding) |
| check_trust_p11kit | true | read p11-kit's live anchors and blocklist too (trust extract), when the trust command is there |
| check_trust_nss_dirs | [/etc/pki/nssdb, /root/.pki/nssdb, /home/*/.pki/nssdb] | NSS databases read with certutil (the system's: curl and other NSS clients on older RHEL; root's and each user's: Chrome, Evolution) |
| check_trust_ca_certificates_fixed | (empty) | the ca-certificates package version that dropped the roots, e.g. '2026.2.82' (empty: not compared - the bundles themselves tell) |
| check_trust_nss_fixed | '3.131.1' | the NSS version that dropped emSign ECC Root CA - G3 (the package nss / libnss3 below it: a finding; empty: not compared) |
| check_trust_java | true | Java: every cacerts / jssecacerts under these folders (JDKs, and the keystores of running java programs), read directly (JKS) or with keytool (PKCS12), with the JDK's version. |
| check_trust_java_dirs | [/usr/lib/jvm, /usr/java, /etc/pki/java, /etc/pki/ca-trust/extracted/java, /opt, /usr/loc… | folders searched for cacerts (seven levels deep) |
| check_trust_python | true | Python: every certifi bundle (certifi/cacert.pem: requests, pip, urllib3 ... carry their own roots, not the system's) under these folders - system Python, venvs, applications. |
| check_trust_python_dirs | [/usr/lib, /usr/lib64, /usr/local, /opt, /srv, /var/lib, /root, /home] | folders searched for certifi/cacert.pem (twelve levels deep) |
| check_trust_node | true | Node: its built-in roots (not the system's: Node carries its own, unless started with --use-openssl-ca), and its version against the one that dropped the root, per major line. |
| check_trust_node_paths | [/usr/local/bin/node, /usr/local/node*/bin/node, /opt/*/bin/node, /opt/node*/bin/node, /r… | where node binaries are (globs); the one in PATH and running ones are found too |

| Setting | Default | What it does |
|---|---|---|
| check_trust_node_fixed | {"22": "22.23.4", "24": "24.22.1", "26": "26.12.1"} | Node's fixed version per major line (an older line than the oldest listed: unsupported, no fix) |
| check_trust_go | true | Go: a Go program carries its own certificate verification. Go 1.27.3 / 1.26.10 reject any chain with one of the compromised keys (CVE-2026-111943) - a program built with an older Go has to be rebuilt. The running programs, and the executables under these folders, with the Go version each was built with. |
| check_trust_go_dirs | [/usr/local/bin, /usr/local/sbin, /opt, /srv] | folders searched for Go programs (four levels deep, executables over 1 MB) |
| check_trust_go_fixed | {"1.26": "1.26.10", "1.27": "1.27.3"} | Go's fixed version per line (a program built with an older line: unsupported - rebuild it with a fixed Go) |
| check_trust_go_max | 200 | at most this many programs looked at per host |
| check_trust_containers | false | Containers: the CA bundle inside every running podman container (root's and each logged-in user's), read with podman exec - a container image carries its own ca-certificates (Red Hat: UBI images are affected too). Off by default (0.34.0): a shell started in each container that reads CA bundles and searches its whole f... (all of it: the role file) |
| check_trust_containers_max | 30 | at most this many containers per host |
| check_trust_find_dirs | [/etc, /opt, /srv, /usr/local, /var/lib, /root, /home] | Certificate files an application carries (its own bundle, a copied root): every *.pem, *.crt, *.cer, *.der, cacert*, ca-bundle*, ca-certificates*, and every Java keystore (*.jks, *.keystore, *truststore*; *.p12 / *.pfx with keytool and the default password) under these folders, on the host's own filesystems - only the... (all of it: the role file) |
| check_trust_find_max | 2000 | at most this many files read per host |

| Setting | Default | What it does |
|---|---|---|
| check_trust_find_exclude | '/(proc\|sys\|dev)/\|/\.cache/\|/containers/storage/\|/docker/overlay2?/\|/ca-trust/extracted/\|… | regular expression on the path: matching files are skipped (container image layers, the copies update-ca-trust extracts, Debian's per-CA files - the bundles stand for them; this repository's own test certificates on the AAP controller) |
| check_trust_timeout | 60 | seconds for each search and each command (keytool, node, podman exec) |
| check_trust_remove | false | Removal (the Trust store report only, playbooks/trust_report.yml; the VMware trust store report has the same switch): take the listed certificates out of the stores that hold them. OFF by default - a read-only report. Check mode shows what would go and changes nothing. The system trust is blocklisted (RHEL / SUSE: a P… (all of it: the role file) |
| check_trust_remove_kinds | [system, java, python, file, nssdb] | which kinds of store to change: system (blocklist), java (every keystore), python (certifi), file (an application's bundle or certificate file), nssdb (certutil -D) |
| check_trust_remove_severities | [critical, warning] | only certificates at these severities go; a notice-level one (AAA Certificate Services) never, unless named here |
| check_trust_remove_backup_dir | /var/backups/site-automation/trust | every file changed is copied here first (a folder per run) |
| check_trust_remove_blocklist_dir | (empty) | the system blocklist folder; empty = the distribution's (/etc/pki/ca-trust/source/blocklist, SUSE /etc/pki/trust/blacklist) |
| check_trust_remove_update_cmd | (empty) | the command that rebuilds the system trust after blocklisting; empty = the distribution's (update-ca-trust, update-ca-certificates) |
| check_trust_keystore_password | changeit | the Java keystores' password (keytool's default) |

## fapolicyd_containers

File: roles/fapolicyd_containers/defaults/main.yml

**Note**
FAPOLICYD AND CONTAINERS (playbooks/fapolicyd_containers.yml, docs/RUNBOOKS.md "fapolicyd and containers"): what fapolicyd blocked that a container caused, the narrowest allow rule for each, and - once approved - those rules added to the hosts. A container's files are not in fapolicyd's trust database (RPM fills it from the host's packages), so a program in a container image can be denied.

| Setting | Default | What it does |
|---|---|---|
| fapolicyd_container_since | today | ausearch -ts: recent (10 minutes), today, yesterday, this-week, boot |
| fapolicyd_container_rules | [] | The approved rules to add, each exactly as the report's "Rules to approve" gives it (or your own allow rule). Written to fapolicyd_container_rules_file on every host of the run, then loaded (fagenrules --load). A rule that is not a plain allow rule stops the change on every host. Route each one through your security a... (all of it: the role file) |
| fapolicyd_container_rules_remove | [] | rules this job added earlier, to take out again |
| fapolicyd_container_rules_file | /etc/fapolicyd/rules.d/35-site-containers.rules | The file this job owns - and only this one. 35: after the deny rules of 30-patterns (ld.so tricks), before 40-bad-elf, 41-shared-obj, 42-trusted-elf and 90-deny-execute that deny a container's untrusted files. |
| fapolicyd_container_compiled | /etc/fapolicyd/compiled.rules | what fagenrules builds: the rules are checked there |
| fapolicyd_container_rules_owner | root | the rules file's owner, as the other rules.d files |
| fapolicyd_container_rules_group | fapolicyd | and its group |
| fapolicyd_container_scope | path | The proposed rules. A file that is on the host at the path the audit log holds (a bind mount): the narrowest rule - the file (path) or its folder (dir: one rule for every file there; never a system folder), with the program that was denied (exe=..., narrower) or for any program (false). path (the file) or dir (its fol... (all of it: the role file) |
| fapolicyd_container_rule_exe | true | the rule names the program denied (false: any program) |

| Setting | Default | What it does |
|---|---|---|
| fapolicyd_container_storage_rules | true | A file inside an image is not on the host at that path (the audit log holds the container's path), and its host path changes at each pull of the image: its rule is container storage's folder, for any program - allow perm=any all : dir=/var/lib/containers/storage/overlay/ (storage.conf's graphroot; a user's own rootles… (all of it: the role file) |
| fapolicyd_container_review | true | Review the rules files in place (rules.d/*.rules, fapolicyd's own left out) and list what to tighten in the report: a wildcard (fapolicyd has none: the rule matches nothing), a system folder or a user-writable folder opened to any program, a rule that allows everything, a repeated rule, a rule for an RPM-owned file (t… (all of it: the role file) |
| fapolicyd_container_labels | '(^\|:)(container_[a-z0-9_]*_t\|spc_t)(:\|$)' | What marks a denial as a container's (any one): an SELinux label on the program or the file, a container program, a path in container storage, or a process in a podman container's cgroup (while it runs). The labels: a regular expression on the program's and the file's SELinux labels. |
| fapolicyd_container_programs | - podman - crun - runc - conmon - buildah - skopeo - catatonit - netavark - aardvark-dns… | The container programs (their base names): what they run or open is a container's. |
| fapolicyd_container_paths | - '^/var/lib/containers/' - '/\.local/share/containers/' - '^/run/containers/' - '^/run/u… | Container storage: regular expressions on the program's or the file's path. |
| fapolicyd_container_names | true | the containers' names (podman ps, root's and each logged-in user's) |
| fapolicyd_container_max | 200 | at most this many groups per host (the most denied first) |
| fapolicyd_container_timeout | 120 | seconds for ausearch |

## horizon

File: roles/horizon/defaults/main.yml

**Note**
VMware Horizon 8 (playbooks/horizon_*.yml): the Connection Server's REST API, from the AAP side, with the "Horizon API" credential (its address, domain, user name and password) - no collection needed. Every job asks each API path at the newest version first, down to v1, so it fits any Horizon 8 release. A dry run (Check) reads and says what it would do; it changes nothing. docs/HORIZON.md

| Setting | Default | What it does |
|---|---|---|
| horizon_ca_path | (empty) | ---- Connection ---------- a CA file for the Connection Server's HTTPS certificate ("" = what the execution environment trusts) |
| horizon_timeout | 60 | seconds per request |
| horizon_page_size | 500 | records per page when reading pools, desktops and sessions (Horizon's most: 1000) |
| horizon_api_newest | 9 | the newest API version to try (v9 down to v1): the first that answers is used |
| horizon_action_version | v1 | the API version of the actions (reset, recover, log off, push image...) |
| horizon_protected_pools | [] | ---- Pools no job changes ---------- Desktop pools no job fixes, logs off or pushes an image to: names as Horizon Console shows them, * and ? allowed, any case (e.g. [VIP-*, Kiosk]). The reports still show them. |
| horizon_cert_warn_days | 30 | ---- Health report (playbooks/horizon_health_report.yml) ---------- a Connection Server's, gateway's or vCenter's certificate expires within this many days -> amber |
| horizon_cert_crit_days | 7 | within this many days (or expired, or not valid) -> red |
| horizon_license_warn_days | 60 | the license expires within this many days -> amber |
| horizon_report_max_rows | 50 | rows in the problems table (the job's artifacts have all) |
| horizon_health_fail | false | The health and capacity reports stay green when they find problems: the report says what. true = mark the job FAILED on a critical finding, for workflows and alerts. |

| Setting | Default | What it does |
|---|---|---|
| horizon_capacity_warn_pct | 80 | ---- Capacity report (playbooks/horizon_capacity_report.yml) ---------- a pool's desktops in use (connected or disconnected) against its maximum, % -> amber |
| horizon_capacity_crit_pct | 95 | in use, % -> red (also red: no free desktop left) |
| horizon_broken_states | [AGENT_UNREACHABLE, AGENT_ERR_DISABLED, AGENT_ERR_INVALID_IP, AGENT_ERR_NEED_REBOOT, AGEN… | ---- Broken desktops (the desktops check -> approval -> the fix job) ---------- A desktop in one of these states takes no user until it is fixed. Left out on purpose: states that pass by themselves (being made, customized, deleted, waiting for the agent at start-up, maintenance). |
| horizon_desktops_mode | report | report = the check lists them and stays green (a schedule). approval = when there is a desktop to fix, the check hands the list on and FAILS on purpose, so the workflow takes its "Run on fail" path to the approval step and then the fix job. Set approval on the workflow's check node. |
| horizon_fix_default_action | recover | what fixes a broken desktop: recover (instant clones: deleted and made again from the image), reset, restart, remove, rebuild |
| horizon_fix_actions | {AGENT_ERR_NEED_REBOOT: restart} | By state, when another action fits better (e.g. {AGENT_ERR_NEED_REBOOT: restart, PROVISIONING_ERROR: remove}). |
| horizon_fix_max | 20 | fix at most this many desktops per run: the rest are listed, the next run takes them |
| horizon_fix_exclude_machines | [] | desktops never fixed automatically: names, * and ? allowed (e.g. [W11-TEST-*]) |
| horizon_session_max_disconnected_hours | 8 | ---- Disconnected sessions (playbooks/horizon_session_cleanup.yml) ---------- log off a session disconnected for this many hours or more |
| horizon_session_cleanup_max | 100 | log off at most this many sessions per run |
| horizon_session_cleanup_exclude_pools | [] | pools whose sessions are never logged off: names, * and ? allowed |
| horizon_session_cleanup_exclude_users | [] | users never logged off: login names (jdoe or CORP\jdoe) or SIDs |
| horizon_session_forced_logoff | true | log off a locked session too (a disconnected session is usually locked) |

| Setting | Default | What it does |
|---|---|---|
| horizon_session_cleanup_report_only | false | true = list them, log off none (a dry run - Check - does the same) |
| horizon_user | (empty) | ---- User help (playbooks/horizon_user_help.yml) ---------- the user who cannot connect: jdoe, CORP\jdoe, jdoe@corp.example.mil or a SID (a survey question) |
| horizon_user_action | look | What to do: look (only show their sessions and desktops), logoff (all their sessions), disconnect (their connected sessions), reset or restart (their desktops), recover (their instant-clone desktops: made again from the image). Never a desktop another user has a session on, nor one in a protected pool. |
| horizon_push_pools | [] | ---- Image push (the push plan -> approval -> the push job) ---------- the instant-clone pools to push to: names (a survey: one per line, or separated by commas) |
| horizon_push_base_vm | (empty) | the golden image VM, as vCenter names it (or its path) |
| horizon_push_snapshot | (empty) | its snapshot to push: the name, or the path when two have that name (e.g. /2026-10/2026-11) |
| horizon_push_start | (empty) | When the push starts: "" = as soon as the approved push job runs; or a time - "2026-11-01 22:00" (UTC) or with an offset, "2026-11-01T22:00-05:00". A time that has passed by the approval starts in a minute. |
| horizon_push_logoff_policy | WAIT_FOR_LOGOFF | WAIT_FOR_LOGOFF (each desktop is replaced when its user logs off) or FORCE_LOGOFF (users are logged off) |
| horizon_push_stop_on_first_error | true | stop the push at the first desktop that fails |
| horizon_test_user | (empty) | ---- API test (playbooks/horizon_api_test.yml) ---------- also look up this user (e.g. jdoe), as the user-help job does |

## log_analysis

File: roles/log_analysis/defaults/main.yml

**Note**
LOG ANALYSIS (playbooks/log_analysis.yml for a group of servers, playbooks/vm_log_analysis.yml for vCenter and its ESXi hosts through vCenter's API). Read-only: each host's warning, error and critical log lines of the window, grouped on the AAP side into patterns (the same line with other numbers, times and names is one pattern, counted across hosts), and ACT (GenAI) explains the top ones - likely cause, evidence, fix, confidence - in evidence mode: ACT runs no command anywhere. Every setting can be set for all hosts, per group or host in the inventory, on the template, or as a survey answer (a list may be typed as "a, b" or one per line). docs/LOG_ANALYSIS.md

| Setting | Default | What it does |
|---|---|---|
| log_analysis_hours | 24 | ---- the window ---------- The last this-many hours ... |
| log_analysis_since | (empty) | ... or a fixed window, e.g. the outage: log_analysis_since "2026-10-23 08:00" and log_analysis_until "2026-10-23 10:30" (UTC, as the report shows times). Since alone = from then until now. |
| log_analysis_until | (empty) | the end of the fixed window (empty = now) |
| log_analysis_source | auto | ---- what to read ---------- What a host is: auto (Windows when its connection is WinRM / PSRP, else Linux), linux, windows, or vcsa (a vCenter appliance over SSH: Linux, with the vcsa profile on). Set it per group in the inventory. |
| log_analysis_profiles | [] | Profiles: one word switches on the logs of a product (roles/log_analysis/vars/main.yml): vcsa, mid_server, iis, sql_server, nginx, httpd, mariadb, podman, sssd, idm, auditd, aap. E.g. for a group of web servers: log_analysis_profiles: [nginx, sssd] |
| log_analysis_profiles_extra | {} | Your own profiles, or a shipped one changed (the same name replaces it), the shape of vars/main.yml: log_analysis_profiles_extra: {stigman: {files: [/opt/stigman/logs/*.log], units: [stigman-api]}} |
| log_analysis_priority | err | Linux (and the vCenter appliance): the journal's lines of this priority or worse (journalctl -p: emerg, alert, crit, err, warning, notice). |
| log_analysis_journal | true | false = no journal (text logs, units and containers only) |

| Setting | Default | What it does |
|---|---|---|
| log_analysis_units | [] | Units and syslog identifiers whose journal is read at log_analysis_units_priority, lower than the rest: e.g. [sssd, nginx, stigman-api] (globs such as dirsrv@* work) and [sudo, sshd]. |
| log_analysis_identifiers | [] | syslog identifiers read the same way, e.g. [sudo, sshd] |
| log_analysis_units_priority | warning | the priority those units and identifiers are read at |
| log_analysis_containers | [] | Podman containers whose output (podman logs) of the window is read, its lines that match log_analysis_match: a name for root's, user/name for a user's (rootless), e.g. [stigman-api, svc_app/web]. |
| log_analysis_files | [] | Text logs (globs), read when they changed in the window: their matching lines, the newest log_analysis_max_lines. Compressed rotations (.gz .xz .bz2 .zst) are left out. log_analysis_files is the base list; log_analysis_files_extra adds to it and to the profiles' files (never replaces them). e.g. [/var/log/messages, /v... (all of it: the role file) |
| log_analysis_files_extra | [] | e.g. [/opt/app/logs/*.log] |
| log_analysis_become | true | read as root (sudo), as the other checks do |
| log_analysis_win_logs | [System, Application] | Windows: the event logs read, their critical (1), error (2) and - log_analysis_win_level 3 - warning events. |
| log_analysis_win_logs_extra | [] | more channels, e.g. ["Microsoft-Windows-TaskScheduler/Operational", "Directory Service", "DNS Server"] |
| log_analysis_win_level | 3 | 1 critical, 2 + error, 3 + warning |
| log_analysis_win_providers | [] | Event providers whose events are read at every level up to information, in the logs above (e.g. [Service Control Manager]); log_analysis_win_providers_only: true = only these providers' events. |
| log_analysis_win_providers_only | false | true = only those providers' events |

| Setting | Default | What it does |
|---|---|---|
| log_analysis_win_event_ids | [] | Event IDs always read whatever their level: "Log:ID", or a number for every log read plus Security, e.g. [4625, 4740, "System:7031", "System:7034"]. Security needs an account that may read it. |
| log_analysis_win_files | [] | Text logs (wildcards), read like the Linux ones; _extra adds to the base list and the profiles', e.g. ['D:\Apps*logs*\.log'] (a MID Server's agent logs: the mid_server profile) |
| log_analysis_win_files_extra | [] | more Windows text logs, added to the list above |
| log_analysis_match | "error\|fail\|fatal\|crit\|panic\|denied\|refused\|timed out\|timeout\|unable to\|cannot\|exception\|… | The words that make a line of a text log or a container's output worth reading (grep -E, any case); _extra adds words, e.g. [SEVERE, 'HTTP/1.1" 5']. |
| log_analysis_match_extra | [] | more words, added to log_analysis_match |
| log_analysis_vcenter_logs | [vpxd] | ---- vCenter and ESXi through vCenter's API (playbooks/vm_log_analysis.yml) ---------- No SSH and no login to the hosts: vCenter hands over its own logs and each host's (the "VMware vCenter" credential; a read-only role plus Global > Diagnostics). Keys as vCenter names them. [] = not vCenter's own log |
| log_analysis_esxi_logs | [hostd, vmkernel, vpxa] | [] = no host logs; more: fdm, vobd, hostd-probe … |
| log_analysis_esxi_hosts | [] | host names (* ? allowed); [] = every host |
| log_analysis_read_lines | 20000 | read back at most this many lines from the end of each log |
| log_analysis_max_lines | 500 | ---- every source ---------- keep at most this many lines per log (the newest) |
| log_analysis_max_line_chars | 400 | a line longer than this is cut |
| log_analysis_ignore | - 'Event 10016:' # Windows: DCOM "application-specific permission settings" (harmless, by… | Lines left out everywhere before they are grouped (regular expressions, any case, matched against "<program> <line>"): noise every server logs. log_analysis_ignore is the shipped list; _extra adds yours, e.g. ['sshd.*Connection closed by authenticating user']. |

| Setting | Default | What it does |
|---|---|---|
| log_analysis_ignore_extra | [] | more lines to leave out, added to log_analysis_ignore |
| log_analysis_min_count | 1 | a pattern seen fewer times than this is left out |
| log_analysis_act | true | ---- ACT ---------- ACT runs on the AAP side (the execution environment) with the site_act_* settings (roles/site_act/defaults/main.yml); the key comes from an "ACT model key" credential. false = the patterns only, no ACT (no key needed). |
| log_analysis_question | (empty) | What you want to know, in your words - ACT looks at it first (a good survey question), e.g. "users report slow logins since 09:00". |
| log_analysis_names_extra | [] | more names ACT replaces with placeholders before the model sees them (host names are always) |
| log_analysis_max_items | 30 | ACT analyzes at most this many patterns, the most severe first (the rest are listed) |
| log_analysis_act_batch | "{{ site_act_batch \| default(5) }}" | patterns per question to ACT, each with only its evidence (0 = all in one question) |
| log_analysis_examples | 3 | example lines per pattern in the evidence, the report and the artifacts |
| log_analysis_max_chars | 60000 | the evidence ACT reads is cut at this many characters |
| log_analysis_check_url | true | first check that the AAP node can reach the model's URL |
| log_analysis_act_model_timeout | "{{ site_act_model_timeout \| default(600) }}" | Seconds the model may take for its answer: site_act_model_timeout (600) unless set here. |
| log_analysis_email_if_clean | false | ---- the report ---------- The report is printed, kept as the job artifact log_analysis, emailed (report_email_to, and the "Also email the report to" box) and published (report_web_dest). A run that finds nothing is not emailed unless log_analys... (all of it: the role file) |
| log_analysis_max_rows | 200 | rows per table in the report (the artifacts have them all) |

| Setting | Default | What it does |
|---|---|---|
| log_analysis_fail | false | The job stays green when it finds patterns. It fails when ACT was wanted and could not run (after the report), and when no host could be read. true = also fail when a critical or error pattern was found. |

## net_certs

File: roles/net_certs/defaults/main.yml

**Note**
NETWORK DEVICE CERTIFICATES (playbooks/net_cert_report.yml, net_cert_renew.yml): the TLS certificate each switch, firewall, printer, iLO / iDRAC, UPS, storage ... presents, read from the controller with a TLS hello - no login to the devices, nothing changed. The renewal: a ServiceNow incident per certificate due; on a Cisco IOS / IOS-XE device, optionally a CSR made on the device (net_cert_cisco_csr).

| Setting | Default | What it does |
|---|---|---|
| net_cert_targets | inventory | ---- which devices ---- inventory (the hosts of net_cert_group) / subnets (net_cert_subnets) / both |
| net_cert_group | net_cert_devices | the inventory group of the devices; a host's own net_cert_ports / net_cert_address / net_cert_servername override |
| net_cert_ports | [443] | the ports read (a host's own net_cert_ports wins), e.g. [443, 8443, 636] |
| net_cert_subnets | [] | CIDR ranges to sweep (net_cert_targets: subnets or both), e.g. ['192.0.2.0/24'] - every address, each port in net_cert_ports |
| net_cert_exclude | [] | never connected to: addresses, CIDR ranges, host name patterns (e.g. ['192.0.2.1', 'ups*']) |
| net_cert_max_addresses | 4096 | a sweep bigger than this (all subnets together) is refused: a /16 by mistake would take hours |
| net_cert_resolve_names | true | a swept address that answers: its DNS name (PTR) in the report, else the address |
| net_cert_timeout | 5 | seconds for each connection, and each handshake |
| net_cert_workers | 32 | connections at the same time |

| Setting | Default | What it does |
|---|---|---|
| net_cert_only | [] | only these devices (names, wildcards), e.g. ['sw01*'] - to try the renewal on one |
| net_cert_warn_days | 30 | ---- what is due ---- a certificate expiring within this many days is due: a warning incident (amber in the report) |
| net_cert_crit_days | 7 | within this many days (or expired): critical |
| net_cert_notice_days | 60 | the report colours these blue (a heads-up, no incident) |
| net_cert_unanswered_finding | false | true = an inventory device whose port gives no certificate gets an incident too (a warning) |
| net_cert_cisco_csr | false | On: for a due certificate of an inventory host with ansible_network_os cisco.ios.ios (and its network_cli credential on the job template), the job makes a key on the device (crypto key generate rsa, not exportable), a trustpoint <prefix>-<expiry> (enrollment terminal) and its CSR (crypto pki enroll), and attaches the... (all of it: the role file) |
| net_cert_cisco_key_bits | 2048 | the new RSA key (2048, 3072, 4096) |
| net_cert_cisco_trustpoint_prefix | SA-RENEW | the trustpoint and key label: SA-RENEW-20270131 (the old certificate's expiry) |

## net_config

File: roles/net_config/defaults/main.yml

**Note**
NETWORK DEVICE CONFIGURATIONS: the backup (playbooks/net_config_backup.yml) keeps each device's running configuration - in a git repository (a commit when one changed) or on a file share (dated copies) - and can open a ServiceNow incident when one changed. The load (playbooks/net_config_load.yml) sends configuration lines, or restores a backup, after an approved change request - always a dry run (Check) first. Cisco IOS / IOS-XE (over its CLI: network_cli, a Machine credential) and Palo Alto Networks PAN-OS (over its XML API from the controller: the "Palo Alto PAN-OS API" credential). docs/NETWORK_CONFIG.md. A backup holds the device's secrets (password hashes, SNMP communities, keys): keep it in a private, internal repository or share only - never in this one.

| Setting | Default | What it does |
|---|---|---|
| net_config_group | net_config_devices | The inventory group of the devices. Each one's platform: ansible_network_os (cisco.ios.ios, paloaltonetworks.panos), or net_config_platform (ios, panos) on the host. |
| net_config_only | [] | Only these devices (names, * and ? allowed), e.g. [rtr-lab*] - to try a job on one first. [] = every one. |
| net_config_store | git | git (a repository), share (a folder on a file server or the AAP node), or both. |
| net_config_git_repo | (empty) | The repository (ssh://git@git.example.mil/netops/configs.git or https://...). Private and internal: the configurations hold secrets. Its login: the "Network config git repository" credential (an SSH key, or a username and token). |
| net_config_git_branch | main | the branch (made when the repository is new and empty) |
| net_config_git_path | configs | the folder in the repository: <path>/<device>.cfg (IOS), .xml (PAN-OS) |
| net_config_git_author_name | site-automation | the commits' author |
| net_config_git_author_email | site-automation@localhost | and its email |
| net_config_git_host_key_checking | accept-new | The git server's SSH host key: accept-new (the first connection learns it), or yes (only the keys in net_config_git_known_hosts - lines as in a known_hosts file). In an execution environment ssh's own known_hosts is empty every job, so accept-new learns the key each time: only net_config_git_known_hosts + yes really c... (all of it: the role file) |
| net_config_git_known_hosts | (empty) | the git server's host keys (known_hosts lines), for net_config_git_host_key_checking: yes |
| net_config_share_dir | /var/lib/site-automation/net-config | The folder for share: on the AAP node (a path AAP exposes to jobs) or on net_config_share_host - <dir>/<device>/<device>.cfg (the latest), and a dated copy beside it each time one changed. |
| net_config_share_host | (empty) | an inventory host to keep them on (Linux, or a Windows file server); its credential on the template |

| Setting | Default | What it does |
|---|---|---|
| net_config_share_keep | 30 | dated copies kept per device (the oldest go) |
| net_config_ios_command | show running-config | ---- what is read ---- IOS: the command whose output is kept |
| net_config_panos_port | 443 | PAN-OS: the XML API's HTTPS port |
| net_config_panos_scheme | https | PAN-OS: always https - http is for the tests' stand-in firewall only (it would send the API key in clear) |
| net_config_panos_validate_certs | true | PAN-OS: check the firewall's certificate |
| net_config_panos_ca_path | (empty) | PAN-OS: a CA bundle (PEM) for that check ('' = the system's) |
| net_config_timeout | 60 | PAN-OS: seconds per request |
| net_config_diff_lines | 60 | lines of a diff shown in the report, the email and an incident (secrets hidden) |
| net_config_fail_on_failed | true | the backup job fails (red) when a device could not be backed up |
| net_config_tickets | false | true = an incident when a device's configuration changed (one per change, with the diff) and when one could not be backed up (noted as cleared at the next good backup). The "ServiceNow API" credential. |
| net_config_ticket_changes | true | an incident for a change |
| net_config_ticket_failures | true | an incident for a failed backup |
| net_config_load_mode | lines | lines (send configuration lines) or restore (put a device back to a backup). |
| net_config_lines | [] | IOS: the lines, as typed in configuration mode, e.g. ["ntp server 192.0.2.123", "banner motd ^Authorized use only^"]; under net_config_parents (e.g. ["interface GigabitEthernet1"]) when they belong to one. |
| net_config_parents | [] | IOS: the parent lines the lines go under (e.g. ["interface GigabitEthernet1"]); [] = global configuration |
| net_config_panos_entries | [] | PAN-OS: the changes - each {xpath, element, action}: action set (adds or replaces the element's children), edit (replaces the element at the xpath), delete (no element). |

| Setting | Default | What it does |
|---|---|---|
| net_config_restore_ref | (empty) | restore: which backup - git: a commit, tag or branch ('' = the newest); share: a dated copy's file name ('' = the latest). |
| net_config_change | (empty) | The change request the load belongs to (CHG0012345). Its state must be one of net_config_change_states; the job writes on it what it did. A dry run without one only warns. |
| net_config_require_change | true | false = a real load without a change request (not recommended) |
| net_config_change_states | [Implement] | the change states a load may run in (as ServiceNow shows them) |
| net_config_change_note | true | a work note on the change: each device, what was sent, the result |
| net_config_save | true | IOS: write memory after a change (false: only the running configuration - a reload undoes it) |
| net_config_ios_file | flash:site-automation-restore.cfg | IOS restore: where the backup is copied first (scp; ip scp server enable on the device) |
| net_config_panos_allow_pending | false | PAN-OS: change even with someone's uncommitted changes (they would be committed too) |
| net_config_max_devices | 10 | a load to more devices at once is refused (net_config_only, the template's Limit - or raise it) |

## net_perf

File: roles/net_perf/defaults/main.yml

**Note**
NETWORK PERFORMANCE (playbooks/net_perf_report.yml): from each Linux host - the latency, jitter and loss to the targets it depends on (ping, and a timed TCP connect), its OCSP responders answering a real request (from the host, where it matters), its interfaces' drops and errors since the last run, and how today compares with the host's own recent runs (kept on the host). iperf3 throughput only between the pairs you name: short, one at a time. Read-only, except the few files of the host's own history (net_perf_dir). docs/NETWORK_PERF.md.

| Setting | Default | What it does |
|---|---|---|
| net_perf_targets | [] | ---- Targets (from every host of the job) ---------- --- name, host; port (optional: a timed TCP connect too); thresholds of its own (optional: rtt_warn_ms, rtt_crit_ms, loss_warn_pct, loss_crit_pct). A target may also be set per group or host (net_perf_targets_extr... (all of it: the role file) |
| net_perf_targets_extra | [] | more targets for some hosts (group_vars / host_vars): added to net_perf_targets |
| net_perf_ping_count | 20 | pings per target |
| net_perf_ping_interval | 0.2 | seconds between them (0.2 is the least a non-root ping may use) |
| net_perf_tcp_tries | 5 | timed TCP connects per target with a port |
| net_perf_rtt_warn_ms | 50 | average round trip from this many ms -> warning |
| net_perf_rtt_crit_ms | 200 | ... from this many -> critical |
| net_perf_loss_warn_pct | 1 | packet loss from this percent -> warning |
| net_perf_loss_crit_pct | 10 | ... from this percent -> critical (100 % = the target does not answer at all) |
| net_perf_timeout | 5 | seconds a single ping or connect may wait |
| net_perf_ocsp | [] | ---- OCSP responders (asked from every host) ---------- Simplest: site_ocsp_certs (below) - a certificate the responder knows, its OCSP address and issuer read from it. net_perf_ocsp, when set, is used instead: a url alone - an OCSP request about a made-up certific... (all of it: the role file) |
| site_ocsp_certs | [] | certificates the OCSP responders know (PEM files from playbooks/), for net_perf_ocsp and vm_sso_ocsp when those are empty, e.g. [files/ocsp/test-user.pem] |
| net_perf_ocsp_warn_ms | 2000 | an OCSP answer slower than this -> warning |
| net_perf_ocsp_severity | critical | an OCSP responder that fails: critical (smart card logins fail) or warning |

| Setting | Default | What it does |
|---|---|---|
| net_perf_nics | true | ---- Interfaces (drops and errors since the last run) ---------- read the interfaces' counters (/sys/class/net/*/statistics) |
| net_perf_nic_exclude | "^(lo\|veth\|virbr\|docker\|podman\|cni\|flannel\|vnet\|tap\|tun)" | interfaces left out (regular expression) |
| net_perf_sysfs | /sys/class/net | where the kernel shows the interfaces and their counters |
| net_perf_drops_warn_per_hour | 100 | dropped packets per hour (received + sent) from this many -> warning |
| net_perf_errors_warn_per_hour | 10 | errors per hour (CRC, frame, carrier ...) from this many -> warning |
| net_perf_dir | /var/lib/site-automation/netperf | ---- History (on the host) and the trend ---- ---------- --- each run's summary and the last counters; root-only. A dry run keeps nothing |
| net_perf_keep | 30 | runs kept on the host |
| net_perf_slower_factor | 2.0 | a target's average round trip this many times its usual (the median of the kept runs) -> warning |
| net_perf_slower_min_ms | 5 | ... and at least this many ms more than usual (2 ms against 1 ms is not news) |
| net_perf_iperf_pairs | [] | ---- iperf3 between named pairs (optional, needs iperf3 on both hosts and port 5201 open) ---------- server and client are inventory names of this job's hosts; the server runs iperf3 for one test only (-1), then exits. e.g. [{server: app01.example.mil, client: db01.example.mil, min_mbps: 900}] |
| net_perf_iperf_seconds | 5 | seconds per test (at most 30) |
| net_perf_iperf_port | 5201 | the TCP port iperf3 listens on, on the server of each pair |

## ontap_report

File: roles/ontap_report/defaults/main.yml

**Note**
NetApp ONTAP health report (playbooks/ontap_health_report.yml). Read-only: GET requests to each cluster's REST API, with the account of a "NetApp ONTAP" credential (docs/NETAPP.md).

| Setting | Default | What it does |
|---|---|---|
| ontap_clusters | [] | The clusters' management addresses (names or IPs), e.g. [cluster1.yoursite.mil, cluster2.yoursite.mil]. The credential's account must exist on each of them. |
| ontap_validate_certs | true | Check the clusters' HTTPS certificates. ONTAP's own certificate is often self-signed: then put its CA in ontap_ca_path (a file in the project, e.g. "{{ playbook_dir }}/files/ca/ontap-ca.pem"), or set false. |
| ontap_ca_path | (empty) | the CA file (PEM) the clusters' certificates are checked with; empty = the execution environment's own trust |
| ontap_timeout | 30 | seconds for one request (no answer: asked once more) |
| ontap_tls_legacy | false | true = connect to older ONTAP whose TLS the execution environment refuses (UNSAFE_LEGACY_RENEGOTIATION_DISABLED, KEY_TOO_SMALL, SHA-1) |
| ontap_volume_warn_pct | 85 | Thresholds (used %, hours, days, seconds): at or above "warn" = amber, at or above "crit" = red. |
| ontap_volume_crit_pct | 90 | volume used % -> red |
| ontap_volume_min_gb | 50 | volumes smaller than this (GB) are left out of the Volumes table - full (root volumes and other small ones run full by design) or not online; 0 = every volume |
| ontap_aggr_warn_pct | 85 | aggregate used % -> amber |
| ontap_aggr_crit_pct | 90 | aggregate used % -> red |
| ontap_inode_warn_pct | 85 | inodes used % of a volume -> amber |
| ontap_inode_crit_pct | 90 | inodes used % -> red |
| ontap_snapmirror_lag_warn_hours | 24 | last good transfer longer ago than this |
| ontap_snapmirror_lag_crit_hours | 48 | last good transfer longer ago than this -> red |
| ontap_snapshot_reserve_warn_pct | 100 | snapshots using this much of their reserve (100 = they spill into the volume) |
| ontap_snapshot_reserve_crit_pct | 150 | snapshots using this much of their reserve -> red |
| ontap_cert_warn_days | 60 | certificates expiring within this many days |
| ontap_cert_crit_days | 14 | certificates expiring within this many days (or expired) -> red |
| ontap_time_drift_warn_seconds | 60 | a node's clock against the AAP server's |
| ontap_time_drift_crit_seconds | 300 | a node's clock this far off -> red |
| ontap_port_errors_warn | 1 | receive errors (CRC and other bad frames) since the node booted |
| ontap_port_errors_crit | 10000 | receive errors on a port since boot -> red |

| Setting | Default | What it does |
|---|---|---|
| ontap_min_spares | 1 | fewer spare disks than this on a node = amber (0 = red) |
| ontap_ems_hours | 24 | the error events of the last this-many hours |
| ontap_ems_max | 500 | read at most this many events per cluster |
| ontap_ems_ignore_ldap | auto | leave out events whose name has "ldap" in it: auto = on a cluster where no SVM has an LDAP client configured; true = always; false = never |
| ontap_ems_ignore | [] | event names to leave out (regular expressions, any case), e.g. ['^secd\.ldap\.', 'callhome\.battery\.low'] |
| ontap_report_max_rows | 50 | rows per table (problems first); the job's artifacts have all |
| ontap_snapshot_top | 25 | the snapshot space table: the volumes whose snapshots use the most |
| ontap_report_fail | false | The job stays green when it finds problems: the report and email say what. true = mark it FAILED when there is a critical finding or a cluster could not be read, for workflows. |

## patch

File: roles/patch/defaults/main.yml

> **Note**
> PATCH: install updates with dnf, reboot if the updates need it, and prove the host came back: every service that was running before must be running after. One host at a time by default; the rollout stops at the first host that fails.

| Setting | Default | What it does |
|---|---|---|
| patch_packages | ['*'] | what to update ('*' = everything that has an update) |
| patch_security_only | false | true = only updates that fix security advisories |
| patch_exclude | [] | never update these here, e.g. ['mariadb*', 'kernel*'] |
| patch_serial | 1 | How many hosts are patched at the same time (1 = one by one: the rollout stops at the first host that fails). ONLY in the job template's Variables, a survey or the launch: Ansible picks the batch size before it reads all.yml or any group or host variables, so set there it does nothing. |

| Setting | Default | What it does |
|---|---|---|
| automatic_restarts | false | NO AUTOMATIC RESTARTS unless you say so: this playbook reboots a host only when automatic_restarts is true (set it in the job template's Variables, or answer the survey), and then only as patch_reboot says. With automatic_restarts false (the default) it installs the updates and tells you which hosts still need a reboo... (all of it: the role file) |
| patch_reboot | when_needed | Reboot (only with automatic_restarts: true): when_needed (needs-restarting says so) \| never \| always |
| patch_reboot_timeout | 1800 | seconds to wait for the host to come back |
| patch_never_reboot_groups | [aap, aap_hosts, sat, netapp, netapp_console_hosts] | Hosts in these groups are patched but NEVER rebooted by this playbook (reboot them by hand, in a maintenance window). Add your own group names; the built-in protection below is on always. |
| patch_never_patch_groups | [no_patch, aap, aap_hosts, sat, netapp, netapp_console_hosts] | Hosts in these groups are skipped entirely (Satellite is upgraded with satellite-maintain, not dnf). |
| patch_min_free_mb | 1024 | BUILT IN, whatever you set above (roles/patch/vars/main.yml): the AAP hosts (groups aap and aap_hosts) are never patched and never rebooted by a job, and neither is the machine the job runs from (a local connection). A package script can restart services under the running job. Set the two lists above only to ADD group... (all of it: the role file) |
| patch_service_wait | 120 | seconds to wait for services to come back after a reboot |
| patch_ignore_services | '^(user@\|user-runtime-dir@\|session-\|dnf-makecache\|packagekit\|fwupd\|run-\|systemd-(hostname… | Services that come and go on their own and are not compared before/after (regular expression) |

## poam

File: `roles/poam/defaults/main.yml`

| Setting | Default | What it does |
|---|---|---|
| `poam_csv` | `"{{ playbook_dir }}/../poam/poam.csv"` | POA&M STATUS: which Plan of Action & Milestones items are overdue or due soon, and (optional) which open CAT I / CAT II findings in STIG Manager have no POA&M item yet. The POA&M list is a CSV file in your work Git (export from eMASS or your spreadsheet, keep the columns below, save as CSV). POA&M data is sensitive: k... (all of it: the role file) |
| `poam_columns` | `id: POAM ID # unique item number weakness: Weakness # what is wrong severity: Severity #…` | Column names in YOUR file for each field this check needs (change the right-hand side). |
| `poam_closed_statuses` | `[Completed, Closed, Risk Accepted, Cancelled]` | items with one of these statuses are done: never reported |
| `poam_due_soon_days` | `30` | an open item due within this many days -> warning (past due -> critical) |
| `poam_date_formats` | `['%Y-%m-%d', '%m/%d/%Y', '%m/%d/%y', '%d-%b-%Y', '%d %b %Y', '%b %d, %Y']` | the date formats your file may use (Python strptime), tried in this order |
| `poam_stigman_api` | (empty) | ---- STIG Manager cross-check (optional) ---------- Needs a Keycloak client that may use the client-credentials grant, with the scope stig-manager:collection:read, and a grant in the collections to read. Your realm owner creates it - this repository never changes the realm. T... (all of it: the role file) |
| `poam_stigman_collections` | `[]` | collection names; empty = every collection the client can read |
| `poam_stigman_severities` | `[high, medium]` | CAT I and CAT II findings must have a POA&M item |
| `poam_stigman_scope` | (empty) | Scope to ask for. Empty = the client's default scopes (they must include stig-manager:collection:read or stig-manager:collection). |
| `poam_validate_certs` | `true` | check STIG Manager's and Keycloak's HTTPS certificates |

## podman_discover

File: `roles/podman_discover/defaults/main.yml`

**Note**
PODMAN DISCOVERY (read-only): every podman container on the host, root's (rootful) and each user's (rootless), with how it is run (a systemd unit or plain podman) and whether it should be running. Used by the Health check `containers` and by Service watch. It changes nothing.

| Setting | Default | What it does |
|---|---|---|
| `podman_discover_enabled` | `true` | false = do not look for containers at all (the containers check then says "skipped", and Service watch watches only its watch_containers list). |
| `podman_discover_rootless` | `true` | Also look at the containers of users (rootless podman). A user is looked at when a podman process (conmon) runs as them, when they have "linger" on (/var/lib/systemd/linger/NAME), or when they are in /etc/subuid AND have container storage (~/.local/share/containers/storage). Their containers are read AS that user (run... (all of it: the role file) |
| `podman_discover_users` | `[]` | More users to look at (names), e.g. an application account the rules above miss. |
| `podman_discover_ignore` | `[]` | Containers to leave out, as regular expressions that must match the WHOLE container name or OWNER/NAME, e.g. ['test-.*'] *(every container whose name starts with test-, any owner), ['alice/.*']* (all of alice's containers), ['root/web'] (root's web only). |
| `podman_discover_quadlet_dirs` | `[/etc/containers/systemd, /usr/share/containers/systemd]` | Where root's Quadlet files are (*.container, and *.kube: a pod from Kubernetes YAML). A user's are always read from ~/.config/containers/systemd, /etc/containers/systemd/users/UID and /etc/containers/systemd/users. |
| `podman_discover_unit_dirs` | `[/etc/systemd/system, /usr/local/lib/systemd/system]` | Where root's own systemd units are looked for: a *.service that runs podman (`podman generate systemd`, `podman run`, `podman kube play`, `podman pod start`, podman-compose / docker-compose), and the podman-kube@ units enabled there (.wants links). A user's: ~/.config/systemd/user. Packaged units (/usr/lib/systemd/sys... (all of it: the role file) |
| `podman_discover_timeout` | `60` | Seconds one podman or systemctl call may take before it is given up (reported as a finding). |

## service_watch

File: `roles/service_watch/defaults/main.yml`

| Setting | Default | What it does |
|---|---|---|
| `watch_discover` | `true` | ---- What to watch ---------- With watch_discover on (the default), service watch finds the containers itself: every podman container on the host that SHOULD be running - root's (rootful) and each user's (rootless) - as the Health check `containers` finds th... (all of it: the role file) |
| `watch_containers` | `[]` | Containers to watch in any case, in dependency order (a container's dependencies first). They are watched before the discovered ones, and one that is missing is reported as down. name: the container name, as `podman ps -a` shows it (or its Quadlet / service name) user: optional: the user that owns it (rootless podman)... (all of it: the role file) |
| `watch_url_ok_status` | `[200, 301, 302, 401, 403]` | HTTP codes that count as "answering" |
| `watch_find_units` | `true` | ---- How each container is run (found by itself) ---------- A container that a systemd unit runs (a Quadlet .container file, `podman generate systemd`, or a unit you wrote) must be started with systemctl, never with `podman start`: the unit owns it, and while the unit is stopped the co... (all of it: the role file) |
| `watch_quadlet_dirs` | `[/etc/containers/systemd, /usr/share/containers/systemd]` | where root's Quadlet files are looked for (to find the unit that runs a container) |
| `watch_mode` | `approval` | ---- What happens when one is down ---------- approval: ACT only diagnoses. Its proposed fix waits for the AAP approval step, then service_fix_approved.yml runs exactly the approved command(s) - no model involved. When ACT proposes nothing (or cannot run), the standard fix... (all of it: the role file) |
| `watch_verify_retries` | `6` | re-checks after a fix (every watch_verify_delay seconds) |
| `watch_verify_delay` | `5` | seconds between those re-checks |

| Setting | Default | What it does |
|---|---|---|
| watch_record_file | /var/log/service-watch/incidents.jsonl | ---- Where incidents are recorded (on the host) ---------- One JSON line per event, plus a syslog line (tag service-watch) that your log forwarding (Splunk, Elastic, rsyslog relay) picks up. AAP's job history keeps its own copy. |
| watch_record_syslog | true | also send each incident to syslog (tag service-watch) |
| watch_act_env | {} | ---- ACT settings ---------- Which provider and model: site_act_provider / site_act_model / site_act_url / site_act_ca, the same settings every runbook uses (roles/site_act/defaults/main.yml). The key comes from the "ACT model key" AAP credential, never f... (all of it: the role file) |
| watch_act_timeout | 1200 | seconds for one ACT run (the model's time per answer: site_act_model_timeout) |
| watch_instructions | (worked out by the role; see its file) | What ACT is told, on top of the act_triage role's standard instructions. |

## servicenow

File: roles/servicenow/defaults/main.yml

**Note**
SERVICENOW - open, update (and optionally resolve) incidents from check findings, and check that ServiceNow itself (the instance and its MID Servers) is healthy. Uses the REST Table API with ansible.builtin.uri: no extra collection to install. Connection: the "ServiceNow API" AAP credential sets SN_HOST (https://<instance>), SN_USERNAME and SN_PASSWORD in the job's environment. Without it, nothing is sent: the job prints what it WOULD have opened (a safe way to try it out).

| Setting | Default | What it does |
|---|---|---|
| servicenow_min_severity | warning | Findings at or above this severity get a ticket: critical \| warning. A ticket already open stays open while its finding is still there, even below this (the certificate renewal opens one for every CSR) |
| servicenow_table | incident | the table tickets are opened in (incident) |
| servicenow_assignment_group | (empty) | Values for new incidents. Names work (the API is asked to accept display values), e.g. servicenow_assignment_group: Linux Operations |
| servicenow_caller | (empty) | user name; empty = the API account |

| Setting | Default | What it does |
|---|---|---|
| servicenow_category | (empty) | e.g. Software |
| servicenow_urgency | {critical: "1", warning: "2"} | a new ticket's urgency, per finding severity (your instance's values) |
| servicenow_impact | {critical: "2", warning: "3"} | a new ticket's impact, per finding severity |
| servicenow_set_ci | true | set Configuration item = the host name (must exist in the CMDB) |
| servicenow_fields | {} | More fields for every new ticket - for an instance that makes fields such as Location, Subcategory or Channel mandatory: {column name: value}. Names and choice labels work, e.g. servicenow_fields: {location: "Building 100", subcategory: "Operating System", contact_type: "Self-service"} A field's column name: right-cli… (all of it: the role file) |
| servicenow_note_cleared | true | When a problem that had a ticket is gone, add a work note (and, if you want, resolve it). |
| servicenow_resolve_cleared | false | true = also resolve the ticket (servicenow_resolved_state, servicenow_close_code) |
| servicenow_resolved_state | "6" | Resolved |
| servicenow_close_code | "Solution provided" | must be a close code your instance has |
| servicenow_resolve_fields | {} | More fields set when a ticket is resolved (servicenow_resolve_cleared, and the test ticket), for an instance that makes them mandatory: {column name: value}, e.g. {resolved_by: svc_aap_sn} |
| servicenow_note_repeats | true | Add a work note to an open ticket each time its finding is seen again. |
| servicenow_correlation_prefix | site-health | Every ticket from these playbooks carries a correlation ID starting with this, so the same problem on the same host always updates the same ticket. |
| servicenow_validate_certs | true | check the instance's HTTPS certificate (see servicenow_ca_path) |

| Setting | Default | What it does |
|---|---|---|
| servicenow_ca_path | (empty) | A CA bundle (PEM) for the instance's TLS certificate, when the execution environment does not trust its CA (e.g. a DoD root CA): a path on the controller, such as a file you add to the project: servicenow_ca_path: "{{ playbook_dir }}/files/ca/servicenow-ca.pem". Empty = the execution environment's normal trust store.... (all of it: the role file) |
| servicenow_timeout | 30 | seconds for one API request |
| servicenow_fail_when_findings | true | Fail the job when it opened/updated tickets, so the workflow shows red while problems exist. |
| servicenow_check_mid | true | servicenow_health.yml: also check the MID Servers (needs read access to the ecc_agent table) |
| servicenow_slow_seconds | 5 | the API answering slower than this -> warning |
| servicenow_mid_names | [] | ---- servicenow_health.yml: the MID Servers ---------- What the instance says about each MID Server (ecc_agent): Up and validated, its heartbeat (last_refreshed) no older than servicenow_mid_stale_minutes, the same version on every MID Server, the ones you expect all there; its... (all of it: the role file) |
| servicenow_mid_expected | [] | these must exist (names): one missing -> critical |
| servicenow_mid_stale_minutes | 15 | Up, but its heartbeat older than this -> critical (it reports every few minutes) |
| servicenow_mid_version_check | true | MID Servers on different versions -> warning (one stuck in an upgrade) |
| servicenow_mid_queue_check | true | read ecc_queue (the account needs read on it, e.g. the mid_server role) |
| servicenow_mid_queue_minutes | 10 | output waiting longer than this for its MID Server counts as stuck |
| servicenow_mid_queue_warn | 1 | this many stuck messages -> warning |
| servicenow_mid_queue_critical | 50 | ... this many -> critical |
| servicenow_mid_issues_check | true | read the open MID Server Issues (ecc_agent_issue) |
| servicenow_check_mid_hosts | true | The MID Server HOSTS (Windows, over WinRM), in the same job and report: the hosts of this group of the job's inventory - the Windows inventory, with its Machine (Windows) credential next to the ServiceNow API one. No such group (or false below): the instance's view only. Read-only. On each host: every MID Server servi... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| servicenow_mid_hosts_group | servicenow_mid_hosts | the inventory group of the MID Server hosts |
| servicenow_mid_service_match | "snc_mid*" | service names (* and ?); the display name "ServiceNow MID Server*" matches too |
| servicenow_mid_hours | 24 | how far back the logs are read for SEVERE lines and JVM restarts |
| servicenow_mid_log_stale_minutes | 30 | the agent log not written for this long while the service runs -> critical (hung) |
| servicenow_mid_severe_warn | 20 | this many SEVERE lines in servicenow_mid_hours -> warning |
| servicenow_mid_jvm_restarts_warn | 1 | this many JVM restarts (crash, hang, out of memory) -> warning |
| servicenow_mid_disk_warn_pct | 10 | free space of the agent's drive under this % -> warning |
| servicenow_mid_disk_critical_pct | 5 | ... under this % -> critical |
| servicenow_mid_url_check | true | config.xml's instance URL not this instance (a clone, a moved MID) -> critical |
| servicenow_test_resolve | true | servicenow_test_ticket.yml: resolve the test incident at the end (false = leave it open to look at) |
| cert_renew_servicenow | incident | How the CSRs the certificate renewals make (Linux, Windows, VMware, network devices) go to the PKI. incident: each one is attached to its certificate's incident, with a work note saying what to do next - a person submits it to the PKI. catalog: each one is ordered from your PKI's catalog item (cert_renew_catalog_item)... (all of it: the role file) |
| cert_renew_csr_memory_days | 90 | incident: a CSR on an incident of the certificate made in the last this-many days counts as made - also when that incident was resolved or closed since (the PKI may still be signing it): vCenter, ESXi and Cisco keep one pending key, and a second CSR would replace it. Older: a new CSR (the certificate is due again). |
| cert_renew_catalog_item | (empty) | The catalog item: its name exactly as the catalog shows it (e.g. "DoD PKI Certificate Request"), or its sys_id. |

| Setting | Default | What it does |
|---|---|---|
| cert_renew_catalog_variables | {} | The item's questions - by their internal names, as the catalog test lists them - and the answers: {name: answer}. A reference (Requested For, Location ...): the record's name, or its sys_id. A choice: its value or its label. In an answer, {host}, {subject}, {names}, {expires}, {issuer} and {where} become the certifica... (all of it: the role file) |
| cert_renew_catalog_csr_variable | (empty) | The attachment question that gets the CSR (its internal name). Empty = the item's only attachment question. |
| cert_renew_catalog_names_variable | (empty) | A question that gets the certificate's names (subjectAltName), e.g. a "List any additional SANs" box. Empty = none. |
| cert_renew_catalog_reorder_states | ["4", "7"] | A request closed in one of these states (a requested item's: 4 Closed Incomplete, 7 Closed Skipped - rejected or cancelled) is ordered again on the next run. In any other state - open, in progress, Closed Complete (the PKI issued it: install it) - the certificate is not ordered again. |
| cert_renew_catalog_upload_table | ZZ_YYsc_cart_item | Where the CSR is uploaded before the order, for the attachment question (ServiceNow moves it to the request). |
| servicenow_catalog_test_order | false | servicenow_catalog_test.yml: true = also order the item once, with a test CSR - the PKI gets a real request (tell them first, or cancel it). false = only read the item and check the settings. |
| cert_install_attachment_exts | [cer, crt, pem, der, p7b, p7c, p7, cert, zip, txt] | ---- Certificate install: the issued certificate, from its request (cert_install_check.yml, win_cert_install_check.yml) ---- The PKI puts the issued certificate on the request (RITM) that carries the CSR: attached to the RITM, to one of its attachment questions, to its REQ or to its catalog task (SCTASK) - one file or... (all of it: the role file) |
| cert_install_max_bytes | 1048576 | a bigger attachment is not read |
| cert_install_journal | true | also look for a certificate pasted into the request's work notes and comments |
| cert_install_min_days | 30 | the issued certificate must be valid at least this many more days (else held for a person) |
| cert_install_fewer_names | false | true = install a certificate that lacks names (subjectAltName) the old one had; false = held for a person |

## site_act

File: `roles/site_act/defaults/main.yml`

| Setting | Default | What it does |
|---|---|---|
| `site_act_level` | `explain` | ACT (GenAI) on top of ANY check playbook. The check roles already did the cheap, certain part: they found WHAT is wrong (site_findings) and know which read-only command shows more (each finding's `hint`). This role hands that to ACT (the act_triage role) to find out WHY, and to propose - or, if you allow it, make - th... (all of it: the role file) |
| `use_act` | `false` | The runbooks (Health check, Database health, Troubleshoot): true = ACT looks at what the job found. A job variable, per template or launch - not all.yml (every check job would then call the model). |
| `site_act_allow` | `[]` | self-heal only: regular expressions of fix commands ACT may run without a person. Each must match the WHOLE command. Keep them narrow, e.g. site_act_allow: ['systemctl restart (chronyd\|rsyslog\|crond)'] |
| `site_act_diagnose_only_groups` | `[aap, aap_hosts, sat, idm, netapp, netapp_console_hosts]` | Hosts where ACT may only diagnose - never fix by itself, whatever site_act_level says: AAP itself (a fix could stop the job that is running it) and vendor-managed appliances. The apply step (act_fix_approved.yml) skips them too: a person applies the fix there by hand. AAP (groups aap, aap_hosts) is ALWAYS in this list... (all of it: the role file) |
| `site_act_provider` | `genai` | ---- Which model ACT talks to ---------- site_act_provider genai GenAI.mil (the default) asksage Ask Sage - RETIRED FOR NOW: do not pick it (kept for when it returns) genai-beta the GenAI.mil beta proxy (preview models) site_act_models the model to use FOR EACH pr... (all of it: the role file) |
| `site_act_models` | `{genai: gemini-3.8-flash, asksage: gpt-5.6-sol-gov}` | the model for each provider (see above): the model follows the provider picked |
| `site_act_urls` | `{}` | the API URL for each provider; a provider not listed uses ACT's default |
| `site_act_model` | (empty) | one model for this run, whatever the provider (e.g. a survey answer); empty = site_act_models |
| `site_act_url` | (empty) | one URL for this run; empty = site_act_urls |

| Setting | Default | What it does |
|---|---|---|
| site_act_ca | (empty) | a CA bundle file ON THE HOSTS for the provider's certificate; empty = the host's trust store |
| site_act_env | {} | Any other ACT environment settings (advanced; these win over the ones above), e.g. {HTTPS_PROXY: "http://proxy.yoursite.mil:8080"}. See the top of vendor/act/act for the list. Every ACT job streams the model's answer (ACT_STREAM=1), so a firewall or proxy idle timeout does not cut a long answer ("connection reset by p... (all of it: the role file) |
| site_act_model_timeout | 600 | Seconds the model may take for ONE answer, retries after a provider hiccup included (ACT's GENAI_TIMEOUT; ACT's own default, 90, is too short for a long analysis or a busy provider). Every ACT job uses it: the runbooks' use_act, the VMware alarms and capacity analyses, service watch. |
| site_act_timeout | 1200 | seconds for one whole ACT run (several answers): keep it above site_act_model_timeout |
| site_act_concurrency | 3 | How many hosts may run ACT at the same time in one job (0 = no limit: as many as the job's forks). Every host in a job uses the same key, and GenAI.mil's default quota is 60 requests and 200,000 tokens a minute per key: a fleet run with many hosts at once gets HTTP 429 (too many requests). |
| site_act_batch | 5 | The jobs where ACT reads evidence (the VMware alarms analysis and capacity planning, the log analysis) ask it about this many items per question: a long answer about many items is what a gateway or proxy cuts off, or what runs past site_act_model_timeout. 15 items are 3 questions of 5, in one report. 0 = all in one qu... (all of it: the role file) |
| site_act_extra_instructions | (empty) | anything ACT should know about this site |
| site_act_evidence_text | (empty) | extra evidence (troubleshoot.yml passes what it collected) |

## site_email

File: roles/site_email/defaults/main.yml

**Note**
EMAIL A REPORT through your mail relay (SMTP). Off until report_email_to has an address. Used by every report job: health checks, certificate reports, database health, POA&M, ServiceNow health, troubleshooting, the VMware jobs (docs/EMAIL_REPORTS.md). The relay settings belong in playbooks/group_vars/all.yml. If the relay needs a login, attach an "SMTP relay" credential (aap/credential_types/smtp_relay.yml) - never put a password in these settings.

| Setting | Default | What it does |
|---|---|---|
| report_email_to | [] | Who gets it: a list, or text with commas / spaces / one per line (a survey answer). Empty = no email. Set it on each TEMPLATE (its Variables, or a survey): in all.yml every report job emails. |
| report_email_cc | [] | copied: a list, or text like report_email_to |
| report_email_also | [] | "Also email the report to": more people for one run, asked when the job is launched - a survey question (playbooks/aap_email_question.yml adds it to the report templates). Added to report_email_to, never in its place; empty = only report_email_to. A list, or text: addresses separated by commas, semicolons, spaces or o... (all of it: the role file) |
| report_email_also_domains | [] | The domains an address in report_email_also may be in (a subdomain too), e.g. [yoursite.mil]; any other is listed in the job and gets no email. Empty = any domain (your mail relay may still refuse some). |
| report_email_from | (empty) | the sender, e.g. aap-noreply@yoursite.mil |
| report_email_smtp_host | (empty) | your mail relay, e.g. smtp.yoursite.mil |
| report_email_smtp_port | 25 | 25 or 587 with starttls, 465 with ssl |
| report_email_security | starttls | starttls = encrypt after connecting (the relay must offer it); ssl = encrypted from the start (port 465); none = not encrypted (only for a relay without encryption; then no login is sent). |
| report_email_ca_path | (empty) | a CA file that signed the relay's certificate (empty = the CAs the execution environment trusts) |
| report_email_subject_prefix | "[AAP]" | put before every subject, e.g. "[AAP] Certificate report: ..." |

| Setting | Default | What it does |
|---|---|---|
| report_email_html | true | Send a formatted (HTML) version too, for reports that come with one (all VMware jobs). The email always carries a plain-text copy for mail clients that do not show HTML. false = plain text only. |
| report_email_xlsx | false | Attach the report's tables to the email as an Excel workbook (.xlsx): a sheet per table. Some reports attach their own whatever this says (VM - inventory report: every VM on one sheet). The website gets no workbook. |
| report_email_theme | standard | The look of the HTML version (docs/EMAIL_REPORTS.md, "The look"): standard = a header band in the report's colour (green, amber, red, blue) with number boxes under it; command = a dark header with your logo and report_email_header_text, the job's last runs as a strip of bars (report_history), the numbers in one row; p... (all of it: the role file) |
| report_email_header_text | Site automation report | command, portal, glass: the line with the logo |
| report_logo_file | (empty) | Your logo in the header (every look but standard): a PNG, JPEG or GIF in your repository - a path from playbooks/ (files/report_logo.png = playbooks/files/report_logo.png) or a full path. Attached to each email, so keep it small: about 160 px tall, under 100 KB. Updating with scripts/update-from-release: list the file... (all of it: the role file) |
| report_logo_height | 84 | pixels in the email (its width follows); portal: at most 56, glass: at most 80 |
| report_history | true | command, portal, glass: the job's last runs, read from AAP - attach a "Red Hat Ansible Automation Platform" credential to the template, whose user can see the template's jobs (Organization Auditor on its organization, docs/EMAIL_REPORTS.md); without it the email goes out without the strip. |
| report_history_runs | 14 | bars in the strip, this run included |
| report_history_aap_ca_path | (empty) | a CA file for AAP's own HTTPS certificate ("" = what the execution environment trusts) |

| Setting | Default | What it does |
|---|---|---|
| report_email_html_template | report.html.j2 | Your own look instead: copy roles/site_email/templates/report.html.j2 (or report_command / _portal / _glass) to e.g. playbooks/files/email/report.html.j2, change it, and set report_email_html_template: "{{ playbook_dir }}/files/email/report.html.j2" report.html.j2 (the default) = the look report_email_theme picks. |
| report_time_zone | (empty) | The time zone of the times in a report, besides UTC: e.g. America/New_York (a name as timedatectl list-timezones prints them) shows "2026-10-25 10:03:12 EDT (14:03 UTC)". vCenter, ESXi and their log files keep UTC; the vSphere Client shows the browser's time zone. "" = UTC only. Used by the VMware health and alarms... (all of it: the role file) |
| report_web_dest | (empty) | THE REPORT WEBSITE (docs/WEB_REPORTS.md): every report job also writes its report as a web page - off until report_web_dest is set. Works without an email (report_email_to empty). Each run: its page, a copy as the template's latest page, the report's data as JSON beside each (for your own site), and an index of every... (all of it: the role file) |
| report_web_url | (empty) | the address people open that folder at, e.g. https://reports.yoursite.mil ("" = dest when it is https://) |
| report_web_theme | portal | the page's look: standard, command, portal or glass (as report_email_theme) |
| report_web_page | "{template}/{date}_{time}_job-{job}.html" | The names inside the folder, with {names} filled in: {template} (the AAP template's name, lower case with dashes; the play's name outside AAP), {template_id}, {job} (the AAP job; local outside AAP), {status} (ok, warning, critical, info), {date} (2026-10-23), {time} (0605), {year}, {month}, {day} - the time the job wr... (all of it: the role file) |
| report_web_latest | "{template}/latest.html" | always the template's newest run ("" = none) |
| report_web_index | index.html | every report's last runs, a tab per area ("" = none) |

| Setting | Default | What it does |
|---|---|---|
| report_web_history | 5 | the runs each report keeps: listed under its line of the index, the pages of older runs deleted |
| report_web_delete_old | true | false = the older runs' pages stay in the folder (only the index forgets them) |
| report_web_tabs | - {tab: Certificates, match: ["*certificate*", "*trust store*"]} - {tab: VMware, match: […] | The index page's tabs: a report is on the first tab one of whose names its AAP template's name matches (any case, * = anything); else on the tab of its name up to " - " (Backup - nightly: Backup); else on Other. A tab with no report is not shown. Set it in all.yml, for every template alike: the index is one page. |
| report_web_names | {} | your own {names}, e.g. {site: "{{ site_code }}"} for "{site}/{template}/..." |
| report_web_json | true | the report's data as .json beside each page and index.json beside the index |
| report_web_smb_auth | negotiate | A share (report_web_dest \\server\share\folder): how the job logs in, and the connection. negotiate (Kerberos - FIPS too: user@YOURDOMAIN.MIL, the server's full name - else NTLM), ntlm, or kerberos (a ticket already in the job) |
| report_web_smb_encrypt | false | true = encrypt (SMB 3) even when the share does not require it; one that requires it is always |
| report_web_smb_port | 445 | the share's port |
| report_web_ca_path | (empty) | a CA file for an https:// report_web_dest, or report_web_host's WinRM certificate ("" = what the EE trusts) |
| report_web_host | (empty) | Or on a Windows server (a file server, or the IIS server itself): report_web_host is that server, report_web_dest the folder on it (D:\reports, there already). The pages are made on the AAP side and written there over WinRM, in one PowerShell script per run (Constrained Language Mode too): nothing to mount, no WebDAV.... (all of it: the role file) |
| report_web_host_connection | winrm | winrm or psrp |
| report_web_host_port | 5986 | WinRM: 5986 (https), 5985 (http) |

| Setting | Default | What it does |
|---|---|---|
| report_web_host_scheme | https | https, or http (Kerberos and NTLM encrypt each message then) |
| report_web_host_transport | ntlm | ntlm, kerberos (Kerberos in the EE), credssp or basic |
| report_web_host_cert_validation | validate | validate (the CA: report_web_ca_path) or ignore |
| report_web_copy | false | THE COPY JOB instead (docs/WEB_REPORTS.md, "The copy job"): for report templates whose execution environment cannot reach the web server (no WinRM or Kerberos in it). true = the report jobs write nothing there - each keeps its report in AAP (the job artifact site_report_page) and the one job template "Report website -... (all of it: the role file) |
| report_web_copy_max_kb | 256 | the kept report's greatest size, packed (a bigger one: its longest tables cut, said on the page) |
| report_web_copy_jobs | [] | The copy job itself (playbooks/report_web_copy.yml): it reaches report_web_host as a host of its own inventory when it is one (that host's connection settings and the template's Machine credential - your Windows inventory), else with the report_web_host_* settings above. Its "Red Hat Ansible Automation Platform" crede... (all of it: the role file) |
| report_web_copy_hours | 24 | its first run (no copy before it): the report jobs of the last 24 hours |
| report_web_copy_max_pages | 50 | pages a run writes at most; the rest are the next run's |

## site_findings

File: roles/site_findings/defaults/main.yml

**Note**
Shared by every check playbook. A "finding" is one thing that is wrong on one host: check: the check that found it (disk, services, mariadb, ...) id: a stable name for this problem on this host, e.g. disk:/var. ServiceNow uses it to update the same ticket instead of opening a new one every run. severity: critical \| warning summary: one line a person understands ("/var is 96% full (critical at 95%)") hint: a read-only command that shows more. A person can paste it into a shell; ACT (when you add it later) runs it to collect evidence. Check roles only ADD findings. This role starts the list, reports it, and decides pass/fail.

| Setting | Default | What it does |
|---|---|---|
| site_fail_on | [critical, warning] | Which severities make the job FAIL. A failed job is what sends an AAP workflow down its "on failure" path (open tickets, ask ACT, wait for approval). [] = report only, never fail. |
| site_unreachable_fails | false | A host that cannot be reached (switched off, no network, ssh or WinRM refused, its login refused) in the health checks, the certificate report and renewal, database health, the service watch and troubleshoot: listed as NOT REACHED and left out, and the job is not failed for it. It still counts as checked-for, so the S... (all of it: the role file) |
| site_syslog | true | One syslog line per finding (tag below), so your log server / SIEM sees them too. |
| site_syslog_tag | site-health | the syslog tag those lines carry (to filter on in the SIEM) |
| site_eventlog_source | site-health | Windows: one Application event log entry per finding instead of syslog (source below). |
| site_os | linux | ---- Linux or Windows (set by the playbook; do not set) ---------- Linux playbooks run the roles check_<name>, the Windows ones (win_*.yml) the roles win_check_<name>. |
| site_check_role_prefix | "{{ 'win_check_' if site_os == 'windows' else 'check_' }}" | check_ (Linux) or win_check_ (Windows): set by the playbook from site_os |
| site_windows_connections | [winrm, psrp, ansible.builtin.winrm, ansible.builtin.psrp] | How a host is recognized as Windows: its connection, or its shell (Windows OpenSSH). |
| site_settings_notice | true | A notice in the job output when a project settings file (playbooks/group_vars/*.yml) defines a setting: it wins over AAP's inventory Variables box. false = no notice. |

## stigman_stack

File: roles/stigman_stack/defaults/main.yml

> **Note**
> STIG Manager stack on podman, managed by systemd through Quadlet unit files: stigman-mysql (MySQL 8.4) <- stigman-api (STIG Manager) <- stigman-nginx (TLS, port 443) Keycloak is EXTERNAL: the realm already exists and this role never touches it. Needs podman >= 4.4 (Quadlet): RHEL 9.2+ / RHEL 8.8+.

| Setting | Default | What it does |
|---|---|---|
| stigman_scope | system | system = rootful (/etc/containers/systemd, systemctl) <- production user = rootless (~/.config/containers/systemd, systemctl --user) <- tests / rootless hosts |
| stigman_images | mysql: docker.io/library/mysql:8.4 api: docker.io/nuwcdivnpt/stig-manager:1.6.17 # pin a… | the container images (pin STIG Manager's version; change it deliberately) |
| stigman_config_dir | /etc/stigman | nginx.conf, TLS files, CA bundle |
| stigman_https_port | 443 | host port nginx publishes |
| stigman_server_name | "{{ inventory_hostname }}" | the name nginx answers to (and the TLS certificate is for) |
| stigman_oidc_provider | "https://keycloak.example.mil/realms/stigman" | Existing Keycloak realm (server-side URL used by the API; browser-side URL for the web client) |
| stigman_client_oidc_provider | "{{ stigman_oidc_provider }}" | the Keycloak realm URL the browser uses (when it differs from the server-side one) |
| stigman_client_id | stig-manager | the Keycloak client the web app logs in with |
| stigman_classification | U | banner: U, CUI, C, S, TS, SCI, NONE |
| stigman_db_schema | stigman | the MySQL database STIG Manager uses |
| stigman_db_user | stigman | the MySQL account STIG Manager uses (its password: the credential) |
| stigman_mysql_buffer_pool | 1G | STIG Manager docs: >= 8G for real deployments |
| stigman_ready_retries | 40 | How long to wait after (re)starting: for MySQL to be healthy, then for the API to answer through nginx (each) |
| stigman_ready_delay | 6 | seconds between those tries |
| stigman_mysql_root_password | "{{ lookup('ansible.builtin.env', 'STIGMAN_MYSQL_ROOT_PASSWORD') }}" | ---- Secrets: NEVER set these in a vars file or inventory ---------- They come from AAP credentials (aap/credential_types/*.yml) as environment variables / files on the controller (execution environment). See docs/SECRETS.md. |

| Setting | Default | What it does |
|---|---|---|
| stigman_db_password | "{{ lookup('ansible.builtin.env', 'STIGMAN_DB_PASSWORD') }}" | from the credential (STIGMAN_DB_PASSWORD), never set here |
| stigman_tls_cert_src | "{{ lookup('ansible.builtin.env', 'TLS_CERT_FILE') }}" | nginx's certificate file, from the TLS certificate credential (TLS_CERT_FILE) |
| stigman_tls_key_src | "{{ lookup('ansible.builtin.env', 'TLS_KEY_FILE') }}" | its private key file, from the credential (TLS_KEY_FILE) |
| stigman_tls_ca_src | "{{ lookup('ansible.builtin.env', 'TLS_CA_FILE') }}" | the CA bundle file, from the credential (TLS_CA_FILE) |
| stigman_replace_secrets | false | A deploy refuses to change a database password secret already deployed: in MySQL the password stays the old one, and STIG Manager would lose its database. true = replace the secrets anyway - ONE launch only, when the database is being created again (Prompt on launch, Variables); never save it in the template. |

## troubleshoot

File: roles/troubleshoot/defaults/main.yml

| Setting | Default | What it does |
|---|---|---|
| ts_area | overview | TROUBLESHOOT: the commands an experienced admin runs first for a kind of problem, run for you (read-only), each with a line saying what it shows. The matching health checks run too, so you also get plain-words findings. Pick the area; for "service" and "network" say what. ts_area: overview \| disk \| performance \| servi... (all of it: the role file) |
| ts_service | (empty) | the systemd unit, for ts_area service (see above) |
| ts_target | (empty) | host:port, for ts_area network |
| ts_since | "-2h" | how far back logs are read (journalctl --since) |
| ts_output_lines | 60 | keep the last this-many lines of each command |
| ts_areas | overview: checks: [disk, mounts, services, performance, time, logging] commands: - {why:… | What each area runs. why is printed above the command's output. Commands must be read-only. |

## vmware_vm

File: `roles/vmware_vm/defaults/main.yml`

> **Note**
> VMWARE VMs: jobs that run against vCenter (not on the VMs), with the certified vmware.vmware collection. vCenter address and account come from a "VMware vCenter" credential on the job template (VMWARE_HOST, VMWARE_USER, VMWARE_PASSWORD). docs/VMWARE.md

| Setting | Default | What it does |
|---|---|---|
| `vm_names` | `[]` | The VMs, by their exact name in vCenter: a list, or (from a survey) one per line. No wildcards. |
| `vmware_datacenter` | (empty) | only needed when the same VM name exists in two datacenters |
| `vmware_validate_certs` | `true` | Check vCenter's certificate. false only if the execution environment does not trust your CA yet. |
| `vmware_vcenters` | `[]` | Several vCenters (e.g. Enhanced Linked Mode), one report: the reports and the ESXi security settings read the credential's vCenter AND these (names, or name:port), with the same account. The same vCenter under two names is read once. docs/VMWARE.md, "Several vCenters". e.g. [vc01.yoursite.mil, vc02.yoursite.mil] |
| `vmware_vcenter_fail_unread` | `false` | true = the job shows failed when a vCenter could not be read (the report lists it in red either way) |
| `vmware_protected_groups` | `[aap, aap_hosts]` | The AAP server is ALWAYS refused (hosts in the groups aap / aap_hosts, matched by VM name, guest host name and IP address), and so are the hosts of these groups and these VM names (* and ? allowed: DC*). List the AAP VM's name here when it differs from its host name, and vCenter's own VM. The snapshot cleanup never de... (all of it: the role file) |
| `vmware_protected_vms` | `[]` | e.g. [AAP01, VCSA01, DC*] (* and ? allowed) |
| `vm_power_mode` | `guest` | guest = ask the operating system (needs VMware Tools running; services stop cleanly) hard = reset / power off, like the button on a physical server (unsaved data is lost) |

| Setting | Default | What it does |
|---|---|---|
| vm_power_timeout | 600 | shut down: seconds to wait for the VM to be off |
| vm_snapshot_name | (empty) | The snapshot's name. Taking one: empty = aap-<date>-<time>. Deleting: the exact name to delete. |
| vm_snapshot_description | (empty) | the AAP job number and user are added |
| vm_snapshot_quiesce | false | true = VMware Tools freezes the file systems first |
| vm_snapshot_delete_all | false | delete: true = every snapshot of the VM (vm_snapshot_name empty) |
| vm_notes | (empty) | The text to write. |
| vm_notes_mode | append | append (a new line, with date and user) \| replace \| clear |
| vm_nic | 1 | Which adapter: 1 = Network adapter 1, 2, ... |
| vm_portgroup | (empty) | the port group (standard or distributed) to move it to |
| vm_secure_boot_folder | (empty) | Only the VMs in this vCenter folder ("" = every VM, or the ones in vm_names). |
| vm_secure_boot_fail | false | The job stays green (successful) when it finds VMs to fix: the report and email say what. true = mark the job FAILED when a VM is BIOS or has secure boot off, so a workflow's "on failure" path can react (open a ticket). |
| vm_datastore_warn_pct | 85 | Used space (% of capacity): a datastore at or above these is listed as a warning / critical. |
| vm_datastore_crit_pct | 90 | a datastore this % used or more is red (critical); amber from vm_datastore_warn_pct |
| vm_datastore_fail | false | The job stays green when datastores are full: the report and email say which. true = mark the job FAILED when one is at or above vm_datastore_warn_pct (or inaccessible), for workflows. |

| Setting | Default | What it does |
|---|---|---|
| vm_hostnet_services | [vmotion] | Read-only: vmkping between the hosts of each cluster on these VMkernel services' adapters (full size, don't fragment), the adapters' MTUs, the storage latency. The pings run on the hosts through vCenter (esxcli): the vCenter account needs Host > CIM > CIM interaction on them (docs/VMWARE.md, "The ESXi network check").... (all of it: the role file) |
| vm_hostnet_targets | [] | pinged from every host, e.g. [{name: NFS array, host: 192.0.2.50, interface: vmk2}] |
| vm_hostnet_hosts | [] | only these hosts (names, * and ?); [] = every host |
| vm_hostnet_clusters | [] | only these clusters |
| vm_hostnet_exclude | [] | never these hosts |
| vm_hostnet_ping | true | false = the adapters and the storage latency only, no pings |
| vm_hostnet_ping_count | 3 | pings per pair |
| vm_hostnet_max_pairs | 400 | pings at most per vCenter (16 hosts in one cluster: 240) |
| vm_hostnet_workers | 4 | pings at the same time |
| vm_hostnet_rtt_warn_ms | 5 | a round trip this slow or slower: warning (vMotion within a site is well under 1 ms) |
| vm_hostnet_rtt_crit_ms | 150 | ... critical (vMotion's own limit) |
| vm_hostnet_latency | true | read the storage latency (the last hour of the hosts' real-time statistics) |
| vm_hostnet_latency_minutes | 60 | minutes of the hosts' real-time statistics read (at most 60) |
| vm_hostnet_latency_warn_ms | 20 | a datastore's average read or write latency from this: warning |
| vm_hostnet_latency_crit_ms | 50 | ... critical |
| vm_hostnet_raw | false | true = the first ping and the first failed one whole, in the job's output: the host's parameters of the command, what was sent, the request, the answer or the fault (the first run, or a host that refuses a parameter) |

| Setting | Default | What it does |
|---|---|---|
| vm_hostnet_fail | false | true = the job shows failed when the report has a warning or worse (for workflows) |
| vm_inventory_templates | false | ---- VM inventory report (playbooks/vm_inventory_report.yml) ---- true = templates counted and listed as VMs (false: left out, their number in the subtitle) |
| vm_inventory_by_folder | true | a table per vCenter and folder; false = one table of every VM (vCenter and folder columns) |
| vm_inventory_max_rows | 2000 | VMs listed in the email and the page at most (2000: about 2.5 MB of email; the workbook always has every VM) |
| vm_inventory_xlsx | true | attach every VM as an Excel workbook (.xlsx) to the email |
| vm_rightsize_days | 30 | ---- VM right-sizing report (playbooks/vm_rightsizing_report.yml) ---- vCenter's own statistics only (statistics level 1, vCenter's default, is enough). Suggestions only: nothing is changed. days of history (at most 30: what vCenter keeps of 2-hour averages) |
| vm_rightsize_year | true | also each VM's busiest day of the last year (daily averages): month-end and yearly peaks |
| vm_rightsize_min_days | 7 | fewer days of statistics than this = 'Not enough history' (no suggestion) |
| vm_rightsize_clusters | [] | only the VMs of these clusters (names; * and ? allowed); [] = every VM |
| vm_rightsize_vms | [] | only these VMs (names; * and ? allowed) |
| vm_rightsize_exclude | [] | never these VMs (names; * and ? allowed) |
| vm_rightsize_cpu_target_pct | 70 | vCPUs suggested so that the 95th percentile of CPU use would be about this |
| vm_rightsize_cpu_high_pct | 85 | CPU at this or more 95% of the time = too small |
| vm_rightsize_ready_warn_pct | 5 | CPU ready (waiting for a physical CPU, per vCPU) at this or more is said |

| Setting | Default | What it does |
|---|---|---|
| vm_rightsize_min_vcpus | 2 | never fewer vCPUs than this |
| vm_rightsize_min_vcpus_windows | 2 | ... for Windows |
| vm_rightsize_cpu_max_cut_pct | 50 | never suggest more than this % fewer vCPUs at once |
| vm_rightsize_mem_headroom_pct | 100 | memory suggested = active memory plus this % (active memory understates real need) |
| vm_rightsize_mem_high_pct | 90 | active memory at this % or more of what the VM has = too small |
| vm_rightsize_min_mem_gb | 2 | never less memory than this (GB) |
| vm_rightsize_min_mem_gb_windows | 4 | ... for Windows |
| vm_rightsize_mem_max_cut_pct | 50 | never suggest more than this % less memory at once |
| vm_rightsize_min_change_pct | 25 | a smaller memory size is suggested only when it is this % (and 2 GB) smaller or more |
| vm_rightsize_taken_mb | 100 | memory ballooned or swapped by the host at this many MB or more (its peak) is said |
| vm_rightsize_disk_warn_pct | 90 | a guest disk this % used or more (VMware Tools' numbers) is listed |
| vm_rightsize_max_rows | 500 | rows per table in the email and the page (the workbook always has every VM) |
| vm_rightsize_xlsx | true | attach every VM with every number as an Excel workbook (.xlsx) to the email |
| vm_rightsize_batch | 10 | VMs per statistics query (vCenter refuses large ones: vpxd.stats.maxQueryMetrics) |
| vm_rightsize_workers | 4 | statistics queries at a time |
| vm_rightsize_act | false | GenAI (ACT) reviews the suggestions: needs an "ACT model key" credential (docs/VMWARE.md, "ACT analysis") |
| vm_rightsize_act_max_items | 40 | suggestions ACT reviews: too small first, then too big (the most freed first) |
| vm_rightsize_act_batch | "{{ site_act_batch \| default(5) }}" | suggestions per question to ACT (0 = all in one) |

| Setting | Default | What it does |
|---|---|---|
| vm_rightsize_act_model_timeout | "{{ site_act_model_timeout \| default(600) }}" | seconds the model may take for its answer |
| vm_rightsize_act_check_url | true | first check that the AAP node can reach the model's URL |
| vm_snapshot_max_age_days | 5 | Snapshots at least this many days old: listed apart by the report, deleted by the cleanup. |
| vm_snapshot_keep_regex | '(?i)keep\|do.?not.?delete' | Never deleted by the cleanup (the report lists them, with the reason): - a snapshot whose name or description matches this |
| vm_snapshot_cleanup_exclude_vms | [] | - a snapshot of these VMs: names, * and ? allowed (e.g. [DC*, VCSA01, AAP01]). The VMs in vmware_protected_vms and the AAP server are never cleaned up either. |
| vm_snapshot_cleanup_exclude_datacenters | [] | - a snapshot of a VM in these vCenter datacenters: names, * and ? allowed, any case (e.g. [DRSITE]) |
| vm_snapshot_cleanup_exclude_clusters | [] | - a snapshot of a VM running in these clusters: names, * and ? allowed, any case (e.g. [SQL*, MGMT*]). A VM on a host that is in no cluster is not matched. |
| vm_snapshot_cleanup_name_words | [] | - with words here, a snapshot whose name contains none of them: only a name with one is deleted (any case; Patch also matches Patching and Patches - and dispatch). E.g. [Patching, Patches, Patch]: only the patching snapshots. [] = any name. Set it in all.yml, so the report lists the same. |
| vm_snapshot_cleanup_max_size_gb | 1024 | - a snapshot bigger than this many GB (1024 = 1 TB), or whose size cannot be read: merging a very big snapshot can take hours and slow the VM - do those by hand, at a quiet time. |
| vm_snapshot_cleanup_confirm | false | The cleanup on its own (not after a workflow approval) deletes only when this is true - e.g. a survey question "Delete them?". false = it lists what it would delete, and deletes nothing. |

| Setting | Default | What it does |
|---|---|---|
| vm_snapshot_cleanup_max | 50 | Refuse to delete more than this many snapshots in one run (a typo in the age would otherwise delete them all). Raise it on purpose for a big cleanup. |
| vm_snapshot_delete_timeout | 1800 | seconds to wait for one deletion (vCenter merges it into the disk) |
| vm_snapshot_cleanup_email_only_if_deleted | false | The cleanup's email (report_email_to): true = only when it deleted something (or a deletion failed). |
| vm_pick_snapshot_templates | [] | vm_pick_refresh.yml (on a schedule) writes what may be done right now as the choices of one multiple-choice question in the survey of these templates: job templates or workflow job templates, by name (or {template: name or ID, kind: job or workflow} when a job template and a workflow share the name). In a workflow onl... (all of it: the role file) |
| vm_pick_snapshot_question | vm_snapshot_picks | the question's Answer variable name |
| vm_pick_reboot_templates | [] | e.g. ["VM - restart picked VMs (workflow)"] |
| vm_pick_reboot_question | vm_reboot_picks | the question's Answer variable name |
| vm_pick_max_choices | 200 | a list never has more choices than this (the rest is counted in the job's output) |
| vm_pick_snapshot_min_age_days | 0 | The snapshots offered: only those the snapshot cleanup could delete (the same holds - a name that says keep, the excluded VMs, datacenters and clusters, the protected VMs and the AAP server, bigger than vm_snapshot_cleanup_max_size_gb; not the name words), at least this many days old. 0 = any age. |
| vm_pick_reboot_vms | [] | The VMs offered for a restart: powered on, never the AAP server or a protected VM (vmware_protected_*), among these (names, * and ?, any case; [] = every VM) ... |
| vm_pick_reboot_clusters | [] | e.g. [PROD-*] |

| Setting | Default | What it does |
|---|---|---|
| vm_pick_reboot_folders | [] | folder paths as vCenter shows them, e.g. ["/DC1/vm/Linux*"] |
| vm_pick_reboot_exclude_vms | [] | ... and never these (e.g. [SQL*, DC*]) |
| vm_pick_reboot_only_hung | false | true = only the VMs that look hung: VMware Tools installed, but no heartbeat (red) or Tools not running, powered on for vm_pick_hung_min_minutes at least (a VM still starting has neither yet). Restart those with vm_power_mode: hard - a guest restart needs VMware Tools running. |
| vm_pick_hung_min_minutes | 15 | minutes powered on before a VM without a heartbeat counts as hung |
| vm_pick_aap_ca_path | (empty) | a CA file for AAP's own HTTPS certificate ("" = what the execution environment trusts) |
| vm_snapshot_picks | [] | The survey's answers (the picked choices: a list, or text one per line). the snapshots to delete |
| vm_reboot_picks | [] | the VMs to restart |
| vm_pick_confirm | false | vm_snapshot_delete_picked.yml / vm_reboot_picked.yml on their own (not a workflow): true = do it now. false = list exactly what would be done (the plan), and do nothing. Inside a workflow it is never used: the first step plans, the step after the approval does exactly the plan. |
| vm_pick_max | 20 | refuse to restart, or delete, more than this many in one run |
| vm_alarm_types | [vcenter, datacenter, cluster, host] | Whose triggered alarms and configuration issues: vcenter, datacenter, cluster, host. Add vm, datastore or network for theirs too (VM alarms can be many). |
| vm_alarm_hours | 24 | The events of the last this-many hours: every error and warning, failed logins, host connection events and VM shutdowns / power-offs / resets / HA restarts. 0 = alarms and host state only. |
| vm_alarm_max_events | 5000 | read at most this many events (the newest) |

| Setting | Default | What it does |
|---|---|---|
| vm_alarm_max_alarm_changes | 2000 | read at most this many alarm status changes (the newest; 0 = no "Alarms that fired") |
| vm_alarm_max_rows | 200 | rows per table in the report (the artifacts have them all) |
| vm_alarm_login_critical | 10 | This many failed logins of one user from one place, or more, is critical (fewer: a warning). |
| vm_alarm_login_event_types | [BadUsernameSessionEvent, com.vmware.sso.LoginFailure, esx.audit.account.locked, esx.audi… | The event types read whatever their category (vCenter's event type names). Add to these lists rather than replacing them; [] = none of that kind. |
| vm_alarm_connection_event_types | [HostConnectionLostEvent, HostNotRespondingEvent, HostDisconnectedEvent, HostReconnection… | Events about a host losing (or failing to get back) its connection to vCenter: the Host connection table. |
| vm_alarm_vm_event_types | [VmGuestShutdownEvent, VmGuestRebootEvent, VmPoweredOffEvent, VmResettingEvent, VmFailove… | Events about a VM stopping, restarting or failing over (guest shutdown, power off, HA restart): the VM events table. |
| vm_alarm_fail | false | The job stays green when it finds alarms: the report and email say what. true = mark the job FAILED when there is a critical alarm or a host is not connected, for workflows. |
| vm_alarm_act_max_items | 40 | ACT analysis (playbooks/vm_alarm_act_analysis.yml). ACT runs on the AAP side (the execution environment), never on the ESXi hosts; the model provider, model, URL, proxy and CA are the site_act_* settings (roles/site_act/defaults/main.yml), the key an "ACT model key" credential. analyze at most this many problems, the… (all of it: the role file) |
| vm_alarm_act_batch | "{{ site_act_batch \| default(5) }}" | problems per question to ACT: a long answer about many problems is what a gateway or proxy cuts off (0 = all in one question) |
| vm_alarm_act_config_issues | true | configuration issues too (false = alarms and events only) |
| vm_alarm_act_email_if_none | false | true = email even when there is nothing to analyze |

| Setting | Default | What it does |
|---|---|---|
| vm_alarm_act_check_url | true | first check that the AAP node can reach the model's URL |
| vm_alarm_act_model_timeout | "{{ site_act_model_timeout \| default(600) }}" | Seconds the model may take for its answer (ACT's GENAI_TIMEOUT, 90 by default, is too short for a long analysis): site_act_model_timeout (600) unless set here. A GENAI_TIMEOUT in site_act_env wins. |
| vm_health_require_ha | true | A cluster of 2 or more hosts without vSphere HA is a warning (false = HA off is on purpose). |
| vm_health_require_drs | true | A cluster of 2 or more hosts without DRS is a warning (false where DRS is off on purpose or not licensed, e.g. vSphere Standard). |
| vm_health_cpu_warn_pct | 85 | CPU and memory used, in percent of a host (or of a cluster's hosts running VMs): CPU used: a warning from this percent |
| vm_health_cpu_crit_pct | 95 | CPU used: critical from this percent |
| vm_health_mem_warn_pct | 90 | memory used: a warning from this percent |
| vm_health_mem_crit_pct | 95 | memory used: critical from this percent |
| vm_health_sensors | true | read the hosts' hardware sensors (false = skip them) |
| vm_health_clock | true | Each connected host's clock against vCenter's, and its NTP setting: a host writes its own logs (hostd, vmkernel) by its clock, vCenter its events by its own - off clocks are why a disconnection's time in a host's log does not match vCenter's. No NTP server set, or the NTP service stopped, is a warning. vCenter's clock... (all of it: the role file) |
| vm_health_clock_warn_seconds | 60 | a host's clock this many seconds off vCenter's (ahead or behind) -> warning |
| vm_health_clock_crit_seconds | 300 | -> critical (Active Directory logins fail from 5 minutes off) |
| vm_health_max_rows | 500 | rows per table in the report (the artifacts have them all) |

| Setting | Default | What it does |
|---|---|---|
| vm_health_fail | false | The job stays green when it finds problems: the report and email say what. true = mark the job FAILED when there is a critical problem (a host not connected, an HA agent isolated ...), for workflows. |
| vm_net_drops | true | Network drops and errors in the health report: packets dropped and network errors of every ESXi host (by physical NIC) and powered-on VM (by virtual NIC), from vCenter's statistics. The past day needs vCenter's 5-minute statistics at level 2 (the default is 1); at level 1 the last hour is read, and the report says so.... (all of it: the role file) |
| vm_net_drops_vms | true | the powered-on VMs too (false = the hosts only: quicker where there are thousands of VMs) |
| vm_net_drops_hours | 24 | hours to read when vCenter keeps the past day (at most 24: what the 5-minute interval keeps) |
| vm_net_drops_warn_per_hour | 100 | dropped packets an hour (received + sent) of one host or VM from this many -> warning |
| vm_net_errors_warn_per_hour | 10 | network errors an hour (received + sent) of one host or VM from this many -> warning |
| vm_watch_window_minutes | 15 | The HA and storage events of the last this many minutes (set it to the schedule's interval). |
| vm_watch_min_up_minutes | 15 | A VM powered on for less than this is not judged (a VM still starting has no heartbeat yet). |
| vm_watch_confirm_checks | 2 | A VM is hung only when it has no guest heartbeat AND does not answer on the network, in every check: a suspect is checked this many times in one run (1 = once), vm_watch_confirm_seconds apart. |
| vm_watch_confirm_seconds | 120 | seconds between those checks |

| Setting | Default | What it does |
|---|---|---|
| vm_watch_probe | true | The network probe, from the AAP node, of the suspects only: ping, and a TCP connect to this port too (e.g. 22 or 3389; 0 = ping only). A refused connection counts as an answer (the guest is alive). false = no probe: no VM is then called hung (one signal only) |
| vm_watch_probe_port | 0 | TCP port also tried (0 = ping only) |
| vm_watch_probe_timeout | 3 | seconds per address |
| vm_watch_max_suspects | 50 | probe and confirm at most this many suspects in one run |
| vm_watch_one_signal_severity | warning | A VM with no heartbeat that answers on the network (or cannot be probed): VMware Tools died, not the guest. warning = in the tickets (servicenow_min_severity warning); info = the job's output only. |
| vm_watch_skip_vms | [] | VMs never judged: names, * and ? allowed (e.g. [LAB-*]) |
| vm_watch_skip_folders | [] | VM folders never judged, as vCenter shows them (e.g. ["/DC1/vm/Lab*"]) |
| vm_watch_protect | [] | Phase 2 (not built yet) would restart the hung VMs - never these (names, * and ?), never the AAP server or the VMs of vmware_protected_vms / vmware_protected_groups: their hung state is a ticket only. e.g. [DC*, SQL*, VCSA*] |
| vm_watch_max_actions | 5 | more hung VMs at once than this = something bigger: none would be restarted past it |
| vm_watch_fail_on | [critical] | The job shows FAILED (a workflow's "On failure" path) when it finds a condition of these severities. |
| vm_watch_report_on | [critical] | The report is emailed (report_email_to) and published (report_web_dest) only when it finds a condition of these severities; a clean run is one line in the job output. |
| vm_watch_sso | true | the watch also asks: is single sign-on down now (vm_sso_*, below)? false = not asked |

| Setting | Default | What it does |
|---|---|---|
| vm_sso | true | ---- Single sign-on and OCSP (vCenter 8.0): the health report (playbooks/vm_health_report.yml) and the watch ---- vCenter's SSO services, the STS signing certificate (health report only), the identity providers, a login by a test domain account (the "vCenter SSO test account" credential), and the OCSP responders smart... (all of it: the role file) |
| vm_sso_services | [sts, lookupsvc, trustmanagement, vmdird, vmafdd] | SSO services that must be started (as vCenter's and the appliance's service lists name them; one not listed is skipped) |
| vm_sso_sts_warn_days | 60 | the STS signing certificate expires within this many days -> warning |
| vm_sso_sts_crit_days | 14 | ... within this many -> critical (expired: critical) |
| vm_sso_test_user | (empty) | the test domain account; empty = the "vCenter SSO test account" credential's (VMWARE_SSO_TEST_USER); none = no login test |
| vm_sso_ocsp | [] | The OCSP responders. Simplest: site_ocsp_certs (below) - a certificate the responder knows, its OCSP address and its issuer read from the certificate itself. vm_sso_ocsp, when set, is used instead: [{url: http://ocsp.example.mil}] asks only that it answers; with cert (a PEM file from playbooks/, e.g. files/ocsp/test-u... (all of it: the role file) |
| site_ocsp_certs | [] | certificates the OCSP responders know (PEM files from playbooks/), for vm_sso_ocsp and net_perf_ocsp when those are empty, e.g. [files/ocsp/test-user.pem] |
| vm_sso_ocsp_severity | critical | an OCSP responder that fails: critical (smart card logins fail) or warning |

| Setting | Default | What it does |
|---|---|---|
| vm_sso_crl | true | The CRL of each certificate above (its CRL Distribution Point, or crl_url in a vm_sso_ocsp entry; crl: false skips one), in the health report only - from the AAP node: downloaded, issued and signed by the CA, not past its next update, not listing the test certificate. vCenter falls back to the CRL when OCSP does not a... (all of it: the role file) |
| vm_sso_crl_severity | warning | a CRL that fails: warning (OCSP still answers smart card logins) or critical |
| vm_sso_crl_timeout | 120 | seconds for one CRL download (DoD CRLs are tens of MB) |
| vm_sso_crl_max_mb | 200 | a CRL larger than this is not read |
| vm_sso_timeout | 30 | seconds per request |
| vm_created_days | 7 | The window: the VMs created in the last this many days, up to the moment the job runs (7 = the past week; run weekly, each VM is in exactly one report). VMs created and deleted since are listed too, as long as vCenter keeps their events (often 30 days). |
| esxi_security_hosts | [] | Which hosts: names (* and ? allowed; the short name matches too). [] = every host of vCenter (or of vmware_datacenter). |
| esxi_security_clusters | [] | only the hosts of these clusters (names, wildcards); [] = all |
| esxi_security_exclude_hosts | [] | Never these hosts (names, wildcards). E.g. a host you opened SSH on for a support case: the nightly run would otherwise close it again. |
| esxi_security_ssh | disabled | The SSH service (TSM-SSH): disabled = stopped, and set to start manually; enabled = started, and set to start with the host; "" = leave it as it is. |
| esxi_security_shell | disabled | The ESXi Shell service (TSM), the host's local shell: the same choices. |

| Setting | Default | What it does |
|---|---|---|
| esxi_security_settings | UserVars.ESXiShellTimeOut: 600 UserVars.ESXiShellInteractiveTimeOut: 600 | Advanced settings, name: value; {} = none. The defaults (seconds): UserVars.ESXiShellTimeOut SSH and the ESXi Shell stop by themselves this long after someone starts them (0 = never) UserVars.ESXiShellInteractiveTimeOut an idle SSH or ESXi Shell session is logged out Others often set: UserVars.DcuiTimeOut (an idle DCU... (all of it: the role file) |
| esxi_security_lockdown | normal | Lockdown mode: normal = the host is managed only through vCenter (the DCUI still works); disabled; "" = leave it as it is. strict is refused, and a host already strict is left alone. |
| esxi_security_lockdown_exception_users | [] | Accounts that must keep logging in to the hosts directly under lockdown (a vulnerability scanner, a monitoring tool): host-local accounts, or Active Directory accounts with their own permission on the host (not vsphere.local, not an AD group). Added before lockdown is turned on; never removed. |
| esxi_security_email_only_if_changed | true | The email (report_email_to): true = only when a host was changed or failed; false = every run. |
| vmware_cert_warn_days | 30 | A certificate of vCenter (its Machine SSL certificate, the one its web pages show) or of an ESXi host that expires within this many days (or has expired) is due: a ServiceNow incident, and a CSR made where its key stays - vCenter, or the host - attached to the incident with what to do next. To try the job on certifica... (all of it: the role file) |
| vmware_cert_crit_days | 7 | within this many days (or expired) the incident is critical, else warning |
| vmware_cert_esxi | true | the ESXi hosts' certificates too; false = vCenter's only |
| vmware_cert_only | [] | Only these vCenters and hosts: names, any case, * and ? allowed, the short name matches too - e.g. one host to try the job on. [] = every vCenter of the job and every host connected to it. |

| Setting | Default | What it does |
|---|---|---|
| vmware_cert_make_csr | true | false = the incidents only, no CSR. A CSR is made once per certificate: the incident is its record, and a second CSR would replace the key that waits on vCenter or the host for the issued certificate. |
| vmware_cert_vmca_csr | false | A certificate signed by vCenter's own CA (VMCA - how vCenter gives ESXi hosts their certificates unless told otherwise) is renewed in vCenter (its Renew button): its incident says so, and no CSR is made. true = a CSR for those too, to replace them with certificates of your PKI. |
| vmware_cert_vmca_issuers | [] | Issuers taken as VMCA besides VMCA's own root (CN=CA with DC=..., or OU=VMware Engineering): any case, * and ? allowed, e.g. ['*O=vc01.yoursite.mil*']. |
| vmware_cert_csr_fields | {} | vCenter's CSR: its subject and names are the current certificate's. vCenter's API wants an organization, organization unit, locality, state and country - one the certificate does not have goes empty. Set any of them here (the PKI sets the subject by its own rules anyway), e.g. vmware_cert_csr_fields: {organization: U.... (all of it: the role file) |
| vmware_cert_csr_without_servicenow | false | Without the ServiceNow API credential no CSR is made (nothing would record it: the next run would make another one). true = make them anyway, and print them in the job output. |
| vmware_trust_esxi | true | ---- VMware trust store report (playbooks/vmware_trust_report.yml) ---- vCenter's trusted roots and each ESXi host's trusted CAs; false = vCenter only (the ESXi hosts left out) |
| vm_capacity_clusters | [] | ---- vSphere capacity planning (playbooks/vm_capacity_report.yml) ---- only these clusters (names; * and ? allowed); [] = every cluster |

| Setting | Default | What it does |
|---|---|---|
| vm_capacity_history_days | 120 | The history: vCenter's own daily statistics (cluster CPU and memory, datastore space), kept a year at statistics level 1. Read this many days; the growth is a straight line over the last vm_capacity_trend_days, with at least vm_capacity_min_samples daily values. |
| vm_capacity_trend_days | 90 | The growth (straight line) is fitted on this many most recent days. Growth comes in jumps (VMs are deployed in batches): over 30 days one build-out becomes the rate and the runway reads days when it is months; 90 days averages the batches out. 30 reacts faster to a real change of rate. |
| vm_capacity_min_samples | 7 | fewer daily values than this in those days = 'not enough history' (no trend) |
| vm_capacity_events_days | 30 | VMs added and removed (vCenter keeps events 30 days by default) |
| vm_capacity_failover_hosts | 1 | Plan for this many failed hosts per cluster (N+1); a larger HA admission control failover level wins. |
| vm_capacity_mem_target_pct | 80 | memory used at most this %, with those hosts failed |
| vm_capacity_cpu_target_pct | 80 | CPU used at most this %, with those hosts failed |
| vm_capacity_vcpu_per_core_max | 4 | vCPUs per physical core (amber from here, red from 1.5 times it) |
| vm_capacity_horizon_months | 12 | Hosts recommended per cluster (of its average host size) to keep memory and CPU at their targets with those hosts failed: now, and at the current growth within this many months. |
| vm_capacity_storage_warn_pct | 85 | datastore used %: amber |
| vm_capacity_storage_crit_pct | 90 | red, and the target of the storage runway |
| vm_capacity_runway_warn_days | 90 | a runway shorter than this is amber |
| vm_capacity_runway_crit_days | 30 | and red |
| vm_capacity_include_local_datastores | false | a host's own (not shared) datastores count too |

| Setting | Default | What it does |
|---|---|---|
| vm_capacity_max_rows | 100 | rows per table (problems first); the artifacts have all |
| vm_capacity_fail | false | The job stays green: the report says what is short. true = mark it FAILED when a cluster or datastore is critical (for workflows). |
| vm_capacity_act | false | GenAI: ACT reads the numbers and makes its own estimate, shown next to the job's (math vs GenAI). Needs an "ACT model key" credential on the template; the same site_act_* settings as the alarms analysis. ACT runs on the AAP side and runs no command. |
| vm_capacity_act_max_items | 30 | items ACT estimates: everything overall, then clusters and datastores, problems first |
| vm_capacity_act_batch | "{{ site_act_batch \| default(5) }}" | items per question to ACT (each question has the whole evidence; 0 = all in one) |
| vm_capacity_act_model_timeout | "{{ site_act_model_timeout \| default(600) }}" | seconds the model may take for its answer |
| vm_capacity_act_check_url | true | first check that the AAP node can reach the model's URL |
| vm_deploy_ticket | (empty) | ---- Deploy a VM from a ServiceNow ticket (playbooks/vm_deploy_from_ticket.yml) ---- The ticket (survey answer): INC..., TASK..., SCTASK..., RITM..., CHG... - or your instance's own prefixes (INCN..., CHGN...): the job asks ServiceNow which table a ticket is in. The ticket says what to build in 'key: value' lines - in... (all of it: the role file) |
| vm_deploy_ticket_table | (empty) | leave it empty (the job finds each ticket's table); a table of your own, e.g. u_vm_request: for numbers whose prefix the job does not know (CTASK stays change_task) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_ticket_prefixes | {} | Your instance's own number prefixes and their tables, e.g. {INCN: incident, CHGN: change_request}. Rarely needed: for a prefix it does not know, the job asks ServiceNow's task table (and takes INCN as INC, CHGN as CHG when it may not read that table). A table as people say it is understood: change / change request ->... (all of it: the role file) |
| vm_deploy_parent_tables | [change_request] | The request may also be on the ticket's parent - e.g. a task (SCTASK, CTASK) whose change holds the requirements: the parent's 'key: value' lines (and, with vm_deploy_plain_words, its plain words) are read too, under the ticket's own (what the ticket says wins). A parent that cannot be read, is closed, or is not ready... (all of it: the role file) |
| vm_deploy_parent_fields | [change_request, parent] | The ticket's fields that point to its parent - the first one set is used: a change task's change_request, any ticket's parent (or your instance's own field, e.g. u_change): |
| vm_deploy_parent_text_fields | [description] | The parent's fields with the request, in this order (a later one wins), e.g. [description, implementation_plan]; comments and work_notes may be listed too (oldest first). GenAI reads them as well. |
| vm_deploy_parent_wins | false | What wins, from the least to the most: the parent (the change), the ticket's short description and description, its work notes, its comments (each oldest first) - a 'key: value' line and plain words alike: the later one wins, however each is written. So the change task wins over its change, and a comment wins over a w... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_notes_also | [] | Every note and comment this job writes on the ticket goes on these too (closing never does: only the ticket itself is closed): parent = the change a task belongs to (when it was read); tasks = when the job is launched with a change, its open change tasks assigned to vm_deploy_assignment_group (and matching vm_deploy_n... (all of it: the role file) |
| vm_deploy_notes_tasks_query | (empty) | An extra ServiceNow query for the change's tasks that get the notes, e.g. "short_descriptionLIKEto deploy" ("" = the group's) |
| vm_deploy_notes_tasks_limit | 50 | the change's tasks that get the notes (and are closed), at most - more is a warning |
| vm_deploy_close_tasks | false | Launched with a change: also close its tasks found that way (with vm_deploy_done_fields_by_table's change_task entry, when every VM is built) - the change itself is not closed by this. They get the notes too. |
| vm_deploy_ready_by_table | change_request: {approval: Approved, state: Implement} | Before anything is built, a ticket of these tables - the ticket itself, or the parent it belongs to (a change task's change) - must have these fields: one value or a list (any of them), written as the ticket's form shows them (upper / lower case ignored). Checked by the dry run, the propose step AND right before the b... (all of it: the role file) |
| vm_deploy_templates | [] | What a ticket may ask for (names; * and ? allowed). Templates MUST be listed: nothing else is deployed. The others: [] = any. e.g. [rhel9-gold, win2022-gold] |
| vm_deploy_clusters | [] | clusters (and standalone hosts) a ticket may use; also the candidates of automatic placement |
| vm_deploy_datastores | [] | datastores (or datastore clusters) a ticket may use; also the candidates of automatic placement |
| vm_deploy_folders | [] | VM folders a ticket may use (paths below the datacenter's VM folder, e.g. Linux/*) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_max_cpu | 16 | the most vCPUs a ticket may ask for |
| vm_deploy_max_memory_gb | 128 | the most memory a ticket may ask for, in GB |
| vm_deploy_max_vms | 10 | A ticket may ask for several VMs ("2 VMs named web01 and web02", "vm_name: web01, web02", or "Hostnames:" and one name per line): one VM each, the same size - or each its own, from a line of its own - built one after the other, the ticket closed when all are built. More names than this: refused, nothing built (the lim... (all of it: the role file) |
| vm_deploy_name_regex | '^[A-Za-z0-9][A-Za-z0-9-]{0,62}$' | The VM names allowed (a regular expression); e.g. '^[a-z0-9-]{1,15}$' for Windows-safe names. |
| vm_deploy_assignment_group | (empty) | Only tickets assigned to this group (its name as ServiceNow shows it), or to one of a list of groups (["Group A", "Group B"]); "" = any group (the pickup needs one). |
| vm_deploy_assignment_group_by_table | {} | The same for the ticket the job is launched with, by its table - in place of vm_deploy_assignment_group for that table; "" = any group. e.g. {change_request: ""}: launched with a change, the change may be assigned to any group (it must still be approved: vm_deploy_ready_by_table); its tasks that get the notes (vm_depl... (all of it: the role file) |
| vm_deploy_defaults | {} | What a ticket does not say: e.g. {cluster: PROD-CL01, datastore: DS-CLUSTER-01, folder: Linux/New}. Not set = the template's own cluster, datastore and folder; its CPU and memory. |
| vm_deploy_sites | {} | Sites: a ticket's 'site: A' line (or its Location field, or with plain words a site's name or its words in vm_deploy_plain_sites - each VM by its own line) picks the placement - over vm_deploy_defaults and the name rules; a cluster / datastore / folder the ticket names itself still wins. A site in another vCenter: its... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_site_required | true | with sites set up: a ticket must say its site (false = vm_deploy_defaults); a name rule counts too |
| vm_deploy_name_rules | [] | Name rules: the VM's name picks where it goes - the first rule whose 'match' fits the name (* and ?, upper / lower case alike) fills in what the ticket does not say: a cluster (and datastores, folder ...) or a site. The ticket's own lines, and a site the ticket names, still win. A name used at several sites: 'sites' -... (all of it: the role file) |
| vm_deploy_pxe_templates | [] | Templates with no operating system yet, that install it themselves: they boot from the network (PXE, e.g. an MECM task sequence) or from their CD / ISO (e.g. a kickstart). No guest customization, no waiting for an IP address; the MAC address and the computer name go on the ticket. e.g. [win2022-pxe] |
| vm_deploy_guestinfo | true | Give a VM from one of those empty templates the ticket's name and number as VM settings its installer can read: guestinfo.hostname and guestinfo.ticket (a kickstart with VMware Tools in the installer: vmtoolsd --cmd "info-get guestinfo.hostname"). Needs the vCenter privilege Change configuration > Advanced configurati... (all of it: the role file) |
| vm_deploy_template_aliases | {} | Short names for long template names - for the ticket's 'template:' line and GenAI's catalog (whose keys may be these short names): {short name: the template's name in vCenter}, e.g. {win2022: "Windows 2022 Standard Template - Site 1", rhel9: "RHEL 9 Server Template"} A ticket may also write a template's own name in ot... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_template_specs | {} | The customization specification (vCenter > Policies and Profiles) for each template, when the ticket, its site and the name rules name none: {template name or pattern: spec}. A Windows template that is not network-boot needs one (Sysprep: administrator password, domain or workgroup), e.g. {"Windows 2022*": Win2022-Sys... (all of it: the role file) |
| vm_deploy_max_disks | 5 | disks a ticket may ask for, disk 1 included (disks_gb: 100, 200 = disk 1 grown to 100 GB, one 200 GB disk added) |
| vm_deploy_max_disk_gb | 2048 | the biggest disk a ticket may ask for, in GB |
| vm_deploy_max_total_disk_gb | 4096 | all its disks together, in GB |
| vm_deploy_mem_target_pct | 80 | Automatic placement - when the ticket, its site and vm_deploy_defaults name no cluster / datastore: the job chooses among a site's 'clusters' / 'datastores' lists (e.g. SiteA: {clusters: [A-PROD-*]}), else among vm_deploy_clusters / vm_deploy_datastores; with neither, the template's own. a cluster fits when its memory... (all of it: the role file) |
| vm_deploy_failover_hosts | 1 | hosts to plan as failed when choosing a cluster |
| vm_deploy_datastore_max_used_pct | 85 | a datastore fits when it stays under this % with the VM at its full size |
| vm_deploy_datastore_rules | [] | Datastores by the VM's name: the first rule whose match fits the name (* and ?, any case) gives the datastores it goes on - automatic placement chooses among them on its cluster, the least used first, over a site's or vm_deploy_defaults' datastore (one the ticket names must be one of them). A VM no rule matches never... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_datastore_balance | false | true = a VM whose ticket, site, name rule and vm_deploy_defaults name no datastore (and no vm_deploy_datastores) goes on the least used shared datastore of its cluster - the location's - the VMs of one ticket spread out (each counts the ones before it); false = the template's own datastore. |
| vm_deploy_resume | true | Run again for the same ticket (a job that stopped: failed, cancelled, timed out), pick up where it stopped: each VM carries its ticket in its notes from the clone on - one it built is left as it is, one it started and did not finish (cloned, not yet powered on) is finished where it is; the work notes say which. false... (all of it: the role file) |
| vm_deploy_domain | localdomain | the DNS domain Linux customization sets (a ticket's domain: wins) |
| vm_deploy_customize | true | guest customization: host name = vm_name, every adapter on DHCP |
| vm_deploy_power_on | true | power the new VM on (false = leave it off, e.g. to check it first) |
| vm_deploy_wait_customization | 900 | seconds to wait for vCenter's customization result |
| vm_deploy_wait_ip | 600 | seconds to wait for an IPv4 address (DHCP, through VMware Tools) |
| vm_deploy_done_fields | {} | Fields set on the ticket when the VM is up, e.g. to resolve an incident: {state: "6", close_code: "Solution provided", close_notes: "VM deployed"} (your instance's values) |
| vm_deploy_done_fields_by_table | {} | The same for one table (the ticket's table: incident, sc_task, change_task ...), in place of vm_deploy_done_fields for it - e.g. close the task, never the change it belongs to: {sc_task: {state: "3", close_notes: "VM deployed by AAP"}} (3 = Closed Complete; your instance's values) Sent on its own, after the result not... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_done_fields_issues_by_table | {} | Over those fields when every VM is built but something did not finish: no IPv4 address in time, no customization result, the guest's host name not applied, no network at power on, guestinfo not set - e.g. {change_task: {close_code: successful_issues}} (your instance's values; {} = closed the same way). What happened i... (all of it: the role file) |
| vm_deploy_done_fields_pxe | true | Also for a VM from a network-boot template (vm_deploy_pxe_templates), which is powered on and still installing its OS from MECM when the job ends. false = that ticket stays open for whoever checks the install. |
| vm_deploy_service | (empty) | ---- The server's service, picked at launch: words in the VM's notes, an AAP inventory and group ---- What the new server is for, picked by whoever launches the deploy: a survey question (multiple choice, NOT required - the scheduled pickup starts the workflow without it), e.g. sql or dc. Every VM of the ticket gets i... (all of it: the role file) |
| vm_deploy_services | {} | The services the survey offers: each one's AAP inventory and group, and the words for the VM's notes (note: left out = "Service: <its name>."). {} = any service is only written in the notes. vm_deploy_services: sql: {inventory: Windows servers, group: win_sql, note: SQL Server} dc: {inventory: Windows servers, group:... (all of it: the role file) |
| vm_deploy_inventory_name | "{name}" | The name a VM gets in the inventory: write it the way your inventory already names servers, or the same server shows up twice in AAP. {name} = the VM's name, e.g. "{name}.yoursite.mil". A service's own name: wins (e.g. {inventory: Windows servers, name: "{name}.corp.yoursite.mil"}). |
| vm_deploy_inventory_lowercase | true | the inventory name in lower case (APP01 -> app01); false = as the ticket writes it |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_aap_ca_path | (empty) | a CA file for AAP's own HTTPS certificate ("" = vm_deploy_poll_aap_ca_path, else what the execution environment trusts) |
| vm_deploy_plain_words | false | ---- A request in plain words, without GenAI (off by default) ---- true = also read the ticket's own words, by the rules below, so a requester can write "a Red Hat 9 server named ab01web05, medium" instead of 'key: value' lines. Never a guess: two different answers in the text are a question to the requester (a commen... (all of it: the role file) |
| vm_deploy_plain_templates | {} | The words for each template (any case; "windows 2022" also matches "Windows Server 2022", "server 2025" also "a new 2025 VM"). An allowed template's own name and its short names (vm_deploy_template_aliases) count too. An OS no template is for (Server 2025, only a 2022 template) is a question - never another version; t... (all of it: the role file) |
| vm_deploy_plain_sizes | {} | Size words and what they mean, e.g. {small: {cpu: 2, memory_gb: 8}, medium: {cpu: 4, memory_gb: 16}, large: {cpu: 8, memory_gb: 32}}. "4 vCPU" and "16 GB RAM" in the ticket are read too, and win in the same text. |
| vm_deploy_plain_sites | {} | The words for each site (vm_deploy_sites), as people write the place - any case, the spaces optional ("Site Two" is also "SiteTwo"). A site's own name counts too. A change that lists its servers one per line ('new VM / ab01web05 / ... Location: Site Two') puts each VM at the site its own line names; the CPU, memory an... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_genai | false | ---- GenAI reads a request written in plain words (a proof of concept, off by default) ---- When the ticket's own 'key: value' lines do not give vm_name and template: ACT (GenAI, evidence mode, no commands) copies out what the ticket says - the names, the operating system as written, CPU, memory, disks - each with the... (all of it: the role file) |
| vm_deploy_genai_in_check | false | A dry run (Check) asks GenAI too (true): it shows GenAI's proposal and where the VM would go, and writes nothing on the ticket - no proposal note, no questions to the requester. false = Check only says GenAI would be asked. |
| vm_deploy_template_catalog | {} | Optional: templates GenAI may name by key, each with a plain description (the key: the template's exact name). Not needed when the templates' names say their OS: the job finds the template from the OS the ticket names. A key for another version than the ticket asks for is dropped. e.g. {win2022-pxe: "Windows Server 20... (all of it: the role file) |
| vm_deploy_genai_approval | true | true: a GenAI proposal waits for a person - the workflow's approval (or a second run) - before it is built. false: one run reads, checks and builds, when every value is traced to the ticket and within the rules (questions or a refusal still stop it). |
| vm_deploy_phase | (empty) | Which step this run is: "" = one job does it all; propose = the workflow's first step (read, ask GenAI if needed, check, write where it would go on the ticket - build nothing); build = the step after the approval (build what the ticket and its proposal say - never ask GenAI). |
| vm_deploy_genai_model_timeout | "{{ site_act_model_timeout \| default(600) }}" | seconds the model may take for its answer |

| Setting | Default | What it does |
|---|---|---|
| vm_deploy_poll_enabled | true | ---- Pick up VM request tickets on a schedule (playbooks/vm_deploy_poll.yml) ---- Every run (an AAP schedule, e.g. every 15 minutes) asks ServiceNow for NEW VM requests - active, assigned to vm_deploy_assignment_group (required), marked by vm_deploy_poll_prefix and/or vm_deploy_poll_query - and starts the deploy workf... (all of it: the role file) |
| vm_deploy_poll_tables | [incident] | the ServiceNow tables to look in: incident, sc_task, task ... |
| vm_deploy_poll_prefix | "VM request" | a VM request's short description starts with this ("" = mark them by vm_deploy_poll_query) |
| vm_deploy_poll_query | (empty) | An extra ServiceNow encoded query, ANDed with the rest, e.g. "category=vm_request" or "cmdb_ci.name=VM Request" ("" = none). ^NQ (a second query) is refused. |
| vm_deploy_poll_states | ["1"] | the ticket states to pick up (state values: "1" = New) |
| vm_deploy_poll_max | 3 | the most tickets started per run; the rest wait for the next run |
| vm_deploy_poll_scan | 100 | the most tickets read per table and run, oldest first (tickets AAP already has count too) |
| vm_deploy_poll_workflow | "VM - deploy from ServiceNow ticket (workflow)" | the workflow job template started per ticket |
| vm_deploy_poll_dry_run | false | true = only list what it would pick up: no note, no workflow (Check does the same) |
| vm_deploy_poll_claim_fields | {} | Fields set on the ticket together with the "Picked up" note, e.g. {state: "2"} (In Progress) - so the ticket leaves the states picked up as well ({} = only the note). |
| vm_deploy_poll_aap_ca_path | (empty) | a CA file for AAP's own HTTPS certificate ("" = what the execution environment trusts) |

## win_act

File: `roles/win_act/defaults/main.yml`

| Setting | Default | What it does |
|---|---|---|
| `win_act_dir` | `'C:\ProgramData\act'` | ACT FOR WINDOWS on the servers (vendor/act-windows/act.ps1, the PowerShell port of ACT). Installed into win_act_dir. That folder gets its own owner and permissions - Administrators and SYSTEM only, nothing inherited - because ordinary users may create folders and files in C:\ProgramData (and C:\temp): a non-admin coul... (all of it: the role file) |
| `win_act_state` | `present` | present \| absent (removes win_act_dir) |
| `win_act_source` | `"{{ role_path }}/../../vendor/act-windows/act.ps1"` | The ACT script this repository ships (vendor/act-windows, version in its VERSION file) |

## win_check_accounts

File: `roles/win_check_accounts/defaults/main.yml`

> **Note**
> WINDOWS ACCOUNTS (local accounts; domain controllers have none and are skipped) - the built-in Guest account is disabled -> critical if not - enabled local accounts: password never expires, password too old, no logon for a long time -> warning - members of the local Administrators group that you did not allow -> warning Names below allow wildcards (* and ?), e.g. 'YOURDOMAIN\Domain Admins', '*\svc_aap'.

| Setting | Default | What it does |
|---|---|---|
| `win_check_accounts_password_max_age_days` | `60` | an enabled local account's password older than this -> warning (0 = off) |
| `win_check_accounts_inactive_days` | `35` | an enabled local account with no logon for this many days -> warning (0 = off) |
| `win_check_accounts_password_never_expires_allowed` | `[]` | accounts allowed "password never expires" (wildcards) |
| `win_check_accounts_inactive_allowed` | `[]` | accounts allowed to stay unused (wildcards) |
| `win_check_accounts_admins_allowed` | `[]` | Empty = list the members in the job output only. Set it to report anyone else, e.g. win_check_accounts_admins_allowed: ['*\Administrator', 'YOURDOMAIN\Domain Admins', 'YOURDOMAIN\svc_aap'] |

## win_check_audit

File: `roles/win_check_audit/defaults/main.yml`

> **Note**
> WINDOWS AUDIT: the audit policy records what the STIG asks for, the event logs are big enough, and the log forwarding agent runs. One finding lists every missing audit setting.

| Setting | Default | What it does |
|---|---|---|
| `win_check_audit_subcategories` | `- {name: Credential Validation, guid: 0CCE923F-69AE-11D9-BED3-505 054503030, need: Success…` | Advanced audit policy subcategories (checked by GUID, so any Windows language works). need: Success \| Failure \| Success and Failure |
| `win_check_audit_security_log_min_k b` | `196608` | STIG |
| `win_check_audit_application_log_mi n_kb` | `32768` | the Application log's maximum size must be at least this (KB) |
| `win_check_audit_system_log_min_kb` | `32768` | the System log's maximum size must be at least this (KB) |
| `win_check_audit_forwarder_services` | `[]` | Log forwarding agents that must run, e.g. [SplunkForwarder] or [nxlog] |

## win_check_certs

File: `roles/win_check_certs/defaults/main.yml`

> **Note**
> WINDOWS CERTIFICATES: every certificate in the machine stores below, and which of them are in use (https bindings of IIS / WSUS / ..., the WinRM HTTPS listener AAP connects through, Remote Desktop). Expiring soon -> warning / critical. Expired: critical when in use, or when it expired less than win_check_certs_expired_report_days ago (older leftovers are only listed in the certificate report).

| Setting | Default | What it does |
|---|---|---|
| `win_check_certs_warn_days` | `30` | a certificate expiring within this many days -> warning (amber in the certificate report) |
| `win_check_certs_notice_day s` | `60` | the certificate report colours these blue (an early heads-up; not a finding) |
| `win_check_certs_crit_days` | `7` | within this many days -> critical |

| Setting | Default | What it does |
|---|---|---|
| win_check_certs_expired_report_days | 30 | an expired certificate NOT in use is a finding for this many days after it expired; older ones only in the report |
| win_check_certs_stores | ['Cert:\LocalMachine\My', 'Cert:\LocalMachine\WebHosting', 'Cert:\LocalMachine\Remote Des… | the machine certificate stores read |
| win_check_certs_trust | false | also every other machine store: Trusted Root, Intermediate CAs, Other People, … (the certificate report turns it on); with its private key checked as above, without it only listed (trust store) |
| win_check_certs_trust_skip_stores | ['Disallowed', 'AuthRoot', 'REQUEST', 'SMS'] | stores never read: the untrusted certificates, the roots Windows Update keeps, pending requests (a CSR's key), the MECM client's own |
| win_check_certs_remote_desktop | false | false = Remote Desktop's certificates are left out of the report and the renewal (Windows or the GPO renews them): the Remote Desktop store, the one RDP uses, those for Remote Desktop only; true = checked like the others |
| win_check_certs_ignore_subjects | [] | wildcards, e.g. ['CN=localhost*'] |
| win_check_certs_endpoints | [] | TLS services whose certificate a client actually gets, e.g. win_check_certs_endpoints: - {name: IIS, host: 127.0.0.1, port: 443, servername: web01.yoursite.mil} |
| win_cert_renew_dir | 'C:\ProgramData\site-automation\renew' | ---- Renewal: a new key and CSR for each certificate that expires soon (playbooks/win_cert_renew_csr.yml) ---- The certificates the Windows certificate report flags (within win_check_certs_warn_days, or expired) that are the server's own - not a trust store's - get a new key and a CSR with the same subject, names and… (all of it: the role file) |
| win_cert_renew_key_bits | 2048 | RSA key size, at least 2048 (a bigger old key keeps its size; an EC certificate gets a new key on the same curve) |
| win_cert_renew_digest | sha256 | the CSR's signature: sha256, sha384 or sha512 |

| Setting | Default | What it does |
|---|---|---|
| win_cert_renew_max | 10 | at most this many NEW CSRs per server in one run (the rest come the next run) |
| win_cert_renew_self_signed | false | true = a self-signed certificate gets a CSR too, for the PKI to issue a real one in its place (never one only for localhost) |
| win_cert_renew_issuer | (empty) | regular expression: only certificates whose issuer matches (empty = any CA), e.g. 'O=U\.S\. Government' |
| win_cert_renew_exclude_issuer | (empty) | regular expression: certificates whose issuer matches are never renewed here (renewed by their own tool) |
| win_cert_renew_only | (empty) | regular expression on the certificate's place, e.g. 'used by https' (empty = every one) |
| win_cert_renew_exportable | false | true = the new key can be exported (to move it to another server); false keeps it on this one |
| win_cert_install_rebind | [https, rdp, winrm] | ---- Install: the issued certificate (playbooks/win_cert_install_check.yml -> approval -> playbooks/win_cert_install_approved.yml) ---- The PKI's certificate, from the request carrying the CSR (cert_renew_servicenow: catalog), is put together with its pending key by certreq -accept -machine (in Personal); its chain's... (all of it: the role file) |
| win_cert_install_verify_wait | 15 | seconds an https port has, after the rebinding, to serve the new certificate (full PowerShell only) |
| win_cert_restore_backup | (empty) | ---- Restore: the bindings back on the old certificate (playbooks/win_cert_restore.yml) ---- a backup's folder name, as the restore's Check lists it |
| win_cert_restore_only | (empty) | or: the newest backup of each certificate whose thumbprint or subject matches (a regular expression) |

## win_check_disk

File: `roles/win_check_disk/defaults/main.yml`

> **Note**
> WINDOWS DISK: how full each local fixed volume (C:, D:, ...) is.

| Setting | Default | What it does |
|---|---|---|
| win_check_disk_warn_pct | 85 | % used -> warning |
| win_check_disk_crit_pct | 95 | % used -> critical |
| win_check_disk_min_free_gb | 0 | also warn below this many GiB free (0 = off): big volumes |
| win_check_disk_overrides | {} | Different limits for particular volumes, e.g. a database volume kept full on purpose: win_check_disk_overrides: 'D:': {warn: 92, crit: 98} |
| win_check_disk_ignore | [] | volumes not to check, e.g. ['E:'] |

## win_check_eventlog

File: roles/win_check_eventlog/defaults/main.yml

**Note**
WINDOWS EVENT LOG, the last win_check_eventlog_hours: - unexpected shutdown or power loss (41, 6008), a crash / blue screen (1001) -> critical - disk and file system errors (disk, Ntfs, storage drivers) -> critical - the Security log was cleared (1102) -> critical - many errors in System + Application (the finding names the top sources) -> warning - a System / Application / Security log that is full and not overwriting -> critical

| Setting | Default | What it does |
|---|---|---|
| win_check_eventlog_hours | 24 | how many hours back the event logs are read |
| win_check_eventlog_error_warn | 50 | this many errors or more in System + Application -> warning |

## win_check_network

File: roles/win_check_network/defaults/main.yml

**Note**
WINDOWS NETWORK, from the server itself: - a default gateway and DNS servers are configured; - the names you list resolve, the TCP services you list answer; - domain members: the computer's trust with the domain works (nltest /sc_query). A broken trust is the classic "the trust relationship between this workstation and the primary domain failed" at logon.

| Setting | Default | What it does |
|---|---|---|
| win_check_network_dns_names | [] | Names every Windows server must resolve, e.g. [dc01.yoursite.mil, ldap.yoursite.mil] |

| Setting | Default | What it does |
|---|---|---|
| win_check_network_tcp | [] | Services every Windows server must reach, e.g. win_check_network_tcp: - {name: LDAPS, host: dc01.yoursite.mil, port: 636} - {name: SIEM, host: siem.yoursite.mil, port: 9997} |
| win_check_network_secure_channel | true | domain members: test the computer's trust with the domain (nltest /sc_query) |
| win_check_network_timeout_ms | 3000 | milliseconds for each TCP test |

## win_check_patching

File: roles/win_check_patching/defaults/main.yml

**Note**
WINDOWS PATCHING (read-only; installing updates is a later runbook) - a restart is pending for installed updates (Windows Update, component servicing, ConfigMgr) - days since the last update was installed (Get-HotFix) - optional: ask Windows Update which updates are missing (slower; needs full PowerShell)

| Setting | Default | What it does |
|---|---|---|
| win_check_patching_warn_days | 35 | last update installed this many days ago or more -> warning |
| win_check_patching_crit_days | 60 | this many days or more -> critical |
| win_check_patching_reboot_pending | true | report a restart pending for installed updates |
| win_check_patching_search_pending | false | also ask Windows Update which updates are missing (slower) |

## win_check_performance

File: roles/win_check_performance/defaults/main.yml

**Note**
WINDOWS PERFORMANCE: CPU (average of three samples a second apart), memory, page file. The finding names the busiest processes, so you know where to look.

| Setting | Default | What it does |
|---|---|---|
| win_check_performance_cpu_warn_pct | 90 | CPU busy % (average of three samples) -> warning |
| win_check_performance_cpu_crit_pct | 98 | CPU busy % -> critical |
| win_check_performance_mem_warn_pct | 90 | memory in use % -> warning |
| win_check_performance_mem_crit_pct | 97 | memory in use % -> critical |

| Setting | Default | What it does |
|---|---|---|
| win_check_performance_pagefile_warn_pct | 80 | page file in use % -> warning |

## win_check_security

File: roles/win_check_security/defaults/main.yml

**Note**
WINDOWS SECURITY BASELINE (a few high-value STIG items; not a full STIG scan: that is your SCAP / vulnerability scanner)

| Setting | Default | What it does |
|---|---|---|
| win_check_security_av | auto | Antivirus: auto = check Microsoft Defender when it is the active antivirus (skipped when it is passive or off because another product protects the server); defender = always; none = never. |
| win_check_security_av_signature_max_days | 7 | Defender's signatures older than this many days -> finding |
| win_check_security_av_services | [] | Services of your endpoint protection that must run, e.g. Trellix / McAfee: [masvc, mfemms] |
| win_check_security_firewall_profiles | [Domain, Private, Public] | must be on |
| win_check_security_smb1 | true | SMBv1 must be off |
| win_check_security_rdp_nla | true | Remote Desktop must require Network Level Authentication |
| win_check_security_uac | true | User Account Control must be on |
| win_check_security_legacy_tls | true | TLS 1.0 / 1.1 must be off for incoming connections |

## win_check_services

File: roles/win_check_services/defaults/main.yml

**Note**
WINDOWS SERVICES - every required service must be running (critical); - an automatic service that stopped WITH AN ERROR (exit code not 0) is a warning. Automatic services that stop cleanly by design (trigger-start, delayed) are not reported; - a service that crashed several times recently (System log events 7031 / 7034).

| Setting | Default | What it does |
|---|---|---|
| win_check_services_required | [EventLog, RpcSs, Dnscache, W32Time, WinRM, mpssvc] | Must be running on every Windows server. Add your own per group with ..._required_extra, e.g. for IIS servers: win_check_services_required_extra: [W3SVC, WAS] |
| win_check_services_required_extra | [] | more services that must run, e.g. for a group of IIS servers (see above) |
| win_check_services_ignore | - gupdate - gupdatem - edgeupdate - edgeupdatem - MicrosoftEdgeElevationService - GoogleU… | Automatic services never to report (wildcards allowed: * and ?) |
| win_check_services_crash_hours | 24 | look back this far for unexpected stops |
| win_check_services_crash_warn | 3 | this many unexpected stops of one service -> warning |

## win_check_time

File: roles/win_check_time/defaults/main.yml

**Note**
WINDOWS TIME: the Windows Time service runs, syncs with a real source (not the local clock), synced recently, and how far off the clock is from its source. Kerberos (domain logons) fails when a clock is 5 minutes off.

| Setting | Default | What it does |
|---|---|---|
| win_check_time_offset_warn_sec | 1 | seconds off its time source -> warning |
| win_check_time_offset_crit_sec | 5 | seconds off -> critical |
| win_check_time_max_hours_since_sync | 24 | last successful sync longer ago than this many hours -> finding |

## win_check_trust

File: roles/win_check_trust/defaults/main.yml

**Note**
WINDOWS TRUST STORES: which still hold a certificate authority that was compromised or that the browsers dropped - the machine certificate stores, the running account's, every Java cacerts and Python certifi bundle found - matched against the trust check's list (roles/check_trust/vars/main.yml, plus check_trust_extra; a severity changed in check_trust_severity). A certificate in the Disallowed store is blocked: listed, no finding. Removal (check_trust_remove: true, on the Windows trust store report): the certificate goes into the Disallowed store and out of the store it was in, Java entries are deleted, certifi bundles cut - each backed up to win_check_trust_backup_dir first. Check mode shows what would go. docs/TRUST_STORES.md

| Setting | Default | What it does |
|---|---|---|
| win_check_trust_stores | ['Cert:\LocalMachine\Root', 'Cert:\LocalMachine\CA', 'Cert:\LocalMachine\AuthRoot', 'Cert… | the certificate stores read (CurrentUser: the account AAP connects with) |
| win_check_trust_block_stores | ['Cert:\LocalMachine\Disallowed', 'Cert:\CurrentUser\Disallowed'] | the Disallowed stores: a certificate there is distrusted (blocked) |
| win_check_trust_java | true | every Java cacerts under the folders below (JKS natively; PKCS12 with the JDK's keytool) |
| win_check_trust_java_dirs | ['C:\Program Files\Java', 'C:\Program Files (x86)\Java', 'C:\Program Files\Eclipse Adopti… | folders searched for cacerts (seven levels deep) |
| win_check_trust_python | true | every certifi cacert.pem under the folders below |
| win_check_trust_python_dirs | ['C:\Program Files\Python*', 'C:\Program Files (x86)\Python*', 'C:\Python*', 'C:\ProgramD… | folders searched for certifi (seven levels deep) |
| win_check_trust_backup_dir | 'C:\ProgramData\site-automation\trust-backup' | the removal's backups: every certificate and file changed, a folder per run |
| win_check_trust_timeout | 60 | seconds for keytool |

## win_patch

File: roles/win_patch/defaults/main.yml

> **Note**
> WINDOWS PATCH through the ConfigMgr client: install what Software Center has for the server (the software updates deployed to it), restart if the updates need it, and prove the server came back: every automatic service that was running before runs again. One server at a time by default; the rollout stops at the first server that fails. Check mode lists and changes nothing.

| Setting | Default | What it does |
|---|---|---|
| automatic_restarts | false | NO AUTOMATIC RESTARTS unless you say so: this playbook restarts a server only when automatic_restarts is true (the job template's Variables, or the survey) and the updates need it. With automatic_restarts false (the default) it installs the updates and tells you which servers still need a restart; a person restarts th... (all of it: the role file) |

| Setting | Default | What it does |
|---|---|---|
| win_patch_serial | 1 | How many servers are patched at the same time (1 = one by one: the rollout stops at the first server that fails). ONLY in the job template's Variables, a survey or the launch: Ansible picks the batch size before it reads all.yml or any group or host variables, so set there it does nothing. |
| win_patch_never_reboot_groups | "{{ patch_never_reboot_groups \| default(['aap', 'aap_hosts', 'sat', 'netapp', 'netapp_con… | Servers in these groups are patched but NEVER restarted by this playbook (restart them by hand). Follows the Linux setting (patch_never_reboot_groups) when you set it, so ONE list covers both; the fallback is the Linux role's default (roles/patch/defaults/main.yml; tests/test_protection.py checks it). |
| win_patch_never_patch_groups | "{{ patch_never_patch_groups \| default(['no_patch', 'aap', 'aap_hosts', 'sat', 'netapp',… | Servers in these groups are skipped entirely (e.g. servers patched only by hand). |
| win_patch_scan_first | true | ask the client to rescan and re-evaluate before installing |
| win_patch_scan_wait_sec | 120 | let that scan settle |
| win_patch_timeout_min | 120 | per round: download + install |
| win_patch_poll_sec | 30 | how often to look at the progress |
| win_patch_max_rounds | 2 | After a restart, look again: servicing-stack and cumulative updates often reveal more updates. |
| win_patch_restart_delay_sec | 30 | shutdown.exe /t |
| win_patch_restart_timeout_min | 30 | how long the server may take to come back |
| win_patch_restart_settle_sec | 30 | after the port answers again, before trying to log on |
| win_patch_reconnect_pause_sec | 20 | between logon attempts after the restart |
| win_patch_services_grace_min | 5 | After the restart, give delayed-start services this long before calling one missing. |
| win_patch_services_ignore | ['gupdate*', 'edgeupdate*', 'MicrosoftEdgeElevationService', 'GoogleUpdater*', 'MapsBroke… | Automatic services that come and go on their own (wildcards); not compared before/after. |

| Setting | Default | What it does |
|---|---|---|
| win_patch_winrm_host | "{{ ansible_host \| default(inventory_hostname) }}" | Where AAP reaches WinRM, to see the server go down and come back during the restart |
| win_patch_winrm_port | "{{ ansible_port \| default(5986) }}" | its WinRM port (5986 = HTTPS) |

## win_troubleshoot

File: roles/win_troubleshoot/defaults/main.yml

| Setting | Default | What it does |
|---|---|---|
| ts_area | overview | WINDOWS TROUBLESHOOT: the PowerShell commands an experienced Windows admin runs first for a kind of problem, run for you (read-only), each with a line saying what it shows. The matching health checks run too, so you also get plain-words findings. ts_area: overview \| disk \| performance \| service \| network \| login \| tim... (all of it: the role file) |
| ts_service | (empty) | the service name, for ts_area service (see above) |
| ts_target | (empty) | host:port, for ts_area network |
| ts_hours | 2 | how many hours back the event logs are read |
| ts_output_lines | 60 | keep the last this-many lines of each command |
| win_ts_areas | overview: checks: [disk, services, performance, time, eventlog] commands: - {why: "Window… | What each area runs. why is printed above the command's output. Commands must be read-only. |

## Settings of one host

Set these on the host itself: its host variables in the inventory, or the host's **Variables** box in AAP. Never in a job template's Variables - there one value would be used for every host. When a host has none, the role works the value out from the host, so no role file has a default for them.

| Setting | Role | When not set | What it does |
|---|---|---|---|
| net_config_address | net_config | the host's ansible_host, else its name | The address the backup and load jobs connect to, when it is not ansible_host (PAN-OS) |

| Setting | Role | When not set | What it does |
|---|---|---|---|
| `net_config_platform` | net_config | from `ansible_network_os` | `ios` or `panos`: the platform of a host that has no `ansible_network_os` (`cisco.ios.ios` / `paloaltonetworks.panos`) |
| `net_cert_address` | net_certs | the host's `ansible_host`, else its name | The address the certificate report connects to (a host of `net_cert_group`) |
| `net_cert_servername` | net_certs | the host's name | The name sent in the TLS hello (SNI): the certificate a device with several names shows for this one |